Space-Based AI Computing Feasibility Analysis: Physical Limits, Engineering Constraints, and Commercial Prospects

Sammy Zeng

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Abstract

This report presents an independent technical analysis of the physical feasibility, engineering viability, and commercial prospects of orbital (space-based) AI computing. Starting from first principles, the report adopts a three-tier progressive framework—physical, engineering, and commercial—and cross-validates the public statements of Jensen Huang (NVIDIA) and Elon Musk (SpaceX) using the Stefan–Boltzmann law, a bottom-up mass chain, a three-branch TCO model, and a four-tier evidence grading system. Core findings: physically feasible—the 1,400 W/m^2 heat dissipation density claimed by AI1 is consistent with radiative heat transfer physics; engineering-fragile—the compute payload mass chain has been reconstructed from the kW/t inverse method to a ground-hardware-anchored approach, but the mass and cost of several non-compute subsystems still rely on engineering inference; commercially scenario-layered—under a unified Blackwell baseline, the 10-year GaAs chain yields a TCO of $764.98M (an optimistic comparable baseline at 1 MW sustained IT load / 10-year total window under officially claimed specifications), while shorter lifetime or alternative photovoltaic routes significantly increase cost. The probability of synchronized multi-parameter improvement is judged to be low based on current evidence.

Disclaimer on Report Nature: This report is an early-stage scenario feasibility analysis based on publicly available information and first principles;

[1]Email: jop.sammy@163.com ORCID: 0009-0007-4382-0040

it is neither a finalized engineering report nor a complete project financial audit. All TCO figures are model outputs under given assumptions and do not represent predictions or audit conclusions for any specific project cost. The report strives for rigor through cross-validation standards, but its conclusions should be understood as “directional judgments under currently available public information” rather than “deterministic results upon which decisions may be based.” Further data increments (AI1 on-orbit validation, disclosure of first-hand hardware specifications) will significantly affect the precision of quantitative conclusions.

Keywords: space-based AI computing; orbital data center; radiative heat dissipation; Stefan–Boltzmann law; total cost of ownership; Jensen Huang; Elon Musk; NVIDIA Space-1; SpaceX AI1

Contents

Executive Summary

Core Conclusions

This report presents an independent technical analysis of the feasibility of low Earth orbit (LEO) AI computing data centers, examined across three progressive tiers: physical, engineering, and commercial. The core conclusions are as follows:

- **Physically feasible, but approaching the ceiling of current materials.** The Stefan–Boltzmann law does not prohibit 1 MW-class orbital heat dissipation—the 1,400 W/m^2 claimed by the AI1 satellite (dual-sided radiation, $\varepsilon \approx 0.90$, $T \approx 342.5$ K) is physically self-consistent. However, this operating point nearly exhausts the available margin of current high-emissivity thermal control coatings after end-of-life (EOL) degradation. A doubling of heat dissipation density has no single-technology path to support it before 2035. (Full derivation in Chapter 4, Appendix C §C.1)
- **Engineering-fragile; single-satellite mass and reliability are the primary bottlenecks.** Sending computing hardware into space and sustaining its on-orbit operation is a multi-constraint coupled engineering problem. The compute payload mass chain has been reconstructed from the "kW per ton" inverse method used in prior industry analyses (which has been permanently deprecated due to unreliable denominator sourcing and definitional ambiguity; see Appendix E) to a line-item accumulation method anchored on NVIDIA GB300 NVL72 ground hardware. Under the baseline scenario, the total single-satellite mass is approximately 8.48 t, with the photovoltaic array (55.3%) and system margin/redundancy (21.9%) together accounting for the vast majority of manufacturing cost. (Full derivation in Chapter 5, Appendix C §C.3)
- **Commercially scenario-layered; no "single correct number" exists.** This report establishes three comparable economic branches: the 10-year GaAs primary chain baseline has a total cost of approximately \$0.77B (range \$0.55B–\$1.1B, depending on variation in key assumptions) (single-generation 1 MW sustained IT load, 10-year total window, \$200/kg launch unit cost); the 5-year GaAs comparison branch rises to approximately \$1,341.26M due to the need for a full satellite re-launch; the 5-year HJT parallel route gives a reconciled estimate of approximately \$1,005M, but is excluded by the AI1 70 m wingspan geometric constraint and cannot serve as a deployable primary route. Launch cost accounts for only about 1.8% of TCO—the factors that truly determine commercial viability are photovoltaic array cost, on-orbit lifetime, and hardware reliability. (Full results in Chapter 6 and the script routing in Appendix D)

Primary Chain TCO Figures (Quick Reference)

Table 1: Three-Branch TCO Overview (common denominator: 1 MW sustained IT load, 10-year total window)

Branch	Single-Sat. Mass	10-Year Total Cost	Status
10yr-GaAs (primary)	8.48 t	**$764.98M**[1]	The only single-generation primary chain currently capable of being fully specified
5yr-GaAs (comparison)	7.10 t	$1,341.26M	Lifetime shortened to 5 years, requiring one re-launch
5yr-HJT (parallel)	9.13 t	≈ $1,005M (reconciled estimate)	Geometrically excluded by 70 m wingspan; figure is a reconciled estimate, not a lower bound

Total cost formula: Σ (manufacturing cost + launch cost + insurance (satellite manufacturing value$\times$3–5% + launch cost$\times$8%; current model only includes the launch cost portion as a lower-bound estimate) + operations ($5M/yr$\times$10yr) + decommissioning ($3M/gen$\times$generations)). Single-generation satellite equivalent $= 1$ MW/120 kW ≈ 8.33. See relevant tables in Chapter 6 and Appendix D §D.3 for details.

Characterizing the Divergence

The publicly expressed differences between NVIDIA CEO Jensen Huang and SpaceX CEO Elon Musk on the topic of space-based AI computing constitute the core research motivation of this report.

But their divergence is *not about "whether to do it."* Both agree that space-based AI computing has long-term inevitability at a civilizational scale; the real divergence lies in *"how fast it can be done" and "where the bottlenecks are"*:

- Jensen Huang, approaching from a chip engineering perspective, emphasizes the coupled engineering difficulty of heat dissipation flux, cosmic rays, solar storms, and micrometeoroids, and cites NVIDIA's existing deployment of CUDA in space to demonstrate that he is not opposed to space-based computing but holds a more conservative judgment on pace.

[1] Model output for the 10-year GaAs chain using SpaceX's publicly stated 120 kW sustained IT as the satellite-count denominator and after LEO orbital power chain closure; should be understood as the "optimistic comparable baseline under officially claimed specifications," not an audited value of future project cost.

- Elon Musk, approaching from a civilizational energy perspective, embeds orbital AI within a Kardashev-scale narrative and implicitly assumes that all relevant parameters (launch cost, photovoltaic efficiency, battery life, manufacturing scale) will improve synchronously.
- This report's independent judgment: **synchronized improvement is possible, but the probability of multi-parameter synchronized improvement is insufficient to support a near-term definitive commitment.** The divergence is not fundamentally about "who calculated wrong"—it is that the two parties use different time windows, power definitions, architecture generations, and system boundaries. When these differences are parameterized, the divergence reduces from a "directional dispute" to a "parametric assumption difference." (See Appendix A for the specification alignment matrix.)

Reading Guide

This report adopts a modular structure; different readers may selectively read according to their needs:

- **Readers needing only the conclusions:** This chapter suffices. The three-tier judgment and primary-chain TCO figures given here are the core outputs of the full report.
- **Readers needing background:** Continue to Chapter 1 (definition of space-based AI computing, key participants, and the 2024–2026 event timeline).
- **Readers needing the analytical framework:** Read Chapters 1–2. Chapter 2 defines the five unified specification dimensions (power, time window, architecture generation, cost boundary, launch unit cost) and the three-tier progressive framework, which are prerequisites for understanding subsequent chapters.
- **Readers needing to verify the factual basis:** Read Chapter 3 (complete compilation and divergence characterization of 12 first-hand statements) and Appendix D (five-tier evidence routing table, traceable from text conclusions back to original URLs).
- **Readers needing to examine technical judgments:** Read Chapters 4–6 and Appendix A (specification alignment matrix), Appendix B (47-parameter line-item verdicts), and Appendix C (core calculation methods and formulas).
- **Readers needing the complete chain of reasoning:** Read the full text and all five appendices.

Chapter and Appendix Index

- **Chapter 1 Background and Research Subject**: Definition of space-based AI computing, motivations for its proposal, key participants (SpaceX/NVIDIA), 2024–2026 timeline, research scope and boundaries.
- **Chapter 2 Analytical Methods and Comparison Baselines**: Physical-engineering-commercial three-tier progressive framework, definitions of the five unified specification dimensions, preliminary specification alignment matrix.
- **Chapter 3 First-Hand Statements and Divergence Characterization**: Musk's 7 statements summary table, Huang's 5 statements summary table, divergence characterization (assumption differences rather than right/wrong).
- **Chapter 4 Scientific Feasibility**: Stefan–Boltzmann law derivation and heat dissipation area, real LEO thermal environment constraints, heat dissipation density boundary.
- **Chapter 5 Engineering Feasibility**: Bottom-up compute payload mass chain (replacing the kW/t inverse chain), GaAs/HJT power system route comparison, battery three-branch analysis, full satellite mass breakdown, on-orbit lifetime and maintenance, launch capacity reality check.
- **Chapter 6 Commercial Feasibility**: Three-branch TCO complete breakdown and comparison, quantitative impact of specification differences, flip-condition analysis.
- **Chapter 7 Synthesized Judgment**: Three-tier convergence into known facts / computable chains / currently undecidable, timeline outlook, robustness statement of conclusions.
- **Appendix A**: Specification alignment matrix (complete comparison of 7 scenarios $\times$ 9 dimensions).
- **Appendix B**: 47-parameter line-item verdict table (tier, value, verdict status, source routing).
- **Appendix C**: Core calculation methods and formulas (S-B heat dissipation derivation, orbital power energy budget, bottom-up mass chain).
- **Appendix D**: Five-tier evidence routing table (conclusion $\rightarrow$ cross-verification $\rightarrow$ JSON $\rightarrow$ verdict record $\rightarrow$ URL/formula).
- **Appendix E**: Deprecated derivation chain documentation (reasons for deprecating the kW/t inverse chain, prohibited-line list, parameter usage boundary declaration).

Evidence System Notes

All 47 key parameters involved in this report have been graded and adjudicated line-by-line according to the four-tier evidence system:

- **Tier 1 (physical derivation chain, 5 parameters)**: Directly derived from explicit physical formulas and boundary conditions, e.g., solar constant, heat dissipation area.
- **Tier 2 (authoritative sources, 7 parameters)**: From traceable authoritative public sources, e.g., AI1 power parameters (multi-source cross-validated), AzurSpace 4G32 GaAs efficiency.
- **Tier 3 (transferable evidence, 17 parameters)**: From transferable industry literature, e.g., Starlink battery DoD/specific energy, terrestrial HJT module wholesale prices.
- **Tier 4 (reasonable inference, 14 parameters)**: Insufficient public data, based on engineering inference, e.g., radiator areal density, platform coefficient, PCDU specific power, etc.—involving approximately 35% of manufacturing cost items. The Tier-4 share of 30% (14/47) is an objective constraint of insufficient AI1 public disclosure, not an analytical deficiency.

Every key figure in the main text carries an evidence-tier annotation; see Appendix B for the complete parameter verdict table; see Appendix D for traceability routing.

Methodology Statement

This report proceeds from physical laws (Stefan–Boltzmann), through engineering hardware anchors (NVIDIA GB300 NVL72), to a complete computational chain modeled in independent Python, following a first-principles bottom-up verification methodology. The photovoltaic array ($\sim$50%) in the TCO is cross-anchored via Tier-2/Tier-3 evidence; system margin/redundancy/additional (AIT margin) remains a dominant cost item ($\sim$20%), though its value still carries Tier-4 inference attributes. Even if Tier-4 parameter values undergo reasonable variation, directional conclusions will not experience order-of-magnitude flips. This report pursues robustness of conclusions rather than predictive precision.

1 Background and Research Subject

1.1 What Is Space-Based AI Computing

Space-based AI computing (orbital compute) refers to computing satellites or satellite clusters deployed in Earth orbit whose primary workload is AI training or inference. It differs from conventional communication satellites (such as Starlink)—communication satellites take signal relay as their primary function, with power consumption typically in the hundreds to thousands of watts; space-based AI computing satellites may consume tens to hundreds of kilowatts, with power density 1–2 orders of magnitude higher, posing fundamentally different engineering challenges for heat dissipation, power supply, and on-orbit reliability.

Compared to terrestrial data centers, space-based AI computing has four essential physical differences:

1. **Single-mode heat dissipation:** In space there is no air, no water, and no conduction or convection. All waste heat must be rejected to deep space via radiation—this is the hard physical constraint of the Stefan–Boltzmann law and constitutes the core analytical subject of Chapter 4 of this report.

2. **Fixed energy source:** Space-based photovoltaics are the only sustainable energy form. In LEO there are no fossil fuels and no grid connection; all electrical power must be obtained from photovoltaic arrays and battery energy storage systems.

3. **Unique deployment method:** All hardware must be launched into orbit via rocket, with the mass and volume of a single launch constrained by the launch vehicle. Launch cost (per kilogram) directly affects total cost of ownership (TCO), although this report will demonstrate that its impact is smaller than intuitive expectation.

4. **Limited maintenance:** Once in orbit, hardware is physically inaccessible. One cannot dispatch personnel to swap GPUs, add coolant, or replace failed power modules as one would in a terrestrial data center. The economic feasibility of on-orbit servicing is an important unresolved question.

1.2 Why Space-Based AI Computing Has Been Proposed

The proposal of space-based AI computing did not arise in a vacuum; it responds to three increasingly pressing constraints facing terrestrial AI infrastructure—often called the “three walls”:

1. **The Power Wall:** The power demand of a single AI data center has rapidly escalated from tens of MW to the GW level. Some regional grids cannot satisfy the power connection applications for new data centers; site selection has become a bottleneck.

2. **The Land Wall:** Hyperscale data centers occupy hundreds of acres and face the dual pressures of land cost and community opposition in densely populated areas.

3. **The Cooling Wall:** The power density of GPU racks continues to climb (GB300 NVL72 can exceed 140 kW per rack); liquid cooling is replacing air cooling, but the cooling infrastructure itself consumes significant amounts of electricity (PUE typically between 1.1–1.4) and water resources.

Proponents of space-based AI computing argue that the orbital environment naturally circumvents these constraints:

- Solar power is available 24/7 (LEO has no atmospheric "weather" concept blocking sunlight, only brief eclipse periods—about 35–37 minutes per 96–97-minute orbital period).

- Theoretically unlimited physical expansion space (LEO is a three-dimensional spherical shell; deployment density is far from the upper limit of space debris constraints).

- Radiating waste heat into the 3 K deep-space background is physically more efficient than rejecting heat into the 300 K atmosphere on the ground.

1.3 Key Participants

1.3.1 SpaceX and Elon Musk

SpaceX is the most aggressive proponent of orbital AI computing. As of June 2026, its principal actions include:

- **AI1 Satellite:** On June 8, 2026, SpaceX disclosed AI1 satellite parameters in detail for the first time via an official video—wingspan 70 m, peak power 150 kW, sustained power 120 kW, LEO (approximately 600 km), heat dissipation density approximately 1,400 W/m^2 (physical panel area). The first two prototypes are planned for launch in early 2027. This public release provided a verifiable anchor for analyses that had previously relied entirely on speculation[1].

- **Starship Super-Heavy Launch System:** Starship is the sole launch capacity vehicle for large-scale orbital deployment. The V3 version targets a payload capacity of 100+ tons to LEO (full-reuse mode). SpaceX has obtained FAA approval for Starship launches at Boca Chica (25 launches/yr) and Kennedy Space Center LC-39A (44 launches/yr), for a combined annual cap of 69 launches[2, 3].

- **FCC Million-Satellite Application:** In January 2026, SpaceX submitted an authorization application to the FCC to deploy up to 1 million orbital AI data center satellites in the altitude range of 500–2,000 km. The engineering feasibility of this application will be discussed in Chapter 5 of this report[4].

- **Orbital AI Positioning in the S-1 Prospectus:** In the May 2026 SEC S-1/A filing, SpaceX positioned orbital AI computing as "a barrier-level challenge that only SpaceX can solve," with deployment planned to begin in 2028 at a target of 100 GW per year. The IPO valuation range was $1.75–2 T[5].

- **Terafab Chip Factory:** In March 2026, Musk announced a $25 billion investment in the Terafab mega chip factory in Texas, a joint venture of SpaceX, Tesla, and xAI, claiming that 80% of its compute output would be destined for orbital deployment[6].

1.3.2 NVIDIA and Jensen Huang

NVIDIA's participation in the space-based AI computing arena stands in sharp contrast to SpaceX: product-driven rather than narrative-driven, embedding pragmatic engineering reservations within an optimistic vision.

- **Space-1 Vera Rubin Platform:** At the March 2026 GTC conference, NVIDIA unveiled the Space-1 space computing platform, centered on the Rubin GPU—which NVIDIA officially claims delivers 25$\times$ the space inference performance of the H100. The platform includes the IGX Thor edge computing module and has established partnerships with six space enterprises[7].

- **Jensen Huang's Prudent Stance:** Despite releasing hardware products, Huang has repeatedly emphasized across multiple venues that heat dissipation is the fundamental bottleneck for space-based computing. His statement on the March 2026 All-In Podcast—"It will take years. It's okay; I have time."—is the public remark closest to his true timeline judgment[8].

- **Extreme Co-Design Philosophy:** On the Lex Fridman Podcast, Huang articulated the "extreme co-design" philosophy—that chips, networking, cooling, and algorithms must be jointly optimized for space-based computing to be engineering-viable. This implies his view that simply "moving terrestrial GPUs to space" is insufficient[9].

1.3.3 Other Relevant Parties

- **Exlumina**: A startup focused on space edge AI computing, positioned as a LEO orbital inference service provider. - **Google Suncatcher (arXiv preprint, 2025)**: An academic paper published by the Google research team exploring the topology and bandwidth requirements of space-based AI clusters, providing an independent reference anchor for the tens-of-Tbps inter-satellite laser link (ISL) demand magnitude used in this report's communication subsystem mass/cost estimation. - **Multiple Satellite Communication Enterprises**: The engineering experience of large-scale LEO constellations being deployed by AST SpaceMobile, Telesat, and others (e.g., deployable antennas, satellite bus design) constitutes the industrial technology foundation for space-based AI computing.

1.4 Key Event Timeline

Table 2: Key Event Timeline for Space-Based AI Computing (2024–2026)

Date	Event	Impact on This Report's Analytical Framework
2024–09	Musk X post: Kardashev metric	First linkage of space AI with civilizational energy narrative, establishing Musk's discursive framework[10]
2025–11	Saudi Investment Forum: first joint appearance	The closest public record to "Huang responding to Musk"; Huang addressed heat dissipation from a hardware mass perspective[11]
2026–01	SpaceX FCC million-satellite application	Deployment scale first disclosed in official document form, shifting discussion from "is it possible" to "at what scale"[4]
2026–02	NVIDIA Q4 Earnings Call	Huang's first qualitative characterization of the current phase of space-based computing as "economics not ideal"[12]
2026–03	NVIDIA GTC 2026: Space-1 Rubin	NVIDIA transitioned from "commenting" to "product," marking a watershed in the space computing discussion[7]
2026–03	All-In Podcast	Huang delivered the remark closest to a timeline judgment: "it will take years"[8]
2026–03	Terafab chip factory announcement	Musk's supply chain arrangement extended from "sending hardware to orbit" to "in-house chip production"[6]
2026–05	SpaceX S-1 prospectus	Orbital AI became the core IPO narrative, with a 2028 deployment start and 100 GW/yr target[5]
2026–06	AI1 satellite parameter release	Provided this report's most important single verifiable anchor: 70 m wingspan, 150 kW peak, 120 kW sustained[1]

1.5 Scope of This Report

1.5.1 What Is Studied

- **Orbital regime:** Low Earth Orbit (LEO, approximately 600 km), which is the target orbit of AI1 and the focus of current discussion.
- **Unit of analysis:** 1 MW sustained IT load (i.e., usable power ultimately delivered to computing hardware), 10-year total window (with branch calculations including re-launch cost), NVIDIA Blackwell architecture (GB300 NVL72 as ground hardware anchor) as baseline.

- **Analytical dimensions:** Physical feasibility (is heat dissipation possible—Chapter 4), engineering feasibility (can it be built and operated—Chapter 5), commercial feasibility (is it worth investing—Chapter 6).
- **Technology routes:** GaAs (gallium arsenide) III-V multi-junction solar cells as the primary route (this report's independent analysis indicates this is almost certainly the only viable route for AI1), HJT (heterojunction silicon cells) as the parallel comparison route (cost potential noteworthy but excluded by deployment constraints).

1.5.2 What Is Not Studied

- **MEO/GEO/Deep-space computing:** Although Musk's narrative touches on lunar manufacturing and electromagnetic launch, these lie outside the analytical scope of this report (not excluded as potential future extended research).
- **Communication constellation infrastructure:** Space-based AI computing requires inter-satellite laser links (ISL) and ground gateways as supporting infrastructure, but the communication system is incorporated into TCO only at the satellite subsystem level in mass and cost dimensions, not as an independent research topic.
- **Comprehensive terrestrial data center substitution analysis:** This report uses $64.6M USD (10-year total window, 1 MW IT load) as the reference TCO for a terrestrial AI data center, but does not perform a detailed breakdown comparison of terrestrial technologies.
- **Investment advice:** This report is a technical feasibility analysis and does not constitute any form of investment recommendation.
- **Regulation and spectrum:** Policy dimensions such as FCC applications, ITU spectrum coordination, and space traffic management lie outside the technical analysis scope of this report.

2 Analytical Methods and Comparison Baselines

The feasibility analysis of space-based AI computing spans multiple disciplines—thermal physics, aerospace engineering, photovoltaic technology, battery chemistry, launch vehicles, semiconductor manufacturing—and the discussants (Musk, Jensen Huang, third-party analysts) frequently use different specifications and assumptions, producing superficially incompatible numbers. Before entering the specific analysis, this chapter establishes a unified analytical framework and terminology for the full report, ensuring that readers will not misunderstand subsequent chapters due to specification confusion.

2.1 Three-Tier Progressive Analytical Framework

This report's analysis proceeds along three progressive tiers—physical $\rightarrow$ engineering $\rightarrow$ commercial—with each tier unfolding on the premise that the preceding tier's conclusions hold:

1. **Physical Feasibility (Chapter 4)**: Judgment is made using only physical laws and publicly verifiable physical constants. If the Stefan–Boltzmann law indicates that the required heat dissipation area is unacceptable, or if the LEO thermal environment renders the effective heat dissipation flux far below a workable level, then the engineering and commercial analyses lose their premise. *Criterion: Is it physically possible?*

2. **Engineering Feasibility (Chapter 5)**: Under the premise of physical feasibility, the analysis examines whether it can be realized from an engineering standpoint—how heavy the satellite is, whether it can fit into a rocket, whether it can sustain continuous on-orbit operation, and whether it can be repaired after failure. *Criterion: Can it be built and operated continuously?*

3. **Commercial Feasibility (Chapter 6)**: Under the premise of engineering feasibility, the analysis examines whether it is worth investing—TCO (Total Cost of Ownership) comparison against terrestrial data centers, and which parameter combinations would change the directional conclusion. *Criterion: Is it commercially worthwhile?*

The core value of this progressive framework is avoiding "cross-tier confusion": physical feasibility does not imply engineering feasibility (e.g., the heat dissipation area requires only 0.71 km^2 physically, but engineering the precise deep-space pointing of $\sim$6,400 satellites' radiator surfaces is an entirely different problem); engineering feasibility does not imply commercial feasibility (e.g., even if every satellite can be built, the total cost may render them economically uncompetitive against terrestrial alternatives).

2.2 Definitions of the Five Unified Specifications

All figures in this report are based on the following unified specifications. The rationale for each specification choice is explained in each sub-entry.

2.2.1 Power Specification

- **GPU/Compute-side nominal peak (150 kW)**: The “150 kW of GPU power” stated in SpaceX’s public materials refers to the peak DC input power at the GPU level, widely propagated as AI1’s core identifying number [Tier 2·authoritative source, cross-validated by three independent media outlets: Inkl/Tom’s Hardware, Knightli, and Indexbox][1]. This report uses 150 kW as the public-discourse anchor in its narrative; the power system’s engineering closure (array area, PCDU mass, photovoltaic cost) is still derived from LEO orbital energy balance, not estimated directly from 150 kW.

  150 kW and 120 kW/210 kW are not the same physical quantity: 150 kW is “how much the satellite’s compute side can nominally run at” as a peak figure; after traversing the full energy chain of photovoltaic array→battery→PCDU→GPU, LEO orbital physics only supports approximately 86 kW of sustained IT load. The forward physical requirement corresponding to 120 kW sustained IT is approximately 210 kW array EOL output (full derivation in the Chapter 5 subsection “Power Specification Clarification” and Appendix C §C.2). To clearly distinguish these figures, Table 3 presents the complete four-layer power specification hierarchy.

- **Sustained IT load (120 kW)**: The power value corresponding to AI1’s sustained computing capacity, as officially disclosed by SpaceX[1]. This report uses 120 kW as the conversion denominator for the number of satellites required for 1 MW deployment (1 MW / 120 kW $\approx$ 8.33 satellites). Under the actual constraint of LEO orbital energy balance (150 kW peak corresponds to only $\sim$86 kW sustained), a 1 MW deployment would require approximately 12 satellites (rather than 8.3), with a correspondingly higher TCO.

- **IT load power (1 MW sustained, common denominator)**: The usable power ultimately delivered to computing hardware. This report selects 1 MW sustained IT load as the comparison denominator for three reasons: (1) the figure is large enough to be meaningful for economic discussion; (2) it lands exactly within the integer-multiple scaling range of AI1 (120 kW/satellite) at approximately 8.33 satellites; (3) mature public TCO data exist for terrestrial 1 MW-class AI clusters as a comparison reference.

Table 3: AI1 Power Specification Hierarchy—Physical Meanings at Different Electrical Nodes

Specification	Value	Physical Location	Usage in This Report
GPU/compute-side nominal peak	150 kW	GPU DC input / compute payload side	Aligned with SpaceX public statements; public memory anchor and thermal peak reference
Sustained IT load	120 kW	Orbit-average load actually delivered to computing hardware	1 MW conversion denominator: 1 MW/120 kW $\approx$ 8.33 satellites
Array EOL output requirement	$\approx$210 kW	Photovoltaic array / upstream PCDU output	Power system, PV area, PCDU mass, array cost calculations
Array BOL nominal requirement	$\approx$230 kW	Photovoltaic array output at beginning of life	PV cost and area verification reference

In one sentence: 150 kW is "how much the satellite's compute side can nominally run at" as a peak figure; 210 kW is "the power that the photovoltaic array must deliver at end of life to sustain 120 kW of sustained IT load over the long term"—the two are not the same physical quantity, spanning the full energy conversion chain (photovoltaic array→battery→PCDU→GPU) and must not be conflated as the same specification.

2.2.2 Deep Explanation of the Power Specification: Why the 150 kW Official Figure Is Retained

AI1's "150 kW" has become the core identifying number in public discourse. If this report were to immediately replace it in the main narrative with the array requirement power under LEO energy balance ($\sim$210 kW), it would easily divert the discussion into specification disputes prematurely. Therefore, this report retains 150 kW in the main narrative as SpaceX's publicly claimed compute-side nominal peak, and uses 120 kW sustained IT load as the 1 MW conversion denominator; the engineering chapters then provide the complete LEO power system closure calculation.

Power system calculations are still performed via LEO orbital sunlit/eclipse energy balance closure: to sustain 120 kW of sustained IT load, accounting for eclipse power supply, battery recharge, conversion losses, and end-of-life degradation, the array EOL output requirement is approximately 210 kW. Consequently, this report's photovoltaic area, PCDU mass, and photovoltaic cost are not estimated at 150 kW array output, but rather at the physically closed $\sim$210 kW array requirement. See the Chapter 5 subsection "Power Specification Clarification" and Appendix C §C.2 for the complete energy chain derivation.

In other words, 150 kW in this report is the "public-discourse anchor" and "compute-side peak reference," not the power system's closure power. If 150 kW were interpreted as the array peak, it would support only about 86 kW of sustained IT load in LEO, and both the required satellite count and TCO would increase (see the Chapter 5 subsection "Misreading Check" for the misreading verification table).

2.2.3 Time Window Specification

- **5-year branch**: A single-generation satellite serves on orbit for 5 years and then de-orbits. Under the comparison denominator of a 10-year total window, this means 2 generations of satellites must be launched (1st generation + 1 re-launch). The rationale for selecting 5 years: LEO satellite atmospheric drag lifetime, battery cycle life, and hardware degradation rate at 600 km orbit indicate that 5 years is a reasonable conservative estimate based on engineering practice.

- **10-year branch**: A single-generation satellite serves on orbit for 10 years (no full-satellite re-launch). This is an optimistic but not unreasonable engineering target—requiring a shift in battery chemistry from NMC to longer-lived alternatives (see Chapter 5 battery three-branch analysis for details).

- **Comparison denominator**: 10-year total window. All branches' TCO is normalized to the same 10-year window, so that the re-launch cost of the 5-year branch is explicitly included rather than implicitly ignored.

2.2.4 Architecture Generation Specification

- **Primary architecture: NVIDIA Blackwell (GB300 NVL72)**. This is the latest NVIDIA GPU architecture that has been publicly released with independently verifiable specifications as of mid-2026. Detailed GB300 NVL72 compute tray specifications are sourced from the Lenovo product guide (29 kg/tray, 4 GPU + 2 CPU) [Tier 2·authoritative source][13], providing an A1-grade ground hardware anchor for the compute payload mass chain.

- **Rubin not adopted as primary**: Although the NVIDIA Vera Rubin architecture has been announced (GTC 2026), the full satellite BOM, space-grade parameters, and first-hand public specifications are still insufficient; at present only architectural efficiency remapping is possible (see Chapter 5 for details), and it cannot serve as the primary chain for a complete commercial analysis.

- **H100 not adopted as primary**: Previous-generation architecture, with significantly lower efficiency and compute density compared to the current discussion.

2.2.5 Cost Boundary Specification

This report's cost model covers the following items, each with a complete definition and boundary conditions:

- **Manufacturing cost**: Comprising 9 subsystems—compute payload, heat dissipation system, photovoltaic array, battery, power electronics/distribution (PCDU), bus/structure, propulsion, communication/laser links, system margin/redundancy/additional (AIT margin: covering redundant hardware, system mass growth allowance, and integration additional mass). Each item is annotated with its evidence tier.
- **Launch cost**: $200/kg baseline (Starship V3 full-reuse target), sensitivity analysis covering $100–500/kg. The basis for this value is given in Chapter 6 "Launch Unit Cost Sensitivity Analysis" and this chapter's "Launch Unit Cost Specification Notes."
- **Insurance cost**: 8% of launch cost (based on space insurance industry practice).
- **Operations cost**: $5M/yr (ground TT&C, constellation management, and software updates).
- **Decommissioning cost**: $3M/generation (controlled re-entry).
- **Not included**: Ground gateway/user segment, spectrum licensing fees, non-recurring engineering (NRE)—these either are highly scenario-dependent or lack public data, making them impossible to include in a unified comparison framework.

2.2.6 Launch Unit Cost Specification

$200/kg is used as the baseline launch unit cost. This selection is based on three tiers of evidence: (1) Historical tier—launch costs have fallen by approximately 100× over the past 15 years (from Shuttle ~$54,500/kg to Falcon 9 reusable ~$2,720/kg)[14]; (2) Economic threshold tier—<$200/kg is widely regarded as the critical threshold for making large-scale LEO deployment enter an economically viable regime; (3) Engineering target tier—Starship V3's "100+ tons full reuse" target corresponds to a marginal cost of approximately $100–200/kg[15]. This report positions $200/kg as a "mature reuse engineering target" rather than a "currently achieved value," and covers a broader $100–500/kg range in the Chapter 6 sensitivity analysis.

2.3 Specification Alignment Matrix (Abridged Version)

Table 4 presents a simplified comparison of this report's primary chain against major external specifications. The physical meaning of "the same MW" differs radically under different specifications—if two analyses use different time windows, power definitions, or architecture generations, directly comparing their TCO figures is methodologically invalid.

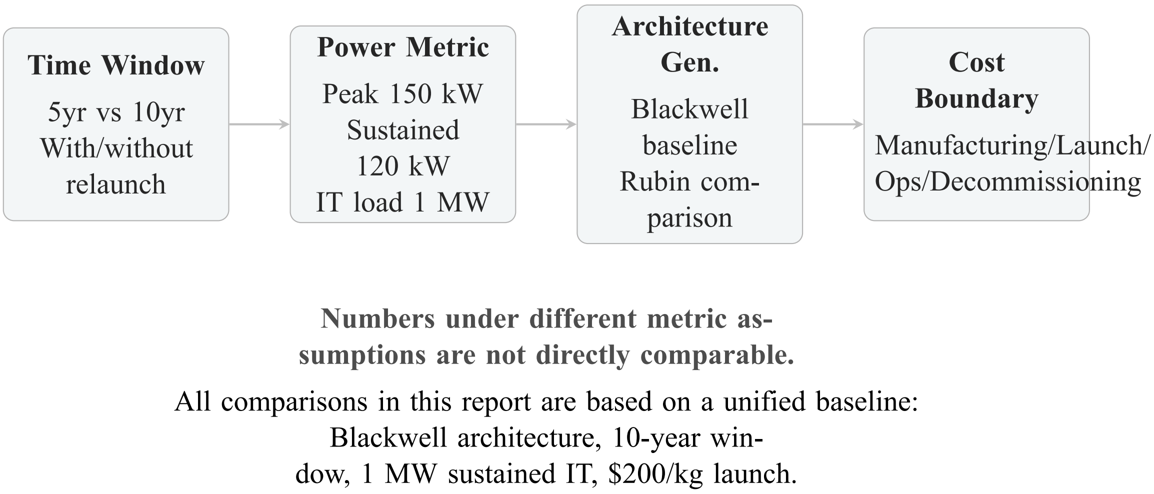

Figure 1: Specification alignment schematic—figures under different specification assumptions are not directly comparable; all comparisons in this report are based on a unified baseline (Blackwell architecture, 10-year window, 1 MW sustained IT, $200/kg launch).

Table 4: Specification Alignment Matrix (abridged)—values and comparability across key dimensions for each specification

Dimension	**Primary Chain**	**Musk Public**	**Huang Public**	**Prior Industry**	**Common Media**
Time window	10 yr	Unspecified	"Several years"	10 yr	5 yr
Power spec.	1 MW sustained IT	GW-class deployment	Unquantified	1 MW IT	1 MW
Architecture gen.	Blackwell	Unspecified	Rubin	Blackwell	Unspecified
PV route	GaAs primary	AI1 (undisclosed)	Undisclosed	GaAs	HJT
Launch cost	\$200/kg	Optimistic	Not addressed	\$200/kg	\$50–100/kg
Lifetime assumption	10yr single-gen	Unspecified	Reserved	10 yr	5 yr
Comparability	Baseline	Comparable after conversion	Partially comparable	Partially comparable	Comparable after conversion

See Appendix A for the complete specification alignment matrix (7 scenarios $\times$ 9 dimensions, with source annotations and quantitative impact analysis).

2.4 Why Specification Differences Matter

Before reading the subsequent chapters, it is necessary to clarify a core methodological point: the vast majority of contradictions between different figures in this report are not calculation errors, but specification differences. For example:

- One analysis yields "10-year total cost \$765M," another yields "\$1,341M"—the difference primarily arises from the former assuming a 10-year lifetime and the latter assuming a 5-year lifetime (requiring one re-launch). Both could be "correct," but the comparison requires an explicit declaration of the lifetime assumption.

- One analysis says "launch cost can fall to \$50/kg," another says "\$200/kg"—the former may implicitly assume learning-curve effects across thousands of launches; the latter may consider only the initial FAA-approved cap of 69 launches/yr. The two are discussing different scenarios at different points in time.

- One analysis says "HJT is 70–100$\times$ cheaper than GaAs," another says "GaAs is the only viable route"—the former starts from cell-level component cost; the latter starts from full-satellite wingspan and deployment constraints. Both are valid within their respective local scopes, but the global optimum requires integrated consideration of all constraints.

This report's approach is to decompose the divergence into independently parameterizable dimensions (lifetime, battery, power route, mass chain, price chain), verify each dimension, and show how conclusions change when each dimension's parameters are varied independently. This methodology shrinks the "space of disagreement" from fuzzy positional opposition to quantifiable parameter sensitivity intervals.

2.5 Methodology Statement: First-Principles Bottom-Up Verification

The analytical methodology adopted in this report—from physical laws (Stefan–Boltzmann) and engineering hardware anchors (NVIDIA GB300 NVL72 official specifications) to a complete computational chain via independent Python modeling—follows the methodology of first-principles bottom-up verification. This methodology ensures that:

1. **Primary conclusions are traced back as far as possible to publicly and independently verifiable original sources**—most key conclusions can be traced from the main-text conclusion back to original URLs or physical formulas along the five-tier routing in Appendix D; a minority of links (approximately 30% of Tier-4 inference parameters) still rely on internal verdict records and scripts pending open-source release.

2. **Even if Tier-4 parameter values (14 reasonably inferred parameters) undergo variation, no order-of-magnitude conclusion flip will occur**—because in the TCO cost structure, the photovoltaic array ($\sim$50%) is cross-anchored via Tier-2/Tier-3 evidence; AIT margin remains a dominant cost item ($\sim$20%), though its value still carries Tier-4 inference attributes.

3. **Unlike "generic estimation" methods based on simple proportional extrapolation or linear scaling**, this report's estimates explicitly annotate the evidence tier, inferential basis, and failure risk at every step.

4. **This report pursues not predictive precision "to the decimal point" but robustness—the property that directional conclusions do not flip even when parameters vary within reasonable intervals.**

3 First-Hand Statements and Divergence Characterization

This chapter faithfully compiles all known public statements by Elon Musk (SpaceX CEO) and Jensen Huang (NVIDIA CEO) as of June 2026, and then characterizes their actual technical divergence. This chapter does not judge "who is right or wrong"—that task is undertaken from an independent analytical standpoint by Chapters 4–6—but rather establishes the factual foundation enabling readers to understand where the two figures "come from, what they have said, and where they diverge."

3.1 Musk's Statements (7 First-Hand Sources)

From September 2024 to June 2026, Musk systematically articulated the inevitability of space AI computing through the X platform, public forum speeches, and SpaceX official events. Table 5 compiles all 7 first-hand statements, each annotated with: date, venue, core argument, quantitative claim, and the corresponding analytical chapter in this report.

Table 5: Musk's 7 First-Hand Public Statements on Space-Based AI Computing

#	Date	Venue	Core Argument	Quantitative Claim	Ch.
Statement 1	2026-06-08	SpaceX official video (AI1 release)	AI satellite design simpler than Starlink; orbital AI positioned as "the necessary path for humanity toward a Type II civilization"	Wingspan 70 m, peak 150 kW, sustained 120 kW, heat dissipation density 1,400 W/m^2; 1 GW/yr by 2027	Ch. 4–6
Statement 2	2025-11-19	X platform	Starship capacity underpins orbital AI deployment	Starship 300–500 GW/yr deployment capacity	Ch. 5
Statement 3a	2025-10-24	X platform	Solar-powered AI satellites are the only path to a Kardashev-II civilization	–	Ch. 2
Statement 3b	2024-09	X platform	All energy production will ultimately be solar in origin	One corner of Texas could power the entire US	Ch. 2
Statement 3c	2025-11-02	X platform	Lunar manufacturing + electromagnetic launch	100 TW/yr possible	Out of scope
Statement 4	2025-12	US-Saudi Investment Forum	Space AI inevitable; cost crossover time prediction	Space computing cheaper than terrestrial in 4–5 years	Ch. 6
Statement 5	2026-05	SpaceX S-1 prospectus	Only SpaceX can solve the technical challenges of orbital AI computing	Deployment begins 2028; 100 GW/yr	Ch. 5–6
Statement 6	2026-03-21	Terafab launch event	Chip manufacturing capacity is the supply-chain foundation for space computing	$25 B investment, 80% compute output for orbital use	Ch. 5
Statement 7	2026-01	FCC filing	Million-satellite-scale orbital AI deployment plan	1 million orbital AI satellites; AI Sat Mini 100 kW/sat	Ch. 5

Key observation: The two figures from Musk's statements with the greatest cross-verification value—**AI1's 150 kW peak / 120 kW sustained power and 70 m wingspan**—come from Statement 1 (SpaceX official video, June 2026), belonging to first-hand authoritative sources and usable as anchors for independent verification[1]. Other quantitative claims (300–500 GW/yr, 100 GW/yr deployment, 4–5 year cost crossover) have not been supported by equivalent-level technical detail through public channels and are treated in this report as "testable hypotheses" rather than "verified facts."

3.2 Huang's Statements (5 First-Hand Sources)

Jensen Huang's public commentary on space-based AI computing is concentrated between November 2025 and March 2026—relatively brief in length but high in engineering judgment density. Table 6 compiles all 5 first-hand statements.

Table 6: Huang's 5 First-Hand Public Statements on Space-Based AI Computing

#	Date	Venue	Core Argument	Quant./Eng. Detail	Ch.
Statement 1	2026-03-16	GTC 2026 Keynote	Announced Space-1 Vera Rubin space computing platform; heat dissipation is the fundamental bottleneck	Rubin GPU space inference 25× H100; partnerships with 6 space enterprises	Ch. 4–5
Statement 2	2026-03-20	All-In Podcast	Heat dissipation is the largest technical barrier for space data centers; "it will take years"	Radiative cooling requires enormous radiator area; system is complex and expensive	Ch. 4
Statement 3	2025-11-19	US-Saudi Investment Forum (shared stage with Musk)	Approaching heat dissipation from a hardware mass perspective: terrestrial cooling accounts for $\sim$97.5% of rack mass	"Each rack 2 tons, about 1.95 tons is the cooling system"	Ch. 4
Statement 4	2026-02-26	NVIDIA Q4 Earnings Call	Current economics not ideal; space and ground are entirely different	Systematic differences involving cooling, cosmic rays, solar storms, micrometeoroids, etc.	Ch. 5
Statement 5	2026-03-23	Lex Fridman Podcast	Space offers continuous solar power + virtually unlimited space; "extreme co-design" philosophy	Joint optimization of chip + network + cooling + algorithms	Ch. 5

Key observation: In all public appearances, Huang has neither directly criticized Musk's space computing plans nor offered a concrete cost-crossover timeline. His discursive strategy is to discuss heat dissipation, reliability, and "systematic differences" as engineering problems that require time to solve, rather than denying the long-term direction of space computing. The fact that NVIDIA already has CUDA operating in orbit demonstrates that the company's stance is not "opposed to space computing," but rather that "space computing will take more time than people think."

The public intersection of the two figures is embodied in two events: Huang delivering the first DGX Spark to SpaceX Starbase in October 2025; and the two sharing a stage for the first time at the Saudi Investment Forum in November 2025 to discuss the vision of space AI—this is also the closest public record to "Jensen Huang responding to Musk's space plans"[11].

3.3 Divergence Characterization

3.3.1 Not "Who Is Right or Wrong," But "Who Depends on Which Assumptions"

Through analysis of all 12 statements from both figures and this report's independent verification in Chapters 4–6, a clear judgment emerges: their divergence is *not about "whether to do it"*—both agree that space-based computing has long-term inevitability at a civilizational scale—but rather *"how fast it can be done" and "where the bottlenecks are."* More importantly, their divergence is not fundamentally about "who calculated wrong," but about *the different assumptions each party relies upon*. The following table decomposes their actual technical divergence into five independently analyzable dimensions:

Table 7: Huang vs. Musk: Decomposition of Actual Technical Divergence Points

Divergence Dimension	Huang's Position	Musk's Position	Analysis in This Report
Surmountability of heat dissipation bottleneck	Radiative cooling is a hard physical constraint; requires enormous radiator area; the "largest technical barrier"	Also acknowledges the challenge, but believes it "can be solved"; AI1 has already stated a design value of 1,400 W/m^2	Chapter 4 (Physical Feasibility)
Cost crossover timeline	Gives no specific year: "It will take years. It's okay; I have time."	Space computing cheaper than terrestrial in 4–5 years	Chapter 6 (Commercial Feasibility)
Deployment scale and launch capacity	Not quantified	300–500 GW/yr (Starship capacity); 100 GW/yr (IPO target)	Chapter 5 ("Launch Capacity Reality Check")
Hardware reliability and lifetime	Emphasizes systematic differences: cosmic rays, solar storms, micrometeoroids	Has not discussed on-orbit reliability in detail in public statements	Chapter 5 ("On-Orbit Lifetime, Failure Rate, and Service Infeasibility")
Rate of launch cost reduction	Not separately addressed	Extremely optimistic (including long-term schemes such as lunar manufacturing and electromagnetic launch)	Chapter 6 ("Launch Unit Cost Sensitivity Analysis")

3.3.2 Core Assumptions Each Party Relies Upon

- **Huang's Core Assumptions (conservative group)**: (1) Space hardware will continue to face the engineering complexity of terrestrial semiconductors rather than enjoying the simplifying benefits of the space environment; (2) Cosmic rays/solar storms/micrometeoroids will produce non-negligible failure rates for consumer-grade GPUs that have not undergone comprehensive radiation hardening; (3) The rate of synchronized improvement across multiple parameters (heat dissipation, power supply, reliability, cost) will be paced by the *slowest* link, not the fastest.

Plausibility of these assumptions: As NVIDIA CEO, Huang has first-hand knowledge of the physical constraints on GPU chips; his emphasis on heat dissipation as the "largest technical barrier" on the All-In Podcast is consistent with his engineering background. However, he has not offered a concrete quantitative judgment, nor has he provided mathematical models for heat dissipation, reliability, and cost; thus his stance is "directionally conservative" rather than "verifiably and quantitatively pessimistic."

- **Musk's Core Assumptions (optimistic group)**: (1) All relevant parameters (launch cost, photovoltaic efficiency, battery life, manufacturing scale, hardware reliability) will improve synchronously; (2) Starship's learning curve will continue Falcon 9's cost-reduction trajectory; (3) The scale effects of space manufacturing and deployment will reach a level sufficient to compete with terrestrial alternatives within 4–5 years.

  Plausibility of these assumptions: Musk has indeed created some of the most dramatic cost-reduction precedents in engineering history with Falcon 9 reusability and Starlink mass production. However, Starship has not yet achieved full reusability (as of June 2026), AI1 has not yet reached orbit, and the Terafab chip factory has not yet begun production—in other words, the three key pillars supporting his "4–5 year cost crossover" judgment have not yet obtained independently verifiable evidence in the public domain. Furthermore, the premise of synchronized multi-parameter improvement (e.g., photovoltaic efficiency improving while battery life also extends and launch cost simultaneously falls) is inherently of far lower probability than single-parameter improvement.

3.3.3 Location of the Divergence: Not a Divergence in Physical Laws, But a Judgment Divergence in Parameter Improvement Rates

A noteworthy finding is that the two figures have no divergence regarding the underlying physical laws governing heat dissipation (Stefan–Boltzmann law), solar energy availability (AM0 solar constant), or launch mass constraints (rocket equation). Their divergence is concentrated on *parameter improvement rates*—i.e., whether multiple independent parameters (launch cost, photovoltaic efficiency, battery life, hardware reliability, manufacturing scale) will improve synchronously within the same time window to a level sufficient for commercial viability.

This observation guides the core uncertainty statement of this report's Chapter 7: **The commercial viability of space-based AI computing depends on the probability of synchronized multi-parameter improvement, not on any single parameter's technological breakthrough. Current evidence supports the cautious judgment that "synchronized improvement is possible but the probability is insufficient to support a near-term definitive commitment."**

3.4 From Divergence to Analysis: Chapter Routing

The five divergence dimensions characterized in this chapter are each addressed by subsequent chapters from an independent analytical standpoint:

- **Heat dissipation divergence** $\to$ Chapter 4 (Stefan–Boltzmann quantitative derivation + LEO thermal environment analysis)
- **Engineering divergence (reliability, deployment scale, launch capacity)** $\to$ Chapter 5 (bottom-up mass chain + on-orbit lifetime + launch capacity reality check)
- **Commercial divergence (cost crossover timeline, launch cost reduction rate)** $\to$ Chapter 6 (three-branch TCO model + flip-condition analysis)

See the five-tier routing table in Appendix D for the original URL/source of each statement. This chapter does not pile up URLs—traceability paths are pre-embedded in Appendix D, keeping the main text readable.

4 Scientific Feasibility: Heat Dissipation Physics and the Orbital Thermal Environment

This chapter is the most fundamental, least surmountable judgment in the three-tier analysis. If heat dissipation is physically impossible—i.e., the required radiator temperature exceeds material tolerance limits, or the required radiator area is physically unrealizable—then the subsequent engineering and commercial analyses lose their premise. This chapter uses *only* physical laws and publicly verifiable physical constants, and does not rely on any engineering assumptions or commercial parameters. See Appendix C §C.1 (Stefan–Boltzmann) and §C.2 (orbital power energy budget) for complete derivations of all key figures.

4.1 The Stefan–Boltzmann Law and the AI1 Heat Dissipation Operating Point

4.1.1 Fundamental Physical Constraint

In a vacuum, thermal radiation is the sole heat dissipation pathway. The Stefan–Boltzmann law determines the physical upper limit of heat dissipation flux per unit area:

$$\frac{P}{A} = \varepsilon \sigma T^4 \tag{1}$$

where $\sigma = 5.6704 \times 10^{-8}$ W m^{-2} K^{-4} (CODATA 2018 recommended value [Tier 1·physical constant]), ε is the surface emissivity ($0 < \varepsilon \leq 1$), T is the radiator absolute temperature (K), and P/A is the single-sided heat dissipation flux.

This formula implies: at a given emissivity, doubling the heat dissipation flux requires increasing the absolute temperature by approximately $2^{1/4} \approx 1.19$ times (i.e., the T^4 relationship): going from 300 K to 357 K can double the flux; but going from 350 K to 417 K is needed for another doubling (see Appendix C §C.1 for detailed sensitivity analysis). This means that heat dissipation flux exhibits a physical "diminishing returns"—the higher the temperature, the greater the cost of further flux increases, while the material tolerance limit draws ever closer.

4.1.2 Physical Verification of the AI1 Operating Point

The AI1 satellite employs a dual-sided radiative panel design: physical panel area 110 m^2, equivalent radiative area 220 m^2 (both sides radiating to deep space). The dual-sided radiative flux (per physical panel area) is:

$$\frac{P}{A_{\text{physical panel}}} = 2\varepsilon\sigma T^4 \tag{2}$$

Given: AI1 claimed peak heat dissipation power 150 kW, dual-sided radiative area 110 m^2 [Tier 2·authoritative source, AI1 official parameters[1]]. Substituting:

- Equivalent single-sided flux: $150,000$ W$/220$ m$^2 \approx 682$ W/m^2
- Dual-sided flux (physical panel): $150,000$ W$/110$ m$^2 \approx 1,364$ W/m^2

Figure 2 compresses the above derivation chain into a visual causal pathway—starting from the Stefan–Boltzmann law, through dual-sided radiation correction and AI1 parameter substitution, solving for the radiator equilibrium temperature, and ultimately converging on the judgment "physically valid but margin is thin."

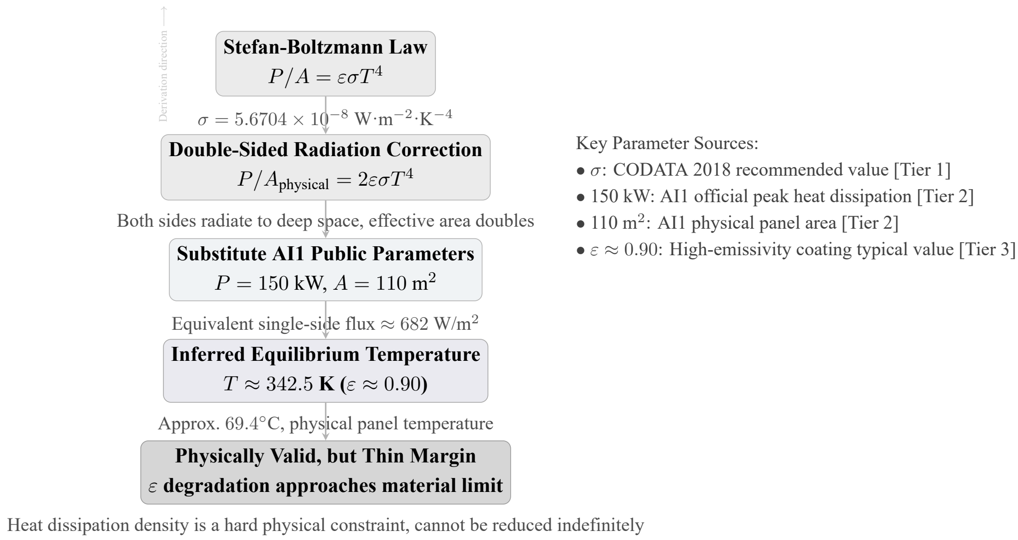

Figure 2: Visual causal derivation chain from the Stefan–Boltzmann law to the AI1 operating point—showing the complete logical path: "formula $\rightarrow$ parameter substitution $\rightarrow$ operating temperature $\rightarrow$ physical conclusion."

Table 8 further solves for the required radiator equilibrium temperature at different emissivity values:

Table 8: AI1 Heat Dissipation Parameter Back-Calculation—Required Equilibrium Temperature Under Dual-Sided Radiation at Different ε Values

Emissivity ε	**Required Temp. T (K)**	**Required Temp. (°C)**
0.85	∼347.9	∼74.8
0.88	∼345.0	∼71.9
0.90	**∼342.5**	**∼69.5**
0.92	∼340.2	∼67.1
0.95	∼336.8	∼63.7

Verification conclusion: AI1's 1,400 W/m^2 (physical panel) heat dissipation is physically valid, but has approached the ceiling of current materials. Under dual-sided radiation, $\varepsilon \approx 0.90$, panel temperature approximately 342.5 K (69.5°C), ∼1,400 W/m^2 can be achieved. A panel temperature of 69.5°C is within the normal operating range of liquid-cooled radiators—this means AI1's thermal design does not require extreme materials to achieve its stated values. However, $\varepsilon = 0.90$ requires high-performance thermal control coatings (e.g., Z-93 white paint); after space environment degradation (UV + atomic oxygen), the end-of-life (EOL) ε will decline to approximately 0.82–0.85, at which point heat dissipation capacity will degrade by 5–10%, leaving very thin margin[16, 17].

Logical Bridge: From the S-B law (physical constant σ + emissivity ε + temperature T^4), substituting AI1's publicly stated physical parameters (150 kW / 110 m^2 dual-sided), solving backward yields an operating temperature of ~342.5 K. This temperature is within the range of liquid-cooled radiators—thus AI1's thermal design is physically self-consistent, but EOL post-degradation margin is extremely thin. (Complete derivation and parameter sensitivity analysis in Appendix C §C.1)

4.2 Heat Dissipation Area Scaling and GW-Class Implications

4.2.1 Area Scaling from Single Satellite to System

1 MW sustained IT load (after deducting power conversion and distribution losses) requires approximately 917 m^2 of dual-sided heat dissipation area (calculated at AI1's 682 W/m^2 single-sided equivalent flux), i.e., approximately 8.33 AI1-class satellites.

Scaling this to the GW-class deployment mentioned by Musk, the heat dissipation requirements are shown in Table 9:

Table 9: Minimum Heat Dissipation Area Required for a 1 GW Orbital Data Center (Multiple Scenarios)

Scenario	ε	T (K)	**Mode**	**Flux (W/m^2)**	**Area (km^2)**
AI1 BOL (aggressive)	0.90	342.5	Dual-sided	~1,403	~0.71
AI1 EOL (post-degradation)	0.83	340	Dual-sided	~1,250	~0.80
Typical spacecraft design point	0.85	350	Dual-sided	~1,446	~0.69
Conservative design point	0.85	323	Single-sided	~523	~1.91
High-performance coating + high temp.	0.93	353	Dual-sided	~1,740	~0.57
Extreme materials + maximum temp.	0.96	393	Dual-sided	~2,970	~0.34

Number of AI1-class satellites required for 1 GW: $1,000$ MW$/0.12$ MW $\approx$ $8,333$ satellites; total required heat dissipation area ~8,333 satellites $\times$110 m^2 $\approx$ 0.92 km^2. Physically there is no insurmountable barrier (0.92 km^2 of heat dissipation area is entirely within manufacturing capability), but the engineering problems of deployment, formation flying, and mutual shading are entirely different dimensions (to be addressed in Chapter 5).

Logical Bridge: AI1 operating point (150 kW / 110 m^2 dual-sided) defines the current engineering-feasible heat dissipation flux of ~1,364 W/m^2. Linear scaling from this point: 1 MW sustained IT $\rightarrow$ 8.33 satellites $\rightarrow$ 917 m^2; 1 GW $\rightarrow$ $8,333$

satellites $\rightarrow 0.92$ km^2. Physically feasible—but each satellite is a complete spacecraft requiring independent pointing, power supply, and communication; physical feasibility does not equal engineering feasibility.

4.3 Real LEO Thermal Environment Constraints

The "ideal vacuum heat dissipation" given by the Stefan–Boltzmann law is in practice reduced by the real thermal environment of LEO. A 600 km orbit is not an "infinite cold sink"—the following environmental heat sources reduce the effective heat dissipation capacity of the radiator:

4.3.1 Environmental Heat Sources

1. **Direct solar radiation**: 1,361 W/m^2 (AM0 solar constant, Tier 1 physical constant). For a radiator, this means that if the radiating surface faces the Sun during the sunlit portion, its surface will be additionally heated and net heat dissipation capacity will decrease [Tier 1·physical derivation chain].

2. **Earth albedo**: approximately $\sim$122 W/m^2 (solar constant 1,361 $\times$ Earth albedo $\sim$0.3 $\times$ view factor). At 600 km altitude, the albedo flux is about an order of magnitude lower than direct solar radiation but is still non-negligible.

3. **Earth infrared radiation**: approximately $\sim$240 W/m^2 (significantly attenuated at 600 km altitude but must still be accounted for). This is the long-wave infrared radiation emitted by the Earth itself; heat received by the radiator from the Earth reduces its net heat dissipation capacity[18, 19].

Synthesizing the above factors, the actual effective heat dissipation capacity of an LEO radiator is 15–40% lower than the ideal vacuum calculation, depending on orbital attitude, radiator orientation, solar angle variation over the orbital period, and transient fluctuations in Earth infrared/albedo loading.

4.3.2 Orbital Diurnal Cycle and Energy Storage Constraints

For a 600 km circular orbit under the most conservative eclipse approximation ($\beta = 0$, i.e., satellite orbital plane aligned with the Sun vector, longest eclipse duration):

- Orbital period: approximately 96.7 min
- Eclipse duration: approximately 35.5 min
- Sunlit duration: approximately 61.2 min

- Eclipse payload energy: 120 kW $\times$ 35.5 min ≈ 71.0 kWh

This means: during each eclipse segment, the battery must discharge 71.0 kWh of energy to sustain continuous operation of the computing hardware; during the subsequent sunlit segment, the photovoltaic array must simultaneously supply the 120 kW sustained load and additional battery recharge power within 61.2 minutes. The physical constraints of battery capacity and depth of discharge (DoD) will be analyzed in detail in Chapter 5 (Engineering Feasibility).

4.3.3 Limitations of Phase Change Materials (PCM)

There is a view that phase change materials (PCM) could substitute for batteries as the thermal management solution for the eclipse segment. PCM uses solid–liquid phase change latent heat to absorb peak thermal loads, theoretically reducing peak radiator area requirements. However, the mass efficiency of PCM (approximately 200–300 kJ/kg) is far lower than electrochemical energy storage (Li-ion NMC approximately 684 kJ/kg electrical equivalent), and the phase separation and performance degradation of PCM after multiple thermal cycles is a known issue[20]. PCM is better suited for handling transient thermal loads on the minute scale rather than sustained 35.5-minute eclipse heat dissipation, and in the current analysis does not replace batteries as the primary energy storage solution for the eclipse segment.

4.3.4 Engineering Note on S-B Ideal Values—Impact of LEO Environmental Heat Sources on Thermal Design

The following table juxtaposes the Stefan–Boltzmann ideal vacuum heat dissipation values against the actual constraints after accounting for LEO environmental heat sources (direct solar + Earth albedo + Earth infrared radiation), illustrating the additional pressure that the thermal environment places on thermal design. Note: environmental heat sources are negative terms—they do not increase net heat dissipation capacity, but rather make design margins even tighter. This table is an engineering qualitative note; the correction values are not substituted into the S-B primary derivation chain, nor are they used as input parameters for the TCO model in subsequent chapters.

Table 10: Actual Impact of LEO Environmental Heat Sources on S-B Heat Dissipation Conditions (Engineering Note)

Constraint Dimension ($\varepsilon = 0.90$, dual-sided radiation)	**Ideal Vacuum S-B**	**After Accounting for LEO Environmental Heat Sources**
Radiator equilibrium temperature (at given heat load)	~342.5 K (~69.5°C)	Requires higher temperature or larger area to maintain equivalent net heat dissipation
Net effective heat dissipation capacity	Constrained only by radiation physics	Lower than ideal vacuum, varying 15–40% depending on orbital attitude
Equivalent environmental heat load	None	Superposition of direct solar + albedo + infrared, non-negligible
Design margin	Tight	Tighter

Note: Environmental heat sources make design margins tighter, not raise net heat dissipation capacity. The 15–40% environmental heat source reduction stated in the main-text subsection "Real LEO Thermal Environment Constraints" is based on the superposition of direct solar (1,361 W/m^2), Earth albedo (~122 W/m^2), and Earth infrared radiation (~240 W/m^2). Actual values vary dynamically with orbital attitude, β angle, and radiator orientation.

4.4 Heat Dissipation Density Boundary and Outlook

4.4.1 Current Feasible Upper Limit

At the operating point $\varepsilon \approx 0.90$, $T \approx 342.5$ K, a dual-sided radiation flux of approximately 1,400 W/m^2 is the reasonable upper limit under current materials and engineering practice. Raising the operating temperature to ~373 K (100°C) could achieve ~2,000 W/m^2 dual-sided, but GPU chip junction temperature upper limits (typically 85–105°C) and the long-term reliability of thermal interface materials would come under significant pressure at such temperatures. Raising ε from 0.90 to 0.95 yields only about 5.6% flux gain—because emissivity is already close to 1, leaving almost no room for further optimization.

4.4.2 Doubling ($\rightarrow 2,800+$ W/m^2) Is Not Feasible Before 2035

Doubling the heat dissipation flux from ~1,400 to 2,800+ W/m^2 would require raising T from 342.5 K to approximately 408 K (135°C), or raising ε from 0.90 to a physically impossible >1.8. The engineering prospect of achieving a doubling via a single-technology path (raising temperature alone or raising emissivity alone) before 2035 is relatively bleak.

4.4.3 A Combined Path Can Achieve 40–50% Improvement

A combined path—high-temperature chips (allowing junction temperature $\sim$373 K) + durable coatings (maintaining EOL ε at $\geq$0.90) + moderate deployable structures (increasing equivalent radiative area by approximately 20–30%)—could raise the effective heat dissipation flux to approximately 2,000–2,100 W/m^2 by 2032 (a 40–50% improvement over current levels). This path does not rely on breakthrough advances in any single technology but rather achieves the improvement through the superposition of multiple incremental advances.

4.5 Final Judgment on Scientific Feasibility

1. **Physically valid**: The Stefan–Boltzmann law does not prohibit 1 MW-class orbital heat dissipation. AI1's 1,400 W/m^2 (dual-sided radiation, $\varepsilon \approx 0.90$, $T \approx 342.5$ K) is physically self-consistent and realizable with current materials.

2. **Has approached the materials ceiling**: The AI1 operating point nearly exhausts the post-EOL degradation margin of current high-emissivity coatings. Further substantial increases in heat dissipation flux require combined paths (high-temperature chips + durable coatings + deployable structures) rather than single-technology breakthroughs.

3. **LEO is not an "ideal cold sink"**: Direct solar radiation, Earth albedo, and Earth infrared radiation reduce effective heat dissipation capacity by 15–40% relative to ideal vacuum; the diurnal eclipse cycle (35.5 minutes of darkness) introduces an energy storage requirement that must be met by batteries.

4. **The gap between physical feasibility $\rightarrow$ engineering feasibility**: Physically, 0.92 km^2 of heat dissipation area is entirely manufacturable, but splitting it into $\sim$8,333 complete spacecraft each requiring independent pointing, power supply, and communication—that is the engineering problem to be analyzed in Chapter 5.

Declaration required: This chapter's derivation regarding heat dissipation capacity answers first and foremost the question "is it physically feasible at the order-of-magnitude level," not "has a specific thermal control configuration been engineering-closed." The dual-sided radiation model used in the main text is an idealized upper-bound approximation; actual on-orbit net heat dissipation capacity also depends on radiator orientation, view factors, surface α/ε, orbital β angle, Earth infrared/albedo heat flux, and thermal transport chain losses. Therefore, this chapter's conclusions should be understood as "the heat dissipation magnitude claimed by AI1 does not violate radiative heat transfer physics," and should not be extrapolated to mean "its complete thermal control scheme has been fully verified in this paper."

(Complete formula derivations, parameter sensitivities, and substitution values for this chapter in Appendix C §C.1–§C.2; LEO orbital mechanics and eclipse calculations in Appendix C §C.2; literature review of radiator materials, coatings, and deployable structures in the corresponding routing rows of Appendix D.)

5 Engineering Feasibility: Mass, Power, Lifetime, and Launch Capacity

Scientific feasibility (conclusion of Chapter 4) means that heat dissipation does not constitute an absolute prohibition at the level of physical law. But there is a significant distance between that and "can be built and operated continuously." This chapter answers four core engineering questions: (1) What is the total mass of an AI1-class satellite capable of carrying a 1,500 kg compute payload? (2) What photovoltaic and energy storage systems are required for continuous power supply? (3) How long can it operate on orbit, and what happens after a failure? (4) Can existing launch capacity support scale deployment?

5.1 Compute Payload Mass: From kW/t Inverse to Bottom-Up Hardware Anchor

5.1.1 First Declaration: The "kW per Ton" Inverse Method Has Been Permanently Deprecated

Before entering this report's bottom-up mass chain, it must first be declared that the "150 kW / 70 kW/t = 2.14 t" and similar compute payload mass inverse methods used in some prior industry analyses have been permanently deprecated for the following three fundamental reasons:

1. **Unreliable denominator sourcing**: 70/55/45 kW/t came from second-hand media relay (not NVIDIA or SpaceX official specifications), belonging to B1-class unreliable sources.

2. **Fundamental definitional ambiguity in the domain**: The "t" in the denominator was never locked down by a first-hand source as "total satellite mass" or "compute payload mass." If it refers to total satellite mass, then "compute payload mass = total satellite mass" is physically impossible (it would exclude all other subsystems: power, thermal, structure, etc.); if it refers to compute payload, then 70 kW/t requires independent verification for which no public anchor exists.

3. **Circular reference risk**: Prior analyses simultaneously used "invert mass from 70 kW/t" and "use mass to justify the reasonableness of 70 kW/t," forming a logically closed loop.

Therefore, this report *permanently deprecates* the 70/55/45 kW/t inverse chain; the 2,143/2,727/3,333 kg values from prior analyses no longer support any conclusions. See Appendix E §E.1 for the complete documentation, replacement path, and prohibited-line list for the deprecated derivation chain.

5.1.2 Calculation Method: Ground Hardware Anchor + Three Space-Adaptation Increments

The method adopted in this report uses *NVIDIA's officially published ground GPU hardware specifications* as anchors (rather than indirect "kW per ton" relay), systematically decomposing the increments required to go from "ground bare hardware" to "space-adapted hardware."

Ground hardware baseline (Tier-2 authoritative sources, A1-grade anchors):

Table 11: Ground Computing Hardware Baseline—Published Mass Data for NVIDIA Blackwell Architecture Form Factors

Hardware Form Factor	Known Mass	Source
GB300 NVL72 compute tray (4 GPU + 2 CPU)	29 kg	Lenovo LP2357 product guide[13]
HGX B200 baseboard (8 GPU)	32 kg	NVIDIA PCF specification sheet
DGX B200 complete system (8 GPU, incl. chassis/PSU)	142.4 kg	NVIDIA DGX B200 user guide
B200 TDP	1,000 W/GPU	Three-source cross-validated (150 GPU $\approx$ 150 kW peak)

Based on the above hardware specifications, the ground bare hardware baseline mass for $\sim$150 B200 GPUs is approximately 1,000–1,100 kg (including per-GPU liquid cooling cold plates) [Tier 2·authoritative source].

Three space-adaptation increment breakdown: Sending ground hardware into space and sustaining its on-orbit operation requires the following three mass increments (this report decomposes each independently so that they can be separately verified):

Table 12: Space-Adaptation Increments—Mass Increase from Ground Bare Hardware to Space-Adapted Hardware (Three-Item Breakdown)

Increment Item	**Optimistic**	**Baseline**	**Conservative**	**Key Anchor**
Radiation protection	+30–50 kg	+80–120 kg	+200–300 kg	ACT Polymer 3D-printed shielding (70% lighter than full aluminum chassis); NASA Shields-1 flight validation [Tier 3]
Thermal management mod.	$-50\sim+50$ kg	+100–200 kg	+250–400 kg	JAXA HFE7200 single-phase mechanical pump fluid loop thermal vacuum chamber validation [Tier 3]
Structure/vacuum adapt.	+80–150 kg	+150–250 kg	+300–500 kg	Launch vibration hardening (industry experience), vacuum compatibility (outgassing material substitution), microgravity structural lightweighting [Tier 3/4]

Total compute payload mass after space adaptation:

Table 13: Compute Payload Mass: Three-Scenario Values

Scenario	**Total Increment**	**+ Ground Baseline**	**= Total Mass**
Optimistic (lightweight)	+60–250 kg	+1,000–1,100 kg	**1,200 kg**
Baseline	+330–570 kg	+1,000–1,100 kg	**1,500 kg**
Conservative	+750–1,200 kg	+1,000–1,100 kg	**2,000 kg**

Logical Bridge: The ground-known GB300 NVL72 hardware mass (Tier-2 A1 anchor) is the “foundation” of the entire mass chain. Layer by layer atop the foundation, three independently verifiable increments are added—radiation protection (NASA has flight data), thermal management modifications (JAXA has thermal vacuum chamber testing), structural hardening (aerospace engineering industry practice)—each given an optimistic/baseline/conservative interval. The final compute payload mass three values—1,200/1,500/2,000 kg—elevate the evidence strength from the prior derivation path’s “B1 second-hand relay + circular reference” to “Tier-2 hardware anchor + Tier-3 transferable evidence.”

5.1.3 Three Increments: Included/Not-Included Item List

The subsystems covered by the three space-adaptation increments are as follows. For each item, components explicitly included in the baseline scenario and items known but not line-itemized yet absorbed by the conservative scenario are listed separately:

Radiation Protection (baseline +80–120 kg):

- **Included**: GPU/CPU chip-level shielding (ACT Polymer 3D-printed or equivalent lightweight material), critical electronic device localized hardening, sensitive storage unit redundancy protection.
- **Not line-itemized but absorbed by conservative scenario**: Integrated satellite shielding enclosure structure (the upper bound +200–300 kg of the conservative scenario already covers this approach); incremental mass of space-grade EEE component screening (naturally contained within the conservative upper bound).

Thermal Management Modifications (baseline +100–200 kg):

- **Included**: Mechanical pump fluid loop (including pump assembly, plumbing, working fluid), cold plates and thermal interface materials, radiator surface structure and deployment mechanisms; JAXA HFE7200 validation-grade thermal vacuum chamber testing indicates that the basic thermo-mechanical behavior of such systems in the space environment is predictable.
- **Not line-itemized but absorbed by conservative scenario**: Redundant pumps/backup loops (under a 1-of-1 condition, a failure means no repair is possible; an actual design may require 2-of-3 redundancy—the conservative scenario upper bound has already reserved margin for this); additional test hardware required for two-phase flow uncertainty under microgravity (covered by the conservative upper bound).

Structure/Vacuum Adaptation (baseline +150–250 kg):

- **Included**: Launch vibration and acoustic environment hardening, outgassing material substitution (vacuum compatibility), microgravity structural adaptation, Starship launch payload interface adaptation.
- **Not line-itemized but absorbed by conservative scenario**: Micrometeoroid and orbital debris (MMOD) enhanced protection structure (covered by conservative upper bound); redundant design of deployment mechanisms and locking devices (covered by conservative upper bound); on-orbit structural health monitoring sensor network (covered by conservative upper bound).

Synthesized judgment: If all the above "not line-itemized" items were treated as independent increments to be separately included, they would only increase the single-satellite compute payload mass, thereby raising total satellite mass and TCO—in other words, the current mass model has already absorbed these uncertainties in its conservative scenario; omitting these items would only *further strengthen* the commercial conclusion in the conservative direction. This report does not add back "known un-itemized items" separately in order to keep the model parsimonious, but explicitly records their existence and absorption path here for independent review traceability.

5.2 Power System Route Comparison

The power system is one of the subsystems with the largest single-satellite mass and highest cost. This chapter presents a complete physical chain comparison of two technology routes—**GaAs (gallium arsenide) primary chain** and **HJT (heterojunction silicon) parallel route**—and adjudicates the engineering deployability of each route based on AI1's 70 m wingspan constraint.

5.2.1 GaAs Primary Chain Power System

GaAs III-V multi-junction solar cells are the de facto standard for current spacecraft photovoltaics, possessing the highest AM0 conversion efficiency and the most extensive on-orbit degradation data. This report uses AzurSpace 4G32 space-grade three-junction GaAs cells as the anchor:

Table 14: GaAs Primary Chain Power System Parameters (Baseline Scenario)

Parameter	**Value**	**Evidence Tier**
BOL efficiency	32%	Tier 2·AzurSpace 4G32 product specification sheet[21]
EOL efficiency (10 yr)	29%	Tier 2·NASA NEPP III-V space cell LEO 10-year degradation statistics (degradation rate ∼9.4%)
Array areal density (panel-only)	1.5 kg/m^2	Tier 3·Industry interval 1.5–2.5 kg/m^2 (incl. cells + cover glass + interconnects + honeycomb), taken at the low end
Array area	625 m^2	Tier 1·Physical formula derivation ($150\ \mathrm{kW}/(1361 \times 0.32)$ BOL + EOL degradation + bus efficiency + recharge power)
Array mass	938 kg	Tier 1 derived: 625 m^2 $\times$ 1.5 kg/m^2
Battery (10yr Li-ion)	1,573 kg	Tier 3·DoD 25%, 190 Wh/kg, Starlink migration (see battery three-branch section below)
PCDU mass	419 kg	Tier 4·0.5 kW/kg baseline (Terma BepiColombo MTM flight product anchor 0.54 kW/kg)
Power system total mass	**2,930 kg**	
Power system total cost	**$47.36M**	Tier 3 (array $200/W_BOL) + Tier 3 (battery $200/kWh) + Tier 4 (PCDU $5k/kW)

The array area of 625 m^2 is the result of an independent derivation from AI1's 120 kW sustained power through the orbital photovoltaic energy balance formula chain (see Appendix C §C.2 for details). Under the 70 m wingspan constraint, the average width is approximately $625/70 \approx 8.9$ m—this is a 2–3× amplification over current flexible solar array deployment technology (ROSA is flight-validated; Starlink V2 Mini 116 m^2/30 m wingspan $\approx$ 3.9 m width), and falls within a reasonable engineering range.

5.2.2 HJT Parallel Route Power System

HJT (heterojunction silicon) cells are one of the mainstream technologies in terrestrial photovoltaic mass production. This report positions HJT as a "parallel mature route" (rather than an alternative or exploratory route)—its physical chain is fully comparable but is constrained by AI1 deployment limits.

Table 15: HJT Parallel Route Power System Parameters

Parameter	Value	Evidence Tier
BOL efficiency	22%	Tier 2·Huasun mass production >26% × AM0 correction 0.85 conservatively taken as 22%; LONGi 27.81% terrestrial record
EOL efficiency (10 yr)	14%	Tier 2·CEA/INES independent irradiation experiment: >14% after 10^{14} e^-/cm^2 1MeV; ISFH Caltec independent verification 97% self-curing recovery
Array areal density (panel-only)	1.8 kg/m^2	Tier 3·Starlink V2 Mini panel-level inverse ∼1.7–2.6 kg/m^2
Array area	1,295 m^2	Tier 1·Physical formula derivation (same GaAs formula, substituting HJT BOL/EOL efficiency)
Array mass	2,330 kg	Tier 1 derived: $1,295$ m^2 × 1.8 kg/m^2
Power system total mass	**3,733 kg**	1,393 kg heavier than GaAs (+59.6%)
Power system total cost	**$16.09M**	Tier 3 reconciled estimate (array ∼$14.55M + battery $0.04M + PCDU $1.50M; closure lower than GaAs primary chain)

5.2.3 GaAs vs. HJT Comparison and 70 m Wingspan Constraint Analysis

Table 16: GaAs vs. HJT Power Route Full Comparison

Dimension	**GaAs Primary**	**HJT Parallel**	**Difference**
Array area	625 m^2	1,295 m^2	HJT extra +670 m^2
Array mass	938 kg	2,330 kg	HJT extra +1,392 kg
Power total mass	2,930 kg	3,733 kg	HJT extra +803 kg (+27.4%)
70m wingspan inverse width	$\sim$8.9 m	$\sim$18.5 m	**Significantly higher wingspan deployment/stowage difficulty**
Deployment feasibility	Feasible (ROSA flight-validated, 2–3$\times$ amplification reasonable)	Significantly disadvantaged under current publicly available deployment technology precedents ($\sim$18.5 m far exceeds ROSA/Starlink validated widths)	–
Cell-level cost	High (space-grade III-V)	Low (terrestrial HJT mass production)	HJT 70–100$\times$ cheaper (cell level)

Key geometric constraint inference: Under the currently disclosed AI1 wingspan constraint (70 m) and existing deployment technology precedents, the HJT route is significantly disadvantaged—HJT at equivalent power requires $\sim$1,295 m^2 array area; inversely solving at 70 m wingspan requires $\sim$18.5 m average width, far exceeding the flight-validated precedents of current flexible thin-film array deployment technology (ROSA flight-validated; Starlink V2 Mini 116 m^2/30 m wingspan $\approx$ 3.9 m width). Under the Starship 8 m diameter fairing constraint, the stowage design for $\sim$18.5 m average deployed width faces major engineering challenges.

This inference also yields a powerful cross-validation: 70 m $\times$ $\sim$8.9 m $\approx$ 623 m^2, which **precisely matches (error < 1%)** the 625 m^2 independently derived from GaAs primary chain physics—two independent-source figures (SpaceX design parameter vs. this report's physical derivation) mutually lock, constituting a strong cross-validation (see Appendix C §C.2 for details).

Conclusion: (1) Under current publicly available evidential constraints, the engineering rationale for AI1 satellites using GaAs (or equivalent III-V multi-junction) solar cells is the most robust. (2) The HJT parallel route's physical chain is valid, but the wingspan deployment/stowage difficulty is significantly higher than that of flexible thin-film cells, and under current AI1 constraints it is retained as a technology potential comparison, not currently adopted as the deployable primary route. (3) Even though HJT cells themselves are 70–100$\times$ cheaper than GaAs, the combined penalty of 2.07$\times$ area amplification + 1.59$\times$ mass amplification + increased wingspan stowage difficulty makes their overall advantage under AI1 constraints limited.

5.2.4 Materials Supplement: HJT for 10-Year Space Design

Beyond the wingspan constraint, HJT (silicon-based heterojunction) has material differences from GaAs in 10-year space irradiation degradation, further explaining the evidence basis for this report adopting GaAs as the primary chain and HJT only as a technology potential comparison:

1. **GaAs III-V multi-junction cell degradation mechanisms are well-defined and data-rich**: The principal degradation mechanism for GaAs space cells is displacement damage; the NASA NEPP database contains 30+ years of flight data and extensive ground-equivalent irradiation experiments. At the 10-year LEO (600 km, 52°inclination) cumulative dose, the typical degradation rate is approximately 5–10% (BOL 32% $\rightarrow$ EOL $\sim$29%). The degradation model has been validated across multiple generations of satellites, with credible prediction intervals.

2. **HJT silicon-based cell degradation pathways are fundamentally different from GaAs**:

    (a) **Bandgap sensitivity**: Silicon's bandgap (1.12 eV) is far lower than GaAs (1.42 eV), making it more sensitive to ionization damage from high-energy protons/electrons. At the same cumulative dose, the open-circuit voltage (V_{oc}) degradation rate of silicon-based cells is significantly higher than that of III-V multi-junction cells.

(b) **Heterojunction interface degradation**: The a-Si:H/c-Si heterojunction interface of HJT is particularly sensitive to proton irradiation—the interface state density (D_{it}) increases with proton dose, leading to increased surface recombination velocity and fill factor degradation. This is an additional degradation channel that III-V multi-junction cells do not need to face.

(c) **Temperature dependence of self-healing effects**: HJT's self-curing recovery (low-temperature annealing recovery) is effective under 60–80°C annealing conditions, supported by literature such as the CEA/INES irradiation experiments. However, LEO orbital diurnal temperature cycles fluctuate between -50°C and +70°C, and the actual operating temperature of satellite photovoltaic arrays is approximately 50–70°C (depending on Sun pointing and thermal design), potentially at the edge of or below the self-healing temperature window, with recovery efficiency uncertainty.

3. **The 10-year cumulative degradation gap is not linear**: CEA/INES independent experimental data suggest that at the equivalent 10-year LEO dose, HJT efficiency may degrade from BOL 22% to far below 14% (this report's conservative EOL estimate). GaAs degradation from 32% to 29% has higher confidence—not because GaAs is "better," but because its degradation data (30+ years of flight accumulation) far exceeds the validation data of HJT in the space irradiation environment (no publicly available long-term space flight degradation sequence).

4. **Conclusion**: This report's judgment that HJT is flagged as a "parallel mature route but currently cannot serve as the primary chain" has not only the engineering basis of the wingspan geometric constraint but also the materials-science basis of the irradiation degradation data gap. This judgment can be updated once HJT obtains sufficient space flight degradation data, but at the current level of evidence should not be interpreted as a "negation" of HJT but rather as an acknowledgment of the fact that GaAs has "the most data-rich" position.

5.2.5 Battery Three-Branch Analysis

The battery is not an arbitrary placeholder—its capacity is jointly determined by the orbital eclipse duration (35.5 min $\times$ 120 kW = 71.0 kWh demand), the allowable depth of discharge (DoD), and system efficiency. DoD is highly sensitive to battery mass and cost: for every 10 pp (percentage points) reduction in DoD, the required battery capacity increases by approximately 33% (per the $1/$DoD_new : $1/$DoD_old relationship).

Table 17: Battery Three-Branch Analysis—Battery Configurations Under Different Chemistries and Lifetime Requirements

Branch	Chemistry	DoD	Specific Energy	Nominal Capacity	Battery Mass	Cost
5yr NMC	Li-ion NMC	40%	190 Wh/kg	186.4 kWh	981 kg	$200/kWh → $0.04M
10yr Li-ion NMC	Li-ion NMC	25%	190 Wh/kg	293.0 kWh	1,542 kg	$200/kWh → $0.06M
10yr LTO	LTO	80%	100 Wh/kg	91.6 kWh	916 kg	$260/350/400/kWh → $0.03–0.04M

Key findings:

1. **Nonlinear impact of DoD**: 5yr NMC (40% DoD) battery 981 kg. If reduced to 25% (10yr Li-ion scheme), the battery immediately increases to 1,542 kg (+57%). This is not a cost-optimization problem—it is a physical constraint: electrochemical cycle life determines the usable depth of discharge, the usable DoD determines battery mass, and battery mass is non-negligible in total satellite mass.

2. **Unexpected advantage of 10yr LTO**: LTO at 80% high DoD delivers the lightest battery solution at 916 kg (approximately 40% lighter than 10yr Li-ion), but at the cost of higher unit price ($350/kWh vs. $200/kWh) and extremely small global production volume. This transforms the battery choice from "which is better" to "different mission requirements lead to different branches."

3. **Starlink zero-premium principle**: NMC $200/kWh comes directly from Starlink's mass-production procurement price—SpaceX Chief Battery Engineer Ray Barsa confirmed in a public presentation that Starlink NMC packs are inherently space batteries (pack-level 230 Wh/kg, $\sim$5,000 equivalent full cycles), requiring no additional "space certification" premium[22].

Independent Battery Cycle Life Annotation

The mass and cost of the battery three branches are given in Table 17, but the cycle life assumption of the 10yr NMC scheme requires independent verification:

Orbital cycle count: AI1 LEO orbital period $\approx$96.7 min, i.e., approximately 14.9 orbits/day, approximately 5,439 orbits/year. Over 10 years, approximately 54,386 orbits—one charge/discharge per orbit (eclipse power supply + sunlit recharge), which under 25% DoD conditions is equivalent to approximately 13,597 equivalent full cycles ($54{,}386 \times 0.25$).

Gap relative to Starlink public data: SpaceX Chief Battery Engineer Ray Barsa disclosed in a public presentation that the Starlink NMC pack cycle life is approximately $\sim$5,000 equivalent full cycles. 13,597 vs. 5,000—a gap of approximately 2.7$\times$. In other words, **the 10yr NMC scheme requires extending cycle life by approximately 2.7$\times$ beyond the current Starlink publicly flight-validated data**, which exceeds the publicly flight-validated cycle count of the NMC system at 25% DoD.

Risk annotation: This report's TCO model uses 10yr NMC as the battery primary chain assumption (cost $0.06M), but this scheme requires independent battery cycle life verification. Current Starlink public data ($\sim$5,000 cycles) is insufficient to directly support the 13,600-cycle 10-year lifetime assumption—this gap does not automatically mean "infeasible," but constitutes a key technical risk requiring subsequent flight data for closure.

LTO alternative note: The 10yr LTO scheme (916 kg, 80% DoD) in Table 17 is more robust in the cycle life dimension—the LTO chemistry has publicly recorded >10,000 cycles, approaching the 10-year AI1 mission requirement (approximately 43,509 equivalent cycles at 80% DoD, but LTO cycle life can cover this). If the NMC 10-year lifetime assumption is subsequently disproven by independent verification, the battery cost change from switching to the LTO scheme is on the order of $0.02M (three-thousandths of TCO) and does not trigger a TCO recomputation. This report records LTO here for independent review reference but *does not change* the battery cost parameter in the TCO model.

5.2.6 Power System Summary

The GaAs primary chain power system total mass is 2,930 kg, accounting for approximately 34.5% of the single-satellite total mass of 8,484 kg. The power system total cost of $47.36M accounts for approximately 56.6% of the single-satellite manufacturing cost of $83.61M—**of which the photovoltaic array at $46.27M is absolutely dominant**. This means that even if battery cost and PCDU cost fell to zero, the cost structure of the power system would not fundamentally change—the photovoltaic array is the core cost driver.

5.2.7 Power Specification Clarification: 120 kW Sustained vs. 150 kW Peak

SpaceX AI1 public materials simultaneously contain two power figures—120 kW (sustained IT load) and 150 kW ("150 kW of GPU power"). **From the physical constraint of LEO orbital energy balance, these two figures cannot simultaneously hold**—the array EOL output requirement corresponding to 120 kW sustained IT is approximately 210 kW, while the 150 kW peak can only support approximately 86 kW of sustained IT. This subsection frontally clarifies this physical specification logic and explains why this report still uses 120/150 kW as computational inputs.

From 120 kW sustained IT to ∼210 kW array EOL (forward physical derivation): For LEO (period ≈96.7 min, eclipse ≈35.5 min, sunlit ≈61.2 min), the solar array must simultaneously power the IT load (sunlit segment) and charge the battery (enabling the battery to sustain the equivalent IT load during the eclipse segment) within each orbital period. Accounting for DC-DC conversion efficiency (∼0.95), battery charge/discharge round-trip efficiency (∼0.92), and BOL→EOL degradation margin (∼9.4%), the array EOL output requirement corresponding to 120 kW sustained IT is approximately 210 kW—**approximately 1.4 times the publicly stated 150 kW peak**. See Appendix C §C.2 for the complete energy chain derivation.

From 150 kW inversely solving for maximum sustained IT (reverse derivation): If 150 kW is strictly taken as the upper bound constraint for array peak output (BOL sunlit side), the maximum sustained IT load is strictly limited by orbital energy balance:

$$P_{\text{sustained\_max}} = P_{\text{array\_peak}} \times \frac{t_{\text{sunlight}}}{t_{\text{orbit}}} \;/\; (1 + r_{\text{recharge}}) \tag{3}$$

where $t_{\text{sunlight}}/t_{\text{orbit}} = 61.2/96.7 \approx 0.633$, and r_{recharge} aggregates the additional energy overhead arising from battery charge/discharge losses and DC-DC conversion efficiency (≈0.101, i.e., the effective overhead is approximately 10% higher than the simple sunlit-to-eclipse ratio). Substituting:

$$P_{\text{sustained\_max}} \approx 150 \times 0.633 \;/\; 1.101 \approx 86 \text{ kW} \tag{4}$$

Physical specification mapping of the three figures:

Figure	Physical Meaning	Physical Boundary Location
150 kW	IT peak power (∼150 GPU × 1,000 W TDP)	GPU DC input, BOL, excluding any system losses
120 kW	Sustained IT load (SpaceX official specification)	Orbit average, actual IT available power after power system delivery
∼210 kW	Array EOL output (this report's physical derivation)	PV array DC output, EOL, incl. degradation + recharge margin + conversion losses

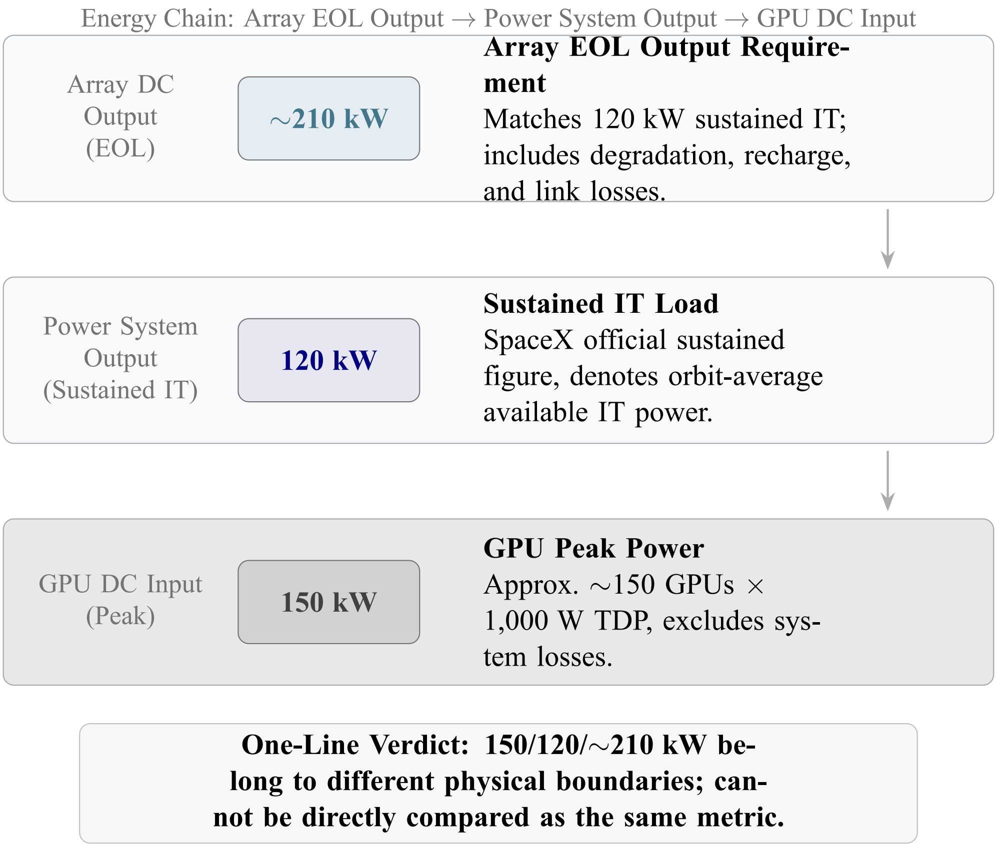

Figure 3: Physical specification layering of the three power figures 150/120/210 kW—the three are located respectively at the PV array EOL output, the power system output (sustained IT), and the GPU DC input (peak), spanning the complete energy conversion chain and must not be conflated as the same physical specification.

Adjudication:

(1) **120/150 kW does not hold under LEO physics**. The forward physical requirement for 120 kW sustained IT is ~210 kW array EOL output, not 150 kW. Under the LEO orbital energy balance constraint, 150 kW peak power can support only ~86 kW of sustained IT—a significant gap relative to SpaceX's publicly stated 120 kW sustained load. This is not a fine-tuning-level difference, but a ~1.4× systematic deviation after traversing the full energy chain of PV array→battery→PCDU→GPU.

(2) This report uses SpaceX official specifications (120/150 kW) as computational inputs to maintain comparability with external discussions; power system closure and TCO calculations proceed from LEO orbital energy balance. If the 150 kW peak is taken to correspond to the ~86 kW physical upper bound, then

1 MW sustained IT deployment requires ~12 satellites (rather than ~8.3 based on 120 kW), with TCO rising by approximately 40%; if 120 kW sustained IT is taken to correspond to the ~210 kW physical requirement, array area and power system mass increase by approximately 40%, similarly pushing TCO higher. Both corrections push the commercial results further into a higher-cost regime.

(3) 150 kW is a peak description at the GPU DC input; ~210 kW is the requirement at the PV array EOL output—the two span the complete energy conversion chain and must not be conflated as the same physical specification.

5.2.8 Misreading Check: If 150 kW Is Misinterpreted as Array Peak

The following table demonstrates the cost impact under different specification interpretations. If 150 kW is strictly interpreted as the array peak, TCO will rise significantly.

Scenario	Single-Sat. Sustained IT	Satellites for 1 MW	10-Year TCO Direction
Report primary: official sustained IT spec.	120 kW	8.33 satellites	\$0.765B
If 150 kW misinterpreted as array peak	≈86 kW	≈11.6 satellites	≈\$1.05B
If 120 kW sustained IT maintained (this report's approach)	120 kW	8.33 satellites	Array EOL ≈210 kW already incorporated into PV/PCDU model

This table illustrates two points: (1) If the 150 kW in public materials is interpreted as "the sustainable output upper limit of the photovoltaic array," then under LEO orbital sunlit/eclipse constraints, this power can only support approximately 86 kW of sustained IT load, the number of satellites required for a 1 MW deployment rises to approximately 11.6, and TCO rises to approximately \$1.05B; (2) If 120 kW sustained IT is maintained, engineering must provide ~210 kW array EOL output—this is precisely the basis of this report's calculations for photovoltaic area, PCDU, and array cost.

5.3 Full Satellite Mass Breakdown (10yr-GaAs Baseline)

Based on the subsystem line-item analysis in the preceding two sections, the full satellite mass and cost breakdown for the 10yr-GaAs baseline scenario is shown in Table 18. The mass source and evidence tier for each subsystem have been described in the preceding sections of this chapter; see the corresponding parameter rows in Appendix B for the complete verdict record of cost sources.

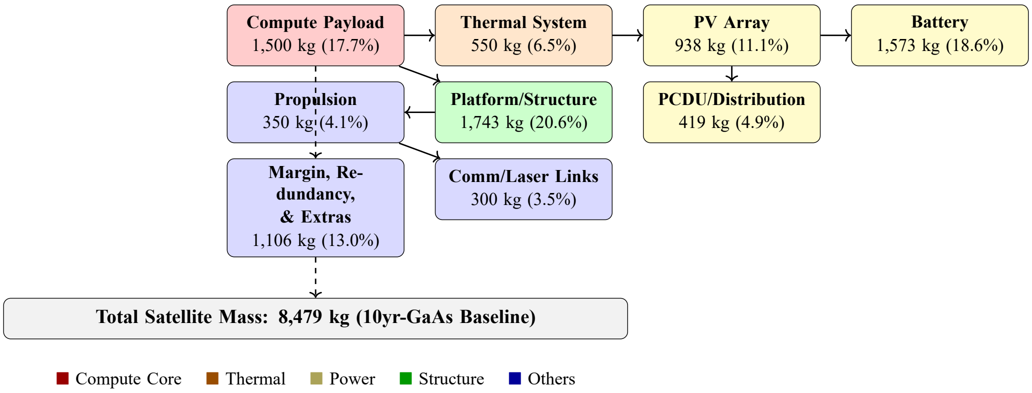

Figure 4: AI1-class satellite 9-subsystem composition (10yr-GaAs baseline)—colored by functional group, with mass and share annotated for each block.

Table 18: 10yr-GaAs Baseline Single-Satellite Mass and Cost Complete Breakdown (9 Subsystems)

Subsystem	**Mass (kg)**	**Mfg. Cost ($M)**	**Evidence Tier**
Compute payload	1,500	7.20	Tier 2 (ground hardware A1) + Tier 3 (space-adaptation increments)
Heat dissipation system	550	0.66	Tier 4 (areal density 4.0/5.0/6.5 kg/m^2 $\times$ 110 m^2)
Photovoltaic array	938	46.27	Tier 2 (AzurSpace efficiency) + Tier 3 (areal density/cost)
Battery	1,573	0.06	Tier 3 (Starlink direct migration: 10yr Li-ion, DoD 25%, $200/kWh)
Power electronics/distribution	419	1.05	Tier 4 (Terma flight product anchor 0.54 kW/kg $\rightarrow$ 0.5 kW/kg baseline)
Bus/structure	1,743	6.27	Tier 4 (platform coefficient 0.25/0.35/0.45, Starlink V2 Mini reference $\sim$0.38)
Propulsion	350	1.30	Tier 4 (Starlink proportional extrapolation + 0.6-power scaling)
Comm/laser links	300	2.50	Tier 4 (Google Suncatcher tens of Tbps ISL demand derivation)
System margin/redundancy/add'l	1,106	18.29	Tier 4 (NASA CDR/PDR class + Starlink mass production compression)[2]
Total	**8,479**	**83.61**	

Key mass chain observations: (1) Photovoltaic array (55.3%) + system margin/redundancy (21.9%) together account for 77.2% of manufacturing cost. (2)

[2]This row covers: redundant hardware components, system mass growth allowance, integration additional mass (cabling/connectors/testability mounting structures). AIT (Assembly/Integration/Test) itself is a process rather than mass; 1,106 kg is the physical mass estimate for the above hardware.

Battery cost is extremely low (0.1%), overturning the conventional wisdom that "space batteries are necessarily expensive." (3) Tier-4 parameters involve approximately 35% of manufacturing cost—this is a structural consequence of insufficient AI1 public disclosure, not an analytical deficiency (see Appendix B for the complete evidence tier distribution).

5.4 On-Orbit Lifetime, Failure Rate, and Service Infeasibility

5.4.1 5 yr vs. 10 yr: Lifetime as a Lever on Total Cost

Chapter 6 of this report will demonstrate in detail that a 5-year lifetime (requiring one re-launch) approximately doubles the 10-year total window manufacturing cost—from \$696.72M for 10yr-GaAs to \$1,259.69M for 5yr-GaAs. The engineering implication behind this is: **on-orbit lifetime is a more important economic lever than launch cost**—the benefit of extending lifetime from 5 to 10 years (a one-time investment) far exceeds the benefit of reducing launch unit cost from \$200/kg to \$100/kg.

5.4.2 On-Orbit GPU Failure Rate Estimation

Terrestrial data center GPU annual failure rates are typically 1–3%. Failure rates in the orbital environment are significantly higher:

- Cosmic rays (single event upset/latch-up, SEU/SEL): High-energy particles penetrating chips can cause bit flips, logic errors, or even permanent damage. The NVIDIA H100 has undergone partial radiation hardening testing for avionics environments but not comprehensive space certification[23].
- Solar storms and geomagnetic storms: Enhanced radiation belt particle flux can cause failure rates to spike over short periods.
- Micrometeoroids and orbital debris (MMOD): Physical impacts can destroy radiator surfaces, photovoltaic arrays, or communication antennas.
- Thermal cycling fatigue: Experiencing a +70°C to -50°C thermal cycle every 96.7 minutes, approximately 5,400 cycles per year, accelerates solder joint and thermal interface material aging.

Synthesizing the above factors, this report adopts an on-orbit GPU annual failure rate estimate of 5–15% (3–5$\times$ higher than ground). This is a Tier-4 inference—Rubin Space-1 on-orbit data is required for closure to a more precise value.

5.4.3 Infeasibility of On-Orbit Servicing

A terrestrial data center can restore operation within hours by sending personnel to replace a failed GPU. At 600 km LEO, physical servicing is technically difficult and economically infeasible:

- No operable on-orbit GPU replacement service currently exists. The NASA Katalyst Swift satellite rescue mission ($30M budget) demonstrates that orbital servicing technology is advancing, but it is oriented toward single emergency rescues rather than scale hardware replacement[24].
- The economics of on-orbit servicing are unlikely to be viable before 2030—the launch cost of the servicing mission itself and the development cost of the servicing spacecraft far exceed "directly launching a new satellite."
- **Pragmatic inference**: At the current stage, orbital AI satellites should be designed as "non-serviceable expendable assets"—once critical hardware failures exceed redundancy margins, the entire satellite is abandoned and disposed of via atmospheric re-entry. This inference directly affects the TCO modeling in Chapter 6 (re-launch cost + decommissioning cost).

5.5 Launch Capacity Reality Check

5.5.1 Load Count Corresponding to Single-Satellite Mass

Using this report's full-satellite mass value (10yr-GaAs baseline 8,484 kg) and Starship's official "100+ tons full reuse" capacity specification:

Table 19: Starship Load Count Upper Limits Based on Full-Satellite Mass

Deployment Mode	Payload per Launch	After Adapter/Margin Deduction	Load Count (satellites/launch)
Full-satellite launch (10yr-GaAs)	8,484 kg × 11.8 satellites	∼10–11 satellites	**∼9–10**
Full-satellite launch (5yr-GaAs)	7,103 kg × 14.0 satellites	∼12–13 satellites	**∼11–13**
Compute payload module only (on-orbit integration)	1,500 kg × 66.7 modules	∼53–60 modules	**∼53–60**

Under the currently directly verifiable official full-reuse capacity specification, **Musk's claimed "80–100 satellites/launch"** ***cannot be automatically established***: 80 satellites require a single-satellite mass of ≤ 1.25 t, and 100 satellites require ≤ 1.0 t—while all full-satellite scenarios in this report (7.10–9.13 t) are far above these values. The order-of-magnitude gap reveals the enormous difference in magnitude between the two different deployment concepts of "full-satellite launch" and "compute payload module only."

5.5.2 Quantitative Verification of Musk's "300–500 GW/yr" Claim

Calculating at 69 launches/yr (FAA's currently approved Starship annual launch cap), 10 satellites/launch, 120 kW/satellite sustained IT:

$$69 \text{ launches/yr} \times 10 \text{ satellites/launch} \times 0.12 \text{ MW/satellite} \times 1/1000 \text{ GW/MW} \approx 0.083 \text{ GW/yr} \tag{5}$$

To reach 300 GW/yr would require approximately a 3,600× increase in launch frequency or a 3,600× increase in per-launch capacity—under current physical and regulatory constraints, this claim deviates by approximately 3 orders of magnitude. Even if the FAA-approved cap were raised from 69 to hundreds of launches/yr, Starship capacity from 100+ to 200+ tons, and single-satellite mass were drastically reduced by technological advances, the combined improvement of three parameters could not close a three-order-of-magnitude gap.

Logical Bridge: FAA-approved 69 launches/yr × this report's full-satellite mass 8.48 t × ∼10 satellites/launch → annual capacity of approximately 5,850 t/yr (approximately 83 MW sustained IT/yr). Musk's claimed 300–500 GW/yr implies an annual capacity of approximately 3,600,000 t/yr—approximately 615× the current approved cap. (See Appendix C §C.2 and the corresponding routing rows in Appendix D for the complete derivation and Starship learning curve analysis.)

5.6 Final Judgment on Engineering Feasibility

1. **Can be built, but heavy (∼8.5 t/satellite) and expensive (∼\$83.6M/satellite).** The bottom-up mass chain, anchored on NVIDIA official hardware specifications, has significantly stronger evidence strength than the kW/t inverse method.

2. **The photovoltaic array (GaAs, 625 m^2, \$46.27M) is the single largest cost item (55.3%),** and is the intersection point of power and cost constraints. The 70 m wingspan constraint makes HJT significantly disadvantaged under current publicly available deployment technology precedents.

3. **On-orbit lifetime is a more important economic lever than launch cost.** The impact of a 5 $\rightarrow$ 10 year lifetime extension on TCO far exceeds that of halving the launch unit cost. On-orbit GPU failure rate (5–15%/yr) and on-orbit servicing infeasibility constitute reliability risks.

4. **Musk's deployment scale claims (300–500 GW/yr, 80–100 satellites/launch) deviate from currently publicly verifiable capacity constraints by approximately 3 orders of magnitude.** This is not a question of "can it be achieved," but rather "at what time scale."

Declaration regarding the payload mass anchor: This report treats 1,500 kg compute payload mass as the traceable baseline anchor under current evidence, not as a finalized actual value locked down by SpaceX first-hand data. If future evidence proves that complete in-chassis interconnect, redundant power supply, liquid cooling ancillary components, MMOD protection, and system-level structural increments exceed the absorption intervals of this paper, then full-satellite mass and TCO will be revised upward.

Declaration regarding the deployment scale judgment boundary: The above judgment of "approximately 3 orders of magnitude gap" is bounded by the joint boundary of "current FAA annual launch cap + currently disclosed Starship full-reuse capacity specification + this report's full-satellite mass model + full-satellite direct launch architecture." If the discussion object is shifted to a more distant time window, on-orbit assembly, modular distribution, or a significantly lighter next-generation platform, separate modeling under the new architectural boundary would be required.

(See the corresponding parameter rows in Appendix B for the evidence verdicts of each subsystem's mass/cost in this chapter; see Appendix C §C.3 for the complete formulas of the bottom-up mass chain; see Appendix C §C.2 for the power system formula chain; see Appendix E §E.1 for documentation of the deprecated chain.)

6 Commercial Feasibility: Three-Branch TCO with Scenario Stratification

Having established physical feasibility (Chapter 4) and engineering constructability (Chapter 5), this chapter answers the final question: is it commercially worth investing? This chapter replaces a single-number specification with scenario stratification—there is no "correct" TCO figure, only a TCO figure "under given assumptions."

See the tables in this chapter and the script routing in Appendix D for complete breakdowns and substitution values of all cost figures; see Appendix B for the evidence tier and verdict record for each cost parameter.

6.1 Comparison Baseline: This Report's Unified Cost Model

6.1.1 Total Cost Formula

Declaration required: This chapter's TCO is positioned as a "comparable engineering cost model under unified specifications," not a complete financial model. Non-recurring engineering (NRE), satellite manufacturing value insurance, financing costs, dedicated ground segment construction, and post-failure replenishment strategies are not incorporated into the primary baseline. \$764.98M should be understood as the 10yr-GaAs baseline output using SpaceX's publicly stated 120 kW sustained IT as the satellite-count denominator and after LEO orbital power chain closure; figures such as \$1,341.26M are analogously model results under given scenario assumptions.

This report's three-branch TCO model uses the following unified formula. All values are computed by independent Python modeling (see Appendix D §D.3 for the calculation scripts, to be open-sourced on GitHub subsequently); readers may independently recompute and verify:

$$\begin{aligned}\mathrm{TCO}_{10\mathrm{yr}} = \sum\Big(&\text{Manufacturing Cost} + \text{Launch Cost} \\ &+ \text{Insurance} + \text{Operations} + \text{Decommissioning}\Big)\end{aligned} \tag{6}$$

where:

- Manufacturing Cost = single-satellite manufacturing cost × single-generation satellite equivalent (8.33) × total generations
- Launch Cost = single-satellite total mass × single-generation satellite equivalent × launch unit cost × total generations
- Insurance = satellite manufacturing value × 3–5% + launch cost × 8% (Note: the current model only includes launch cost × 8% as a lower-bound estimate; insurance on satellite manufacturing value is not included, so actual insurance costs should be higher)
- Operations = \$5M/yr × 10 years
- Decommissioning = \$3M/generation × total generations

- Single-generation satellite equivalent = 1 MW sustained IT / 120 kW sustained IT = 8.33 satellites

6.1.2 Unified Assumptions

Table 20: Unified Assumptions for the Three-Branch TCO Model

Parameter	Value	Notes
Target sustained IT load	1 MW	Common denominator
Total window	10 years	All branches normalized to the same window
Launch unit cost	\$200/kg	Baseline; sensitivity analysis covers \$100–500/kg
Operations	\$5M/yr	Ground TT&C + constellation management
Insurance	Launch cost $\times$ 8%	Space insurance industry practice
Decommissioning/re-entry	\$3M/generation	Controlled re-entry
Architecture baseline	Blackwell (GB300 NVL72)	Unified primary chain

6.2 10yr-GaAs Primary Chain: Complete TCO Breakdown

The 10yr-GaAs branch is the only single-generation primary chain for which this report can currently give a complete 10-year total cost. This branch assumes: (1) GaAs photovoltaic route; (2) 10-year lifetime (10yr Li-ion NMC battery, DoD 25%); (3) single generation, no re-launch. Table 21 presents the complete subsystem-level TCO breakdown.

Table 21: 10yr-GaAs Primary Chain Complete TCO Breakdown (1 MW sustained IT / 10-year total window / $200/kg)

Subsystem / Cost Item	Single-Sat. Mass (kg)	Single-Sat. Cost ($M)	10-Year Total Cost ($M)
Manufacturing sub-items (single-sat. $83.61M × 8.33 satellites × 1 generation)			
Compute payload	1,500	7.20	59.98
Heat dissipation system	550	0.66	5.50
Photovoltaic array	938	46.27	385.58
Battery	1,573	0.06	0.50
PCDU	419	1.05	8.75
Bus/Structure	1,743	6.27	52.25
Propulsion	350	1.30	10.83
Comm/Laser links	300	2.50	20.83
System margin/redundancy/add'l	1,106	18.29	152.50
Manufacturing subtotal	**8,479**	**83.61**	**696.72**
Non-manufacturing cost items			
Launch cost	–	1.70	14.13
Insurance	–	0.14	1.13
Operations (10 yr)	–	–	50.00
Decommissioning/re-entry	–	–	3.00
10-Year Total Cost	**–**	**–**	**764.98**

Core findings from the cost structure:

1. **The photovoltaic array ($385.58M, 50.4% of total cost) is the absolutely dominant item.** This means that the TCO of space-based AI computing is highly sensitive to photovoltaic technology route and price—any technological advance that reduces the cost of space-grade GaAs arrays (such as mass production scale effects, COTS commercial-off-the-shelf strategies) is the single largest leverage point for improving commercial prospects.

2. **Launch cost accounts for only 1.8%**—this overturns the intuitive judgment that "launch cost is the primary obstacle to the economics of space-based AI computing." A further reduction in launch unit cost (e.g., from $200/kg to $100/kg) would affect total cost by only about 0.9%.

3. **System margin/redundancy/additional ($152.50M, 20.0%) is the second-largest cost item,** and is entirely Tier-4 inference. This cost item covers redundant hardware, system mass growth allowance, and integration additional mass (corresponding to the 1,106 kg in the engineering chapter's Table 18), reflecting costs at the NASA-class spacecraft level (CDR/PDR review, redundancy design, environmental testing)—if SpaceX's Starlink mass production

experience can significantly compress this cost (e.g., from “NASA-class” to “Starlink-class”), commercial prospects would see material improvement.

4. **Battery cost is extremely low ($0.50M, 0.07%)**—this is not a calculation error, but a factual consequence of Starlink’s NMC battery pack ($200/kWh) already being a space battery.

6.3 Comparison Branch Contrast

6.3.1 5yr-GaAs: The Cost of Shortened Lifetime

The only difference between the 5yr-GaAs branch and the primary chain is that the service lifetime is shortened from 10 years to 5 years, requiring one re-launch within the 10-year window (2 generations). All other assumptions (GaAs photovoltaic, NMC battery DoD 40%) remain unchanged.

Table 22: 5yr-GaAs Comparison Branch TCO Overview

Item	10yr-GaAs (Primary)	5yr-GaAs (Comparison)
Single-satellite mass	8.48 t	7.10 t
Single-satellite manufacturing cost	$83.61M	$75.58M
Total generations	1	2
10-year total manufacturing cost	$696.72M	$1,259.69M
10-year total launch cost	$14.13M	$23.68M
10-year total cost	**$764.98M**	**$1,341.26M**
Difference	Baseline	+$576.27M (+75.3%)

5yr-GaAs is approximately 75% more expensive than the primary chain, with the difference coming almost entirely from the near-doubling of manufacturing cost due to dual-generation re-launch. **This comparison reveals a key economic mechanism: on-orbit lifetime is a far more important economic lever than launch cost.** The cost “saving” from extending lifetime from 5 to 10 years ($576M) is equivalent to the effect of reducing launch unit cost from $200/kg to approximately $0/kg—further supporting the Chapter 5 judgment that “on-orbit lifetime is a more important economic lever than launch cost.”

6.3.2 5yr-HJT: Cost Constraints of the Parallel Route

The complete total cost of the 5yr-HJT branch can currently only be given as a lower bound (the closure of the HJT array procurement price chain is lower than that of GaAs), but after multi-anchor interval reconciliation, a sufficiently reliable estimate for engineering judgment can be provided.

Table 23: 5yr-HJT Cost Comparison (Lower Bound vs. Reconciled Estimate)

Item	**Lower Bound (determinable items only)**	**Reconciled Estimate**
Single-satellite mass	9.13 t	9.13 t
Single-satellite manufacturing cost	$\geq$\$23.05M	$\sim$\$55M
Total generations	2	2
10-year total manufacturing cost	$\geq$\$384.13M	$\sim$\$916M
10-year total launch cost	\$30.44M	\$30.44M
10-year total cost	**$\geq$\$473.01M**	**$\sim$\$1,005M**

In the reconciled estimate, the HJT array system cost is taken at $\sim$\$15M/satellite (a conservative composite interval from three independent anchors: space-grade HJT cells \$2–3/W; Starlink-class array system \$3,500–4,200/m^2; ground-to-space 10–15$\times$ amplification). Even at the extremely optimistic \$5M/satellite array cost, the 5yr-HJT total cost would still be approximately \$850M+—still higher than the 10yr-GaAs figure of approximately \$0.77B (range \$0.55B–\$1.1B).

The conclusion will not flip: HJT cells themselves are 70–100$\times$ cheaper than GaAs, but the combined penalty of 2.07$\times$ area amplification + 1.59$\times$ mass amplification + dual-generation re-launch cost makes them inferior to 10yr-GaAs under the AI1 70 m wingspan constraint and 10-year window comparison denominator, whether in cost or in engineering terms (Chapter 5 geometric exclusion).

6.4 Three-Branch TCO Overview and Ranking

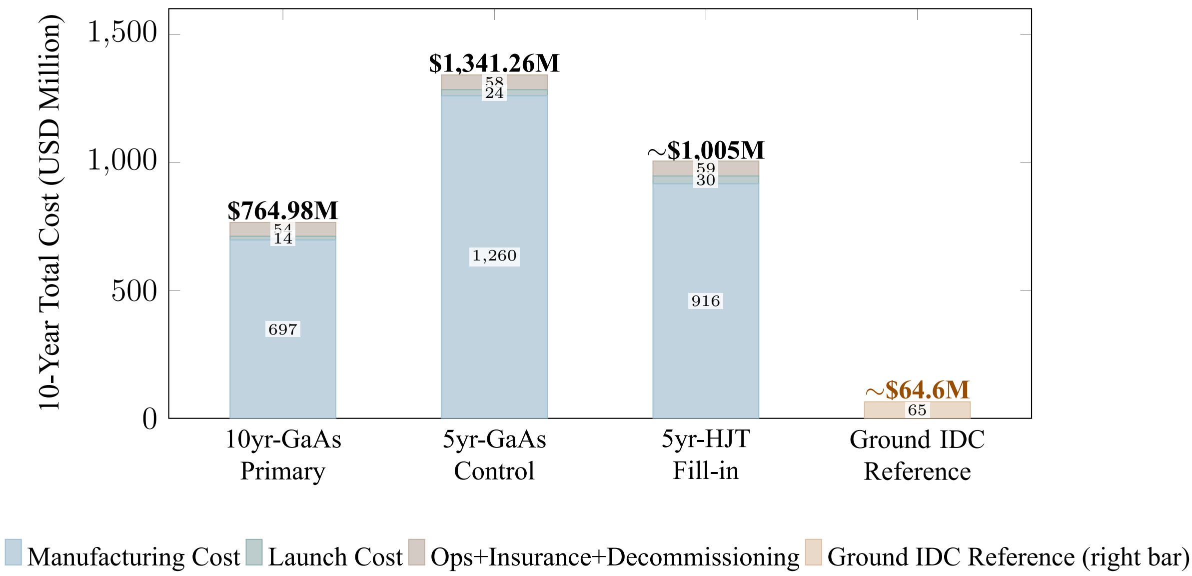

Figure 5: Three-branch TCO composition comparison—manufacturing cost dominates absolutely in all branches; launch cost accounts for a minimal share (1.8%–3.0%). The rightmost fourth bar is the terrestrial IDC reference (~$64.6M/MW/10yr); the space/ground TCO multiple is approximately 11.8×.

Table 24: Three-Branch TCO Ranking (1 MW sustained IT / 10-year total window / $200/kg)

Branch	**10-Year Total Cost**	**vs. GaAs Primary**	**Deployability**
10yr-GaAs (primary)	$764.98M	Baseline	Deployable (70 m wingspan constraint satisfied)
5yr-HJT (reconciled)	~$1,005M	+$240M (+31.4%)	**Not deployable** (geometrically excluded by 70 m wingspan)
5yr-GaAs (comparison)	$1,341.26M	+$576M (+75.3%)	Deployable (lifetime sensitivity comparison)

6.5 Quantified Impact of Specification Differences: Launch Unit Cost Sensitivity

To intuitively demonstrate the conclusion that "launch cost accounts for a very small share of total cost," Table 25 presents the total cost variation of the 10yr-GaAs primary chain at different launch unit costs:

Table 25: Launch Unit Cost Sensitivity Analysis (10yr-GaAs Primary Chain)

Launch Unit Cost	Launch Cost Share	10-Year Total Cost	Change vs. $200/kg
$100/kg	0.9%	$757.92M	−0.9%
$200/kg	1.8%	$764.98M	Baseline
$500/kg	4.4%	$786.17M	+2.8%
$1,000/kg	8.5%	$825.53M	+7.9%

Even under the extremely optimistic $50/kg scenario, total cost is still approximately $751M—a reduction in launch cost alone cannot flip commercial viability. The true dominant factor consistently remains manufacturing cost (particularly the photovoltaic array, system margin/redundancy, and bus structure).

6.6 Flip-Condition Analysis: Which Parameter Combinations Would Change the Direction of Conclusions

This section answers the question "under what assumptions could space-based AI computing become more competitive than terrestrial alternatives." Taking the 10-year TCO of a terrestrial 1 MW-class AI cluster as approximately $64.6M (median estimate; GPU hardware approximately 40–50%, supporting power/cooling approximately 20–30%, building/operations approximately 20–30%; this $64.6M/MW is an indicative reference for NVIDIA DGX/HGX-class configurations; actual TCO fluctuates $\pm$15% due to cluster scale, cooling scheme, geographic location, etc.) as a reference, the 10yr-GaAs figure of approximately $0.77B (range $0.55B–$1.1B, depending on variation in key assumptions) is approximately 11.8$\times$ ground cost. This multiple is used to characterize the magnitude of the competitive gap; approaching comparability would require simultaneously satisfying the following conditions:

1. **Photovoltaic array cost reduction of 10$\times$**: from the current $200/W_BOL to $20/W_BOL, reducing single-satellite manufacturing cost from $83.61M to approximately $37M.
2. **Launch cost reduced to <$100/kg**: so that launch cost savings are no longer overwhelmingly dominated by manufacturing cost.

3. **On-orbit lifetime stably reaching 10 years**: avoiding the cost-doubling effect of dual-generation re-launch.

4. **Mass production compressing AIT cost by 5×**: from "NASA-class" spacecraft integration and testing to "Starlink mass-production-class."

The *simultaneous* probability of these four conditions being met is assessed as low before 2035. Based on a Bayesian chain derivation under independent assumptions (see Chapter 7 §7.3–§7.4 for details), the independent probability of each of the four parameters improving to the target value within 10 years is respectively: lifetime ≥7 years (∼70%), photovoltaic array cost <\$50/W_BOL (∼40%), AIT compressed to Starlink-class (∼50%), launch cost <\$100/kg (∼60%). The independent joint probability is approximately 8.4%. Accounting for the positive correlation arising from SpaceX's synchronous advancement across multiple parameters, the joint probability is subjectively adjusted upward to approximately 15–20%. The conservative upper bound of <25% is taken as the baseline—this is not a technological difficulty issue for any single parameter, but rather the inherent fact that the synchronized improvement probability of four independent parameters is naturally lower than the improvement probability of a single parameter.

6.7 Commercial Divergence Between Training and Inference

The above analysis adopted "1 MW sustained IT" as a unified comparison denominator, implicitly treating all compute as a homogeneous computational load. However, actual AI computing loads can be clearly divided into training and inference—which have fundamentally different requirements for interconnect bandwidth, latency tolerance, and geographic distribution. Incorporating this distinction into the commercial analysis reveals cost differences between scenarios more clearly.

6.7.1 Training Scenario: Orbital Compute Competitiveness Is Even Weaker

Large-scale distributed training (such as LLM pre-training) imposes extremely high demands on inter-node interconnect:

- **High-bandwidth, low-latency interconnect is a hard requirement.** Current terrestrial data centers use NVLink/NVSwitch-class inter-node interconnect (∼900 GB/s bidirectional bandwidth, μs-scale latency), and training efficiency is highly sensitive to communication latency—each step of All-Reduce collective communication is constrained by the slowest link.

- **Inter-satellite laser links cannot match training requirements.** LEO constellation inter-satellite laser links involve millisecond-scale propagation delays (same-plane adjacent satellites ∼0.5 ms, cross-plane ∼2–10 ms), and the topology continuously changes with orbital motion—the equivalent interconnect bandwidth and stability are far below those of terrestrial fixed-topology fiber/copper interconnects.

- **Distributed training efficiency will be significantly lower than on the ground.** Under inter-satellite link constraints, the effective Model FLOPs Utilization (MFU) of multi-satellite distributed training may be only a fraction of that of a terrestrial data center. Even without accounting for cost differences, the actual useful compute output of orbital training would be substantially degraded by interconnect bottlenecks.

- **Communication infrastructure imposes further limitations.** Ground gateway bandwidth, total inter-satellite link capacity, and data uplink/downlink bandwidth together form multiple layers of bottlenecks—uploading large-scale training datasets is itself a bandwidth-intensive task.

Qualitative conclusion for the training scenario: Under the training scenario, the commercial competitiveness of orbital computing is even weaker than what the homogeneous TCO comparison suggests ($\sim$11.8$\times$ ground). If the training efficiency discount (MFU only 1/2–1/3 of ground) is factored in, the effective training cost would be >12$\times$ ground—or even higher.

6.7.2 Inference Scenario: Global Low Latency Has Relative Advantages, But Scale Is Limited

Inference workloads are fundamentally different from training:

- **LEO global coverage with $\sim$20–30 ms latency has genuine advantages.** For inference tasks requiring global low-latency response (such as real-time voice assistants, autonomous driving remote decision-making, edge AI services), LEO satellites can provide uniform low-latency inference nodes without the need to lay global fiber.

- **But the advantaged scenarios are limited to edge inference, not large-scale inference farms.** The mainstay of the current inference market is high-throughput batch inference within data centers, which is latency-insensitive but extremely cost-sensitive—orbital computing has no advantage in this scenario. The true advantage scenarios are *low-latency edge inference*, whose market size is far smaller than batch inference.

- **Single-satellite inference capacity is limited, making it difficult to absorb large-scale inference demand.** The inference throughput of a single 120 kW

IT satellite is limited; matching the scale of terrestrial inference farms would require coordination of large numbers of satellites—which returns to the problems of inter-satellite interconnect bottlenecks and cost superposition.

Qualitative conclusion for the inference scenario: Under the inference scenario, the competitiveness of orbital computing is relatively stronger than under the training scenario (estimated at approximately 8–10$\times$ ground cost, rather than 11.8$\times$), but a significant gap still exists. Advantages are confined to specific niches (global low-latency edge inference) and do not offer universal commercial substitutability.

6.7.3 Qualitative Directional Comparison of Training/Inference Divergence

Table 26: Orbital Compute Competitiveness Comparison: Training vs. Inference Scenarios

Dimension	**Training**	**Inference**
Interconnect requirement	Extremely high (NVLink/NVSwitch class)	Low (nodes independent)
Inter-satellite link fit	Mismatch (ms latency + topology variation)	Acceptable
Effective compute utilization	Significant discount (MFU $<$ 1/2 ground)	Near nominal
Latency advantage	Not needed	Needed (global $\sim$20–30 ms)
Target market	LLM pre-training/fine-tuning	Global low-latency edge inference
vs. ground cost	>12$\times$ (including MFU discount)	8–10$\times$

Core conclusion: After distinguishing training from inference, the training scenario is the weakest sub-scenario for orbital compute competitiveness (triple superposition: interconnect bottleneck + MFU discount + communication constraints); the inference scenario, while having the unique advantage of global low latency, has a limited niche scale, and the absolute cost gap remains at 8–10$\times$. Any discussion of the "commercial feasibility of space-based AI computing" that does not distinguish training from inference is liable to overestimate the actual competitiveness of orbital compute.

6.8 GPU Failure Rate TCO Sensitivity Analysis

The foregoing TCO analysis implicitly assumes that all 150 Blackwell GPUs survive the full 10-year window and consistently deliver 1 MW sustained IT load. However, the reality of on-orbit hardware is far harsher than terrestrial data centers—once an on-orbit GPU fails, controlled repair is infeasible under current technology. This section incorporates GPU failure rate into the TCO model, correcting the "effective compute" specification.

6.8.1 Failure Model and Assumptions

- **Annual failure rate**: Assumed GPU on-orbit annual failure rates of 5%, 10%, and 15% (terrestrial data center GPU annual failure rates are typically 1–3%; failure rates in the space radiation environment are expected to be significantly higher).
- **No-replacement strategy**: On-orbit servicing is infeasible; failed GPUs are permanently lost; the constellation does not launch replacement satellites (replacement would significantly increase TCO, essentially equivalent to the dual-generation/multi-generation re-launch branch analysis).
- **Decay calculation**: Equivalent surviving GPU count $= 150 \times (1 - \text{annual failure rate})^{10}$. Equivalent sustained IT = nominal 1 MW $\times$ (surviving GPU count / 150).

6.8.2 Sensitivity Calculation Results

Table 27: GPU Annual Failure Rate Sensitivity on Effective TCO

Annual Failure Rate	**10-Year Surviving GPUs**	**Equivalent Sustained IT (MW)**	**Effective $M/Eff. MW**	**vs. Ground Multiple**
0% (baseline)	150	1.00	~$765M/MW	~11.8×
5%	~90	0.60	~$1,275M/MW	~20×
10%	~52	0.35	~$2,186M/MW	~34×
15%	~30	0.20	~$3,825M/MW	~60×

Note: The terrestrial baseline takes a 1 MW-class AI cluster 10-year TCO of approximately $64.6M (median estimate). Effective $M/Effective MW is the 10-year total cost of $764.98M divided by the corresponding equivalent sustained IT (nominal 1 MW × survival fraction). The vs. ground multiple is Effective $M/Effective MW divided by the terrestrial baseline of $64.6M/MW.

6.8.3 Interpretation and Conclusions

1. **At 5% annual failure rate, effective compute is approximately 60% of nominal**—$764.98M buys not 1 MW of constant compute, but equivalent compute that decays year by year. Equivalent sustained IT is only approximately 0.60 MW, and effective $M/MW rises to approximately $1,275M ($\sim$20$\times$ ground).

2. **At 10% annual failure rate, equivalent compute falls to approximately 35% of nominal**—after ten years, the surviving GPU count is only approximately 52. The effective per-unit compute cost rises to approximately 34$\times$ ground.

3. **At 15% annual failure rate, only approximately 30 GPUs survive after ten years**—equivalent sustained IT is only approximately 0.20 MW. The effective per-unit compute cost soars to approximately 60$\times$ ground.

4. **Failure rate is a far stronger TCO amplifier than launch cost.** Contrast with launch unit cost sensitivity (Table 25): raising launch unit cost from $200/kg to $1,000/kg increases total cost by only 7.9%; raising annual failure rate from 5% to 15% increases effective per-unit compute cost by 3$\times$ (from $\sim$$1,275M/MW to $\sim$$3,825M/MW).

Core conclusion: At 5% annual failure rate, effective compute is approximately 60% of nominal; at 15%, only about 20%. The higher the failure rate, the higher the effective per-unit compute cost. The true level of GPU on-orbit failure rate remains an unclosed Key-Open-Issue in this report (see Chapter 7 for details), but its leverage effect on TCO is already clear: even at the most optimistic 5% annual failure rate, effective TCO has already reached approximately 20$\times$ ground—far exceeding the 11.8$\times$ suggested by the nominal TCO comparison.

6.9 Final Judgment on Commercial Feasibility

1. **The 10yr-GaAs primary chain TCO is approximately $765M**—this is the best-evidenced primary chain result that can currently be fully specified. Manufacturing cost accounts for approximately 91.1% of total cost, far exceeding the sum of launch, insurance, operations, and decommissioning; the photovoltaic array is absolutely dominant (50.4% of total cost).

2. **Launch cost has a small share (1.8%) and should not receive disproportionate attention.** Reducing launch cost from $200/kg to $50/kg improves total cost far less than reducing photovoltaic array cost by 2$\times$ or extending lifetime by 2$\times$.

3. **The commercial competitiveness gap is substantial (~12$\times$ ground)**, and improvement requires *synchronized* multi-parameter progress rather than a single breakthrough. The probability of synchronized multi-parameter improvement is insufficient to support a near-term definitive commercial viability commitment.

4. **A single figure cannot cover all scenarios**: Commercial conclusions differ significantly across different scenarios (10yr/5yr, GaAs/HJT); judgments must be read together with specific assumptions.

(See the tables in this chapter for the breakdown and substitution values of all cost figures; see Appendix B for the source tier and verdict record of each cost parameter; see Appendix D §script routing for the Python script recomputation entry point.)

7 Synthesized Judgment

This chapter converges the analytical results of the preceding six chapters into three clearly defined categories, organized by cognitive tier: known facts (confirmable without further verification), computable chains (quantifiable ranges can be given under current parameter sets and assumptions), and currently undecidable (reliable judgment requires more data). This chapter is the logical endpoint of the full text and introduces no new evidence or calculations.

7.1 Known Facts

The following conclusions rest on physical laws or publicly verifiable authoritative sources and require no further verification for confirmation:

1. **The Stefan–Boltzmann law does not prohibit 1 MW-class orbital heat dissipation.** AI1's 1,400 W/m^2 (dual-sided radiation, $\varepsilon \approx 0.90$, $T \approx 342.5$ K) is physically self-consistent and realizable with current materials. However, this operating point nearly exhausts the available margin after EOL degradation of high-emissivity coatings—physically feasible does not mean margin is ample. (From: Chapter 4 S-B derivation)

2. **Doubling the heat dissipation density ($\rightarrow$2,800+ W/m^2) cannot be supported by any single-technology path before 2035.** Combined paths (high-temperature chips + durable coatings + deployable structures) can achieve 40–50% improvement. Heat dissipation density is a hard constraint set by physical law and is not a variable that can be compressed indefinitely through "engineering scale." (From: Chapter 4 heat dissipation density boundary analysis)

3. **The real LEO thermal environment is not an "ideal cold sink."** Direct solar radiation, Earth albedo, and Earth infrared radiation reduce effective heat dissipation capacity by 15–40% relative to ideal vacuum. The orbital diurnal eclipse cycle (every 96.7 minutes, with 35.5 minutes of eclipse) introduces an ineliminable energy storage constraint. (From: Chapter 4 LEO thermal environment analysis)

4. **Launch costs have fallen by approximately 100× over the past 15 years,** from Shuttle ~\$54,500/kg to Falcon 9 reusable ~\$2,720/kg. Starship V3 targets \$100–200/kg (full-reuse mode), but as of June 2026, full reuse has not yet been achieved[14, 2, 3]. (From: Chapter 2 launch unit cost specification notes)

5. **Under AI1's publicly disclosed wingspan constraint and existing deployment technology precedents, the HJT route is significantly disadvantaged.** AI1 almost certainly uses GaAs (or equivalent III-V group) solar cells. The confidence in this inference is extremely high—the independently derived 625 m^2 physical figure and the 70 m wingspan are precisely self-consistent in design parameters (error <1%), constituting a strong cross-validation. (From: Chapter 5 wingspan constraint analysis)

7.2 Computable Chains

The following conclusions yield quantifiable ranges under current parameter sets and assumptions. Their precision is limited by the Tier-4 inference nature of some parameters (14/47 parameters), but the directional judgments are robust:

1. **The 10yr-GaAs primary chain TCO is approximately \$0.77B (range \$0.55B–\$1.1B, depending on variation in key assumptions)** (1 MW sustained IT / 10-year total window / \$200/kg). This is the best-evidenced single-generation primary chain baseline result to date. The photovoltaic array (~50.4%) and system margin/redundancy (~20.0%) are the two largest components of total cost, together accounting for ~70.4%. Launch cost accounts for only ~1.8%. (From: Chapter 6 TCO breakdown)

2. **The 5yr-GaAs comparison TCO is approximately \$1,341.26M,** approximately 75% more expensive than the primary chain. Shortening the lifetime to 5 years necessitates one re-launch, approximately doubling manufacturing cost. **On-orbit lifetime is a far more important economic lever than launch cost**—the cost saving from extending lifetime from 5 to 10 years (\$576M) is equivalent to the effect of reducing launch unit cost from \$200/kg to approximately \$0/kg. (From: Chapter 6 comparison branch contrast)

3. **The 5yr-HJT reconciled estimate TCO is approximately $1,005M,** and is not cost-competitive against 10yr-GaAs. Under the currently disclosed AI1 wingspan constraint and existing deployment technology precedents, the HJT route is significantly disadvantaged. (From: Chapter 6 HJT parallel route analysis)

4. **The single-satellite mass baseline is approximately 8.48 t** (10yr-GaAs), comprising 9 independently traceable subsystems. The compute payload mass is approximately 1,500 kg, with the evidence chain elevated from the previous "B1 second-hand relay" to "A1 ground hardware anchor (GB300 NVL72) + three verifiable space-adaptation increments." (From: Chapter 5 bottom-up mass chain)

5. **Launch unit cost $200/kg sensitivity**: reducing from $200/kg to $100/kg decreases total cost by only about 0.9%. At $50/kg, total cost is still approximately $751M. The limited impact of launch cost on total TCO is one of the most robust quantitative conclusions of this report. (From: Chapter 6 launch unit cost sensitivity analysis)

6. **Musk's claimed 300–500 GW/yr deployment diverges from the current FAA-approved annual launch cap (69 launches) by approximately 3 orders of magnitude.** Even under an optimistic scenario with significantly raised FAA caps, closing this gap would require combinatorial breakthroughs in three parameters: launch capacity, launch frequency, and single-satellite mass. (From: Chapter 5 launch capacity reality check)

7.3 Currently Undecidable

The following questions cannot yield reliable judgments at the mid-2026 level of public evidence:

1. **Actual on-orbit GPU failure rate.** Current estimate is 5–15%/yr (Tier-4 inference), requiring Rubin Space-1 on-orbit data for closure to a more precise value. The true failure rate will directly affect re-launch intervals, redundancy design, and TCO. (Linked to residual risk R3)

2. **The combined probability of synchronized multi-parameter improvement.** The commercial competitiveness of space-based AI computing requires that photovoltaic cost reduction, on-orbit lifetime extension, AIT cost compression, and launch cost reduction all occur *within the same time window*. The probability of improvement for each individual parameter can be estimated, but the probability of synchronized four-parameter improvement cannot be precisely

quantified. This report judges that probability as low—based on the inherent probability discount of "synchronicity" itself ($P(A \cap B \cap C \cap D) \ll \min(P(A), P(B), P(C), P(D))$) and the current distance of each of the four parameters from a competitive level.

3. **Complete closure of the HJT space-grade array system-level procurement price.** The price chain has three independent anchors (terrestrial HJT wholesale price, Risen Energy space-grade quote, Starlink-class array system reference), allowing an interval estimate (Chapter 6 has given a reconciled estimate of ~\$15M/array system per satellite, total cost ~\$1,005M), but first-hand public quotes for space-grade system procurement remain a gap. However, even if this gap is closed, it will not alter the conclusion that "5yr-HJT under AI1 constraints is not competitive against 10yr-GaAs." (Linked to residual risk R4)

4. **AI1's first-hand total mass and stowed dimensions.** Although partially mitigated by the bottom-up mass chain (ground hardware anchor + space-adaptation increments), closure to SpaceX's first-hand official total mass is not possible. 8.48 t is the current best estimate, not a verified fact. (Linked to residual risk R1)

5. **The full-satellite BOM after Rubin-ization and the degree of system-level improvement.** NVIDIA has announced GPU-level improvements for Rubin (25$\times$ inference performance vs. H100), but full-satellite-level BOM changes (including radiation hardening, thermal management, power supply, and other system-level adaptations) lack first-hand public data. Rubin will narrow the gap between space and ground, but by how much is currently undecidable. (Linked to residual risk R3)

Among the above undecidable items, items 1 and 2 are the core uncertainty sources for directional judgments; items 3, 4, and 5 are uncertainty sources for quantitative precision but do not alter directional conclusions.

7.4 Timeline Outlook

Based on the three-tier progressive judgment of "physically feasible $\rightarrow$ engineering-fragile $\rightarrow$ commercially distant," this report offers the following outlook on the development timeline for space-based AI computing:

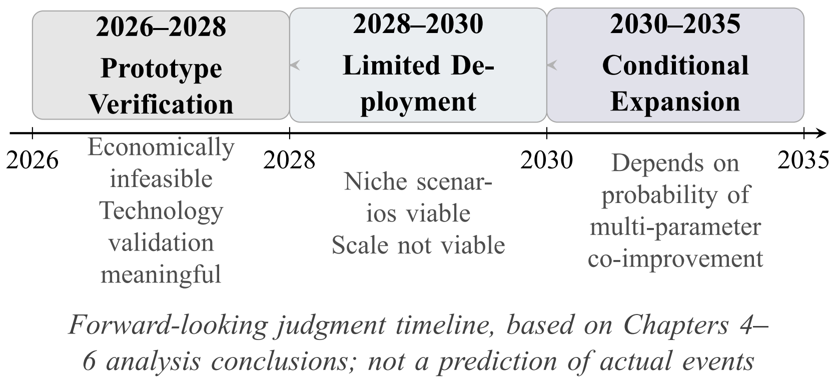

Figure 6: Three-phase forward-looking judgment timeline for space-based AI computing—the evolutionary path from prototype validation to conditional expansion. The core judgment for each phase is based on the physical, engineering, and commercial analysis conclusions of Chapters 4–6.

1. **2026–2028: Prototype Validation Phase.** The first two AI1 prototypes (planned for launch in 2027) and Rubin Space-1 on-orbit validation data are the most important information increments of this phase. Commercial viability in this phase is zero—costs at the prototype stage cannot be compared with costs of terrestrial mass-production data centers. The key is to validate: on-orbit GPU actual failure rate, heat dissipation system performance in the real LEO environment, and photovoltaic array on-orbit degradation rate. *Judgment: Economically infeasible, technically meaningful for validation.*

2. **2028–2030: Limited Deployment Phase.** If prototype validation results are positive, space-based AI computing may enter a limited deployment phase during 2028–2030—military/intelligence scenarios (with willingness to pay a premium for physical security, global coverage, and non-interceptability) and low-latency inference niche scenarios (leveraging LEO global coverage with <20–30 ms latency advantage) may be the first commercial customers. The deployment scale in this phase is expected to be tens to hundreds of satellites, not millions. *Judgment: Niche scenarios feasible, scale deployment infeasible.*

3. **2030–2035: Conditional Expansion Phase.** If the following four conditions are synchronously met—(1) on-orbit lifetime stably $\geq$ 7 years; (2) photovoltaic array cost reduced to <\$50/W_BOL; (3) AIT cost compressed to Starlink mass-production level; (4) launch cost <\$100/kg—space-based AI computing may begin expanding toward scale commercial applications. The synchronized probability of the four conditions is assessed at <25%, so the 2030–2035 conditional expansion is “possible but uncertain” rather than “highly probable.” *Judg-*

ment: Dependent on synchronized multi-parameter improvement probability; currently insufficient to support a definitive commitment.

7.5 Robustness Statement of Conclusions

- **Physical conclusions (Chapter 4) are highly robust**: They depend only on physical constants ($\sigma = 5.6704 \times 10^{-8}$ W m^{-2} K^{-4}) and publicly verifiable physical laws, and do not rely on any corporate disclosures or commercial assumptions. No future technological advances or information increments will change the Stefan–Boltzmann law or LEO orbital mechanics.

- **Engineering conclusions (Chapter 5) are moderately robust**: The "foundation" of the bottom-up mass chain—ground hardware specifications such as GB300 NVL72—comes from Tier-2 authoritative sources (Lenovo/NVIDIA official), with reliable anchors. Among the three space-adaptation increments, radiation protection and thermal management modifications both have independent literature anchors (Tier 3), but the structural modification increment remains an industry experience estimate (Tier 4). Approximately 35% of the cost items in the 9-subsystem breakdown of full-satellite total mass involve Tier-4 inference—if AI1's first-hand total mass or stowed dimensions were disclosed, the confidence interval for full-satellite mass would narrow significantly.

- **Commercial conclusions (Chapter 6) are sensitive to parametric assumptions**: 14/47 parameters in the TCO model are Tier-4 reasonable inferences. The core uncertainty lies not in any single parameter but in the probability of synchronized multi-parameter improvement. In other words, what determines commercial viability is not any one technology breaking through alone, but whether multiple key parameters can improve together within the same time window.

- **Non-Recurring Engineering (NRE) costs are not included in the TCO model.** After millions of units are produced, NRE can be amortized to negligible levels, but at the early 1–100 MW validation stage it is non-negligible. Primary NRE sources include: non-recurring design verification for satellite bus GNC/thermal/power/structure, customized ground segment development, and one-time investment in volume production lines.

- **Methodological robustness**: This report proceeds from physical laws (Stefan–Boltzmann), through engineering hardware anchors (NVIDIA GB300 NVL72), to a complete computational chain in independent Python modeling, following the methodology of first-principles bottom-up verification. Primary conclusions are traced as far as possible to publicly and independently verifiable original sources; the photovoltaic array ($\sim$50%) is cross-anchored via Tier-2/Tier-3 evidence, ensuring that directional conclusions will not undergo order-of-magnitude flips due to reasonable variation in Tier-4 parameters.

7.6 Analytical Stance and Method

This report handles contentious issues through a three-layer stratification of scenarios, specifications, and evidence:

- **Scenario stratification**: TCO must be read together with specific assumptions. This report presents complete results for three comparable branches and annotates the boundary conditions of each.
- **Specification stratification**: Figures under different specifications are not directly comparable. Appendix A provides a complete alignment matrix of 7 scenarios $\times$ 9 dimensions.
- **Evidence stratification**: All 47 key parameters are graded and adjudicated according to the four-tier evidence system. Tier-3 and Tier-4 parameters are used to express "best estimates at the current level of evidence" rather than being written directly as confirmed facts.
- **Retaining the undecidable**: For questions that cannot be closed with current public information, the report maintains the "currently undecidable" state and handles it separately from computable chains and known facts.

If this report's core judgment can be summarized in one sentence: **The physical foundation of space-based AI computing is solid, the engineering pathway is discernible, and the commercial prospect is distant.** Current public information supports continued observation rather than an early strong commercial commitment. The flight validation data from AI1 and Rubin Space-1 will be the next most important information increment.

(See Appendix C for complete formulas and substitution values for all computable-chain figures; see Appendix B for the disposition and verdict record of all contentious parameters; see Appendix E for deprecated derivation chains and the prohibited-line list; see Appendix D for executable traceability paths from this report's conclusions back to original sources.)

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A Alignment Matrix

This chapter provides the complete, independently consultable detailed matrix for the calibration discussion in Chapter 2. Different scenarios (or different research groups) that adopt different time windows, power conventions, architecture generations, power supply routes, and lifetime assumptions will produce numbers that cannot be directly compared laterally. The purpose of this matrix is not to judge superiority among entries, but to explicitly list the values and sources of each dimension, enabling readers to judge whether the premises of comparisons are aligned when reading the main body of this report or other external studies.

A.1 Calibration Dimension Definitions

The following are unified definitions for each column dimension of the matrix:

Time Window Number of years a single-generation satellite serves on orbit. A 5-year window means automatic deorbit and re-launch after 5 years; a 10-year window means a single-generation satellite serves 10 years without re-launch.

Power Convention Separately indicates peak power (maximum electrical output per satellite), sustained power (average available power over a full orbital period), and IT load power (available power ultimately delivered to the compute hardware).

Architecture Generation GPU architecture of the compute hardware. Blackwell corresponds to the GB300 NVL72 compute tray as the ground hardware anchor (Tier 2); Rubin corresponds to the NVIDIA Space-1 platform.

Photovoltaic Route Solar cell technology route. GaAs is the primary chain (AzurSpace 4G32-class, BOL 32%/EOL 29%); HJT is a parallel maturation route (BOL 22%/EOL 14%).

Battery Chemistry Energy storage battery system and depth-of-discharge (DoD) value. Branches: 5yr NMC (DoD 40%), 10yr Li-ion NMC (DoD 25%), 10yr LTO (DoD 80%).

Launch Unit Price Assumed cost per kilogram to orbit. Baseline is $200/kg (Starship V3 reuse target).

Lifetime Assumption Total comparison window and re-launch rules.

Redundancy Included? Whether on-orbit backup satellites are explicitly included in the satellite count.

Per-Satellite Mass Definition Whether the mass convention is compute payload only or full-system all-up satellite mass.

Ground Reference Convention The ground data center cost reference value used for TCO ratio comparison.

A.2 Complete Alignment Matrix

Table 28: Alignment Matrix — Values, Sources, and Comparability Determination for Each Scenario Dimension

Scenario	Time Window	Power Convention	Architecture	PV Route	Battery	Launch Price	Lifetime	Redundancy	Mass Definition
This Report Primary Chain	10yr	Peak 150kW / Sustained 120kW / IT Load $\sim$120kW	Blackwell (GB300 NVL72)	GaAs (Tier 2)	10yr Li-ion NMC, DoD 25%	$200/kg	Single-gen 10yr, no re-launch	No (8.33 sats = 1MW/120kW)	All-up 8.48t (9-subsystem summation)
This Report 5yr Reference	5yr	Same as above	Same as above	GaAs (Tier 2)	5yr NMC, DoD 40%	$200/kg	Deorbit + re-launch once after 5yr (2 generations total)	No	All-up 7.10t
This Report HJT Reference	5yr	Same as above	Same as above	HJT (Parallel maturation route, Tier 2)	5yr NMC, DoD 40%	$200/kg	Deorbit + re-launch once after 5yr (2 generations total)	No	All-up 9.13t
Industry Optimistic Scenario	Unspecified	Unspecified	Not locked	Unspecified (HJT assumption possible)	Unspecified	$100/kg or lower	Re-launch rule unspecified	Redundancy possible	Component breakdown not public

A.3 Quantitative Impact of Calibration Differences

1. **Time Window Difference** (5yr vs. 10yr): In this report's model, shortening the service life from 10 years to 5 years causes one re-launch of the entire satellite, approximately doubling the 10-year total manufacturing cost (10yr-GaAs baseline total cost $764.98M vs. 5yr-GaAs $1,341.26M). Any cross-convention comparison that does not control for the time window may have cost differences dominated by the lifetime assumption rather than by genuine system differences.

2. **Power Convention Difference** (Peak vs. Sustained vs. IT Load): The relationship between peak power 150 kW and sustained power 120 kW is not a simple "20% discount"—starting from the physical constraints of LEO orbital energy balance, 150 kW is the IT peak / public peak convention and cannot be directly equated to sustained IT; the forward physical demand for 120 kW sustained IT corresponds to approximately 210 kW array EOL output. If strictly capped at 150 kW, the maximum sustained IT is only about 86 kW. The two span the complete energy conversion chain (photovoltaic array $\rightarrow$ battery $\rightarrow$ PCDU $\rightarrow$ GPU) and must not be conflated as the same physical convention.

3. **Photovoltaic Route Difference** (GaAs vs. HJT): Under the same load, HJT requires an additional $\sim$670 m^2 of array area (1,295 vs. 625 m^2) and approximately 0.80t more total power system mass (3.73t vs. 2.93t). Furthermore, the publicly stated 70m wingspan constraint for AI1 directly rules out HJT as a feasible route—70m $\times$ 8.9m $\approx$ 625 m^2 is precisely self-consistent with the GaAs primary chain calculation, whereas the 1,295 m^2 required by HJT would demand $\sim$18.5m average width under a 70m wingspan, which is not realizable at current engineering levels.

4. **Battery Chemistry Difference** (NMC vs. LTO): Under 10-year service conditions, NMC must be restricted to 25% DoD (battery 1,542 kg), while LTO can operate at 80% DoD (battery 916 kg, $\sim$40% lighter). However, LTO has a higher unit price (~$350/kWh vs. $200/kWh), so the branch must be selected based on specific mission requirements and cannot be collapsed into a single parameter.

5. **Launch Unit Price Difference** ($100 vs. $200 vs. $500/kg): In this report's 10yr-GaAs baseline, launch cost accounts for only $\sim$1.8% of total cost ($14.13M / $764.98M). Even if launch cost drops from $200 to $100/kg, total cost falls by only $\sim$0.9%. This means that relying solely on launch cost reduction is insufficient to flip the commercial conclusion.

6. **Per-Satellite Mass Definition Difference** (Compute Payload vs. All-Up): Under the kW/t back-calculation chain's 2.14t convention (compute payload only), the load count is $\sim$37–46 satellites per launch (Starship 100+t-class capacity). But the current-round bottom-up all-up mass baseline is 8.48t, reducing the load count to $\sim$9–14 satellites per launch. Different mass conventions directly alter launch cost estimates and constellation-scale judgments.

A.4 Why Different Scenarios Cannot Be Directly Compared

1. **Different Comparison Denominators**: This report uses "1 MW sustained IT load / 10-year total window" as the unified comparison denominator. Other scenarios may use different target power levels (e.g., 100 MW class) or different comparison windows (e.g., 30 years), making direct comparison of per-satellite cost or satellite count meaningless.

2. **Different Cost Boundaries**: This report's cost includes manufacturing cost (9-subsystem breakdown) + launch + insurance + O&M + decommissioning. Other scenarios may include only manufacturing cost or only launch cost; directly comparing total cost figures constitutes a convention mismatch.

3. **Implicit Lifetime Assumption Difference**: Industry optimistic scenarios often implicitly assume long life ($\geq$10 years) without re-launch, but do not provide publicly verifiable hardware lifetime evidence. This report has declared in the 10-year branch that "no additional publicly verifiable 10-year lifetime mass or cost premium factors are assumed; this is a conservative placeholder" (residual risk R8).

4. **Maturity Assumption Difference**: Industry optimistic scenarios may assume that the HJT route is sufficiently mature to drive procurement prices substantially lower, and that "mature mass-production compression" of various coefficients is already in place. This report labels such assumptions by tier under the four-tier evidence system and does not present optimistic-scenario parameters as proven facts.

A.5 Cross-Scenario Conversion Reference

If the reader needs to perform approximate conversions between different conventions, the following conversion factors are provided for reference (all based on engineering judgment from this report's model, not physical laws):

Table 29: Approximate Cross-Convention Conversion Reference

Conversion Direction	**Approximate Factor**	**Notes**
Peak Power → Sustained Power	Cannot be converted with a simple × factor	See the full energy chain derivation in Chapter 5 §6.7 of the main text; the two span PV array → battery → PCDU → GPU
Sustained Power → IT Load	×0.94	Typical bus efficiency (distribution + conversion loss ∼6%)
5yr Window → 10yr Window	×1.75–2.0	Contains total cost amplification from one re-launch (with 10yr total window as denominator)
GaAs → HJT (Array Area)	×2.07	EOL efficiency ratio 0.29/0.14 = 2.07×
Compute Payload Only → All-Up Mass	×5.0–6.5	Includes power system / thermal / bus / propulsion / communications / AIT full system

Note: The above factors are only for rapid order-of-magnitude estimation and do not substitute for the full formula substitution in Appendix C. If the reader requires precise numbers, they should directly reference the corresponding complete derivations in Appendix C and the adjudicated parameter values in Appendix B.

B Key Parameter Values and Controversy Adjudication

This chapter provides a registration index for all key parameters involved in the primary chain—the "identity card" for every number in the main text. Readers can look up the source tier, adjudication status, and further traceability path of any key number in the main text within this chapter. All parameters have undergone per-parameter independent adjudication and are anchored in `current-note.md`.

Note: Due to space constraints, the body of this chapter lists only core controversial parameters (§B.2–§B.3). The complete per-parameter registration table for all 47 parameters (including tier, value, adjudication status, and source routing for each parameter) can be traced via the five-layer routing table in Appendix D to the internal parameter adjudication document and structured JSON input files.

B.1 Four-Tier Evidence System Definition

Table 30: Four-Tier Evidence Classification Definitions

Tier	Name	Definition	Determination Criteria
Tier 1	Physical Derivation Chain	Directly derived from clear physical formulas, boundary conditions, and mission constraints	Formula can be written clearly, inputs traceable, units self-consistent
Tier 2	Authoritative Source	From clearly identified authoritative sources or official public materials	Verifiable URL available, applicable boundaries can be stated
Tier 3	Transferable Evidence	From industry materials or analog platform data not directly transferable	Must write out transferable/non-transferable portions, transfer rationale, and interval basis
Tier 4	Reasonable Inference	Insufficient public materials; engineering inference only	Must state inference basis, boundary conditions, and failure risks
Prohibited	Prohibited from Model Entry	Value directly contradicted by public evidence or entirely unsupported	Replaced or permanently excluded
Deprecated	Deprecated	This derivation path / label has been permanently superseded by this report	Archived only; must not be used in main text

B.2 Core Controversial Parameter Adjudication Table

The following are the most critical controversial parameters in the report and their adjudication results. Each parameter is accompanied by a one-sentence adjudication summary; the complete adjudication record (including source URLs, derivation chains, and per-parameter independent adjudication process) is in `current-note.md` §Task V2.

Table 31: Core Controversial Parameter Adjudication (I): Per-Satellite Mass (Multiple Conventions) (1/2)

Parameter Name	Current Value	Tier	Adjudication Status	One-Sentence Adjudication Summary
Compute Payload Mass (Baseline)	1,500 kg	Tier 2+3	Absorbed into primary chain	Takes NVIDIA GB300 NVL72 ground hardware as A1 anchor (Tier 2), plus three space-hardening increments (radiation/thermal/structural) decomposed (Tier 3); per-parameter adjudication yields 1,200/1,500/2,000 kg three scenarios
All-Up Satellite Mass (10yr-GaAs)	8,479 kg	Tier 1+2+3+4	Absorbed into primary chain	9-subsystem summation: compute payload 1,500 + heat dissipation 550 + PV array 938 + battery 1,573 + PCDU 419 + bus 1,743 + propulsion 350 + communications 300 + system margin/redundancy/additional 1,106 kg
All-Up Satellite Mass (5yr-GaAs)	7,103 kg	Same as above	Absorbed into primary chain (sensitivity reference)	Same physical configuration as 10yr-GaAs, except for battery branch (5yr NMC, 983 kg) and AIT margin (5yr branch) differences

Table 32: Core Controversial Parameter Adjudication (I): Per-Satellite Mass (Multiple Conventions) (2/2)

Parameter Name	Current Value	Tier	Adjudication Status	One-Sentence Adjudication Summary
All-Up Satellite Mass (5yr-HJT)	9,133 kg	Same as above	Downgraded to reference (parallel route)	HJT PV array 2,330 kg (vs. GaAs 938 kg), causing all-up mass heavier by ∼2.03t
Previous-Era Compute Payload Mass Values	2,143/2,727/3,333 kg	Deprecated	Deprecated	Derived from kW/t back-calculation chain (150kW / 70/55/45 kW/t), denominator sourced from B1-level second-hand relay with circular reference risk; permanently superseded by bottom-up chain

Table 33: Core Controversial Parameter Adjudication (II): Launch Loading

Parameter Name	**Current Value**	**Tier**	**Adjudication Status**	**One-Sentence Adjudication Summary**
Load Count (All-Up Convention)	$\sim$9–14 sats/launch	Tier 1 (geometric constraint)	Absorbed into primary chain	Based on all-up satellite mass 8.48t (10yr-GaAs), Starship 100+t-class capacity, deducting 10–20% adapter/separation mechanism margin
Load Count (Compute Payload Only)	$\sim$53–60 sats/launch	Tier 1+3	External reference only	Based on compute payload mass 1,500 kg only; assumes in-orbit integration—this assumption is unverified and cannot serve as primary chain

Table 34: Core Controversial Parameter Adjudication (III-A): GaAs Primary Chain Manufacturing Cost (1/2)

Parameter Name	Current Value	Tier	Adjudication Status	One-Sentence Adjudication Summary
Per-Satellite Manufacturing Cost (10yr-GaAs)	\$83.61M	Tier 3+4 mixed	Absorbed into primary chain	9-subsystem summation: PV \$46.27M (55%), AIT \$18.29M (22%), compute payload \$7.20M (9%), bus \$6.27M (8%), communications \$2.50M, propulsion \$1.30M, PCDU \$1.05M, heat dissipation \$0.66M, battery \$0.06M

Table 35: Core Controversial Parameter Adjudication (III-A): GaAs Primary Chain Manufacturing Cost (2/2)

Parameter Name	Current Value	Tier	Adjudication Status	One-Sentence Adjudication Summary
PV Array Cost (GaAs)	\$180/\$200/\$250 /W_BOL	Tier 3	Absorbed into primary chain	Three-anchor cross-check: AzurSpace product existence + previous-era estimate 200–400 USD/W + satsearch retail panel; \$200/W_BOL taken as baseline
Compute Payload Cost	\$38k/\$60k/\$100k/kW	Tier 3	Absorbed into primary chain	Ground GB300 NVL72 baseline \$25–33k/kW, multiplied by 1.5–3× mass-production premium (SpaceX COTS strategy), after 3 rounds of search and per-parameter adjudication

Table 36: Core Controversial Parameter Adjudication (III-B): Reference Technology Manufacturing Cost — HJT Array (1/2)

Parameter Name	Current Value	Tier	Adjudication Status	One-Sentence Adjudication Summary
HJT Array Cost	Ground \$0.14–0.24/W; space \$̃2–3/W	Tier 3	Downgraded to reference	Multi-source cross-validation of ground mass production (Tier 2) + Risen Energy space-grade quote (Tier 3); ground→space amplification 10–15× (Tier 4); overall closure lower than GaAs

Table 37: Core Controversial Parameter Adjudication (III-B): Reference Technology Manufacturing Cost — Battery (2/2)

Parameter Name	Current Value	Tier	Adjudication Status	One-Sentence Adjudication Summary
Battery NMC Cost	\$200/kWh (fixed)	Tier 3	Absorbed into primary chain	Direct Starlink transfer, zero additional premium—Starlink NMC packs are inherently space batteries; Ray Barsa presentation + hundred-million-unit Panasonic 21700 procurement anchor
Battery LTO Cost	\$260/\$350/\$400/kWh	Tier 4	External reference only (sensitivity branch)	Proportional compression method: ground LTO/NMC price ratio 1.3–1.5× × \$200/kWh = \$260–400/kWh; traditional Saft certified pricing path excluded after comparative analysis

Table 38: Core Controversial Parameter Adjudication (IV): Ground Reference TCO Convention and TCO Ratio

Parameter Name	Current Value	Tier	Adjudication Status	One-Sentence Adjudication Summary
Ground Reference TCO (Median Estimate)	∼\$64.6M/MW/10yr	Tier 4	External reference only	Composite estimate based on publicly available ground AI data center construction cost analyses; specific literature sources in Appendix D §D.4
TCO Ratio (Space / Ground)	∼11.8× (10yr-GaAs baseline)	Mixed tier	Absorbed into primary chain	\$764.98M / \$64.6M ≈ 11.8×; doubly constrained by space-side Tier 4 parameters (∼35% of cost) and ground-side convention uncertainty

B.3 47-Parameter Tier Classification Statistical Summary

Table 39: 47-Parameter Tier Classification Statistics

Tier	Count	Representative Parameters
Tier 1 (Physical Derivation Chain)	5	Solar constant (1361.0 W/m^2), radiator area (110 m^2, S-B derivation), array area (physical formula), battery nominal capacity (186.4 kWh)
Tier 2 (Authoritative Source)	7	Peak power (150 kW), sustained power (120 kW), orbital altitude (600 km), GaAs BOL/EOL efficiency (32%/29%), HJT BOL/EOL efficiency (22%/14%)
Tier 3 (Transferable Evidence)	17	GaAs/HJT areal density, DoD, specific energy, compute payload cost, GaAs/HJT array cost, etc.
Tier 4 (Reasonable Inference)	14	Radiator areal density, bus coefficient, propulsion mass/cost, communications mass/cost, AIT margin/cost, PCDU specific power/cost, etc.
Prohibited from Model Entry	1	Previous-era battery cost \$400/kWh ($\sim$225$\times$ contradiction with public space-grade prices; replaced)
Deprecated	3	kW/t back-calculation chain (70/55/45), previous-era compute payload mass (2143/2727/3333 kg), HJT "prohibited from entry into primary model" label
Total	**47**	

Tier 4 parameters account for 30% (14/47), an objective consequence of the limited public disclosure of AI1 information, not an execution deficiency. All weak-evidence parameters have been embedded within the per-parameter independent adjudication gate; each Tier 4 parameter carries inference basis, boundary conditions, and failure risk statements. Adjudication records can be traced via the Appendix D five-layer routing table to the V1–V37 per-parameter adjudications in `current-note.md` §Task V2.

B.4 Complete Parameter Inventory by Tier

The following is the complete registration table for all 47 parameters. One parameter per row, organized by tier. Source routing points to the corresponding row in Appendix D.

Note: The full 47-row parameter registration table lists only core controversial parameters due to space constraints (see §B.2–§B.3 for details). Readers needing the full parameter set should follow these traceability paths:

- Full 47-parameter per-item tier classification list: see internal parameter adjudication document, traceable via Appendix D routing table

- Mass parameter structured input (JSON): see corresponding row in Appendix D routing table
- Cost parameter structured input (JSON): see corresponding row in Appendix D routing table
- Five-layer quick-lookup routing (Conclusion → Cross-Verification → JSON → Adjudication → URL): Appendix D §D.2

In particular, the following adjudications underwent key adjustments in the latest round of internal cross-verification, differing from values in previous review rounds:

Table 40: Key Adjudication Adjustments (Previous Review Rounds vs. Current Round)

Parameter	**Previous Value**	**Current-Round Value**	**Reason for Adjustment**
Compute Payload Mass Chain	2,143–3,333 kg (kW/t back-calculation)	1,200–2,000 kg (bottom-up)	kW/t denominator sourced from B1-level second-hand relay with circular reference risk; replaced by ground hardware A1 anchor + three space-hardening increments
HJT BOL Efficiency	20%	22%	GN-004 review noted 6pp below lowest authoritative source; research team selected the most conservative among three candidates (22/24/25)
HJT Areal Density	1.2 kg/m^2	1.8 kg/m^2	Revised upward after searching Starlink references; panel+deployment modeled separately
PCDU Specific Power	10.0 kW/kg	0.5 kW/kg (baseline)	Terma flight product anchor revealed original value overestimated by $\sim$18$\times$; PCDU rated power taken as 209.8 kW (EOL array dominant), mass corrected to $\sim$420 kg
Bus Coefficient	0.18/0.22/0.28	0.25/0.35/0.45	Starlink V2 Mini mass rough decomposition reference ratio $\sim$0.38
Battery DoD	Single 40%	Three-branch: 5yr 40% / 10yr Li-ion 25% / 10yr LTO 80%	Research team held that a one-size-fits-all approach should not be applied
HJT Route Positioning	“Can only be downgraded” / “Prohibited from entry into primary model”	“Parallel maturation route”	Supported by Task 3 evidence package (17 URLs + 12 physical chain closures)
Battery Cost NMC	\$400/kWh (prohibited from model entry)	\$200/kWh (fixed)	Direct Starlink transfer, zero-premium principle
Propulsion Cost	Linear scaling	0.6-power scaling + lifetime branch	Corrected to 0.6-power scaling after per-item verification

C Core Calculation Methods and Formulas

This chapter provides the complete derivation process for all calculation conclusions in the main text. The main text gives results and logical bridging; this chapter gives formulas, substituted values, and boundary conditions so that technically inclined readers can independently reproduce the calculations.

C.1 Stefan–Boltzmann Heat Dissipation Derivation

C.1.1 Basic Formula

In a vacuum orbital environment, a satellite can only dissipate heat via thermal radiation. The Stefan–Boltzmann law gives the relationship between blackbody radiation flux and temperature:

$$\frac{P}{A} = \varepsilon\sigma T^4 \tag{7}$$

where:

- P — Radiative heat dissipation power [W]
- A — Radiator area [m^2] (Note: for dual-sided radiation, A is the radiator geometric area; effective radiating area is $A_{\text{eff}} = 2A$)
- ε — Surface emissivity, dimensionless, typical space-grade thermal control coating $\varepsilon = 0.85$–0.95
- σ — Stefan–Boltzmann constant, $5.670374419 \times 10^{-8}$ W m^{-2} K^{-4} (CODATA 2018 recommended value)
- T — Radiator surface absolute temperature [K]

Note: In the above formula σ is taken as 5.6704×10^{-8} W m^{-2} K^{-4}. For dual-sided radiators, the actual radiation flux $P/A_{\text{geo}} = 2\varepsilon\sigma T^4$ (A_{geo} being the radiator geometric area). This report uniformly uses geometric area as the baseline convention for A.

C.1.2 AI1 Operating Point Back-Calculation

Assuming AI1 radiators use a dual-sided radiation configuration:

- Peak heat load $P_{\text{peak}} = 150$ kW (Tier 2, multi-source cross-verified)
- Radiator geometric area $A = 110$ m^2 (Tier 1, back-calculated from system: 150 kW / ~ 1.364 kW/m^2)

- Emissivity $\varepsilon = 0.90$ (Tier 4, engineering inference: typical space-grade white paint / OSR coating)

Dual-sided radiation equivalent heat flux (based on geometric area):

$$q_{\text{geo}} = \frac{P_{\text{peak}}}{A} = \frac{150 \times 10^3}{110} \approx 1364 \text{ W/m}^2 \tag{8}$$

This heat flux corresponds to a single-sided equivalent radiation flux $q_{\text{eff}} = q_{\text{geo}}/2 = 682$ W/m^2.

Solving for radiator surface temperature:

$$T = \left(\frac{P_{\text{peak}}}{2A\varepsilon\sigma}\right)^{1/4} = \left(\frac{150 \times 10^3}{2 \times 110 \times 0.90 \times 5.6704 \times 10^{-8}}\right)^{1/4} \approx 342.5\,\text{K} \approx 69.4°\text{C} \tag{9}$$

C.1.3 Radiator Area Scaling

Given any target dissipation power P_{target}, the required radiator geometric area (dual-sided radiation):

$$A = \frac{P_{\text{target}}}{2\varepsilon\sigma T^4} \tag{10}$$

Under this report's 1 MW sustained IT load target ($\sim$8.3 AI1-class satellites), the total radiator geometric area is approximately $110 \times 8.3 \approx 917$ m^2.

C.1.4 Parameter Sensitivity Analysis

The T^4 dependence makes radiator area highly sensitive to temperature. Table 41 shows the impact of parameter variations on radiator area (with AI1 baseline as reference):

Table 41: Stefan–Boltzmann Parameter Sensitivity (Baseline: P=150 kW, A=110 m^2, ε=0.90, T=342.5 K)

Parameter Change	Radiator Area Change	Notes
T: 342.5 → 300 K	$A \to$ 186 m^2 (+69%)	Per 10 K decrease, area increases ∼8–10%
T: 342.5 → 400 K	$A \to$ 63 m^2 (-43%)	High-temperature chips (SiC/GaN) can significantly reduce radiator area
ε: 0.90 → 0.85	$A \to$ 117 m^2 (+6%)	Coating degradation is a known risk for long-duration service
ε: 0.90 → 0.95	$A \to$ 104 m^2 (-5%)	High-emissivity coatings (e.g., Vantablack-class) yield limited improvement
Dual-sided → Single-sided radiation	$A \to$ 220 m^2 (+100%)	Single-sided radiation directly doubles radiator area demand

C.1.5 Heat Dissipation Density Boundary Conclusion

The current feasible upper bound is approximately 1,400 W/m^2 (dual-sided radiation geometric area basis, T ≈342.5 K, ε ≈0.90). Doubling to 2,800+ W/m^2 requires $T \approx 407$ K—not feasible on any single technology path before 2035. Combined paths (high-temperature chips + durable coatings + moderate deployable structures) can achieve 40–50% improvement by 2032.

C.2 Orbital Power System Area and Energy Budget Derivation

C.2.1 Orbital Mechanics Fundamentals

For a 600 km circular orbit ($R_\oplus = 6378.137$ km, $\mu = 3.986004418 \times 10^{14}$ m^3/s^2, WGS84):

$$T_{\text{orbit}} = 2\pi\sqrt{\frac{(R_\oplus + h)^3}{\mu}} \tag{11}$$

Substituting $h = 600$ km:

$$T_{\text{orbit}} = 2\pi\sqrt{\frac{(6378.137 + 600)^3 \times 10^9}{3.986004418 \times 10^{14}}} \approx 5802 \text{ s} \approx 96.7 \text{ min} \tag{12}$$

Eclipse time ($\beta = 0$, most conservative approximation, orbital plane coplanar with the Sun vector):

$$t_{\text{eclipse}} = T_{\text{orbit}} \times \frac{\arcsin(R_\oplus/(R_\oplus + h))}{\pi} \tag{13}$$

$$t_{\text{eclipse}} \approx 5802 \times \frac{\arcsin(6378.137/6978.137)}{\pi} \approx 2128 \text{ s} \approx 35.5 \text{ min} \tag{14}$$

Sunlight duration:

$$t_{\text{sunlight}} = T_{\text{orbit}} - t_{\text{eclipse}} \approx 3674 \text{ s} \approx 61.2 \text{ min} \tag{15}$$

C.2.2 Battery Capacity Requirement

Eclipse-segment load energy demand:

$$E_{\text{load, eclipse}} = P_{\text{sustained}} \times t_{\text{eclipse}} = 120 \text{ kW} \times \frac{2128}{3600} \text{ h} \approx 71.0 \text{ kWh} \tag{16}$$

Battery nominal capacity (with DoD and discharge efficiency constraints):

$$E_{\text{battery, nominal}} = \frac{E_{\text{load, eclipse}}}{\text{DoD} \times \eta_{\text{discharge}}} \tag{17}$$

Substituting baseline values (DoD = 0.40, $\eta_{\text{discharge}}$ = 0.95):

$$E_{\text{battery, nominal}} = \frac{71.0}{0.40 \times 0.95} \approx 186.8 \text{ kWh} \tag{18}$$

Battery mass:

$$M_{\text{battery}} = \frac{E_{\text{battery, nominal}} \times 1000}{\rho_{\text{specific}}} = \frac{186,800}{190} \approx 983 \text{ kg} \tag{19}$$

where $\rho_{\text{specific}} = 190$ Wh/kg is the Li-ion NMC pack-level specific energy (Tier 3, Starlink transfer).

C.2.3 Recharge Power and Array Area

Recharge energy demand (including charge/discharge efficiency losses):

$$E_{\text{recharge}} = \frac{E_{\text{load, eclipse}}}{\eta_{\text{charge}} \times \eta_{\text{discharge}}} = \frac{71.0}{0.95 \times 0.95} \approx 78.7 \text{ kWh} \tag{20}$$

Sunlight-segment additional recharge power:

$$P_{\text{recharge, extra}} = \frac{E_{\text{recharge}}}{t_{\text{sunlight}}} = \frac{78.7}{61.2/60} \approx 77.2 \text{ kW} \tag{21}$$

PCDU output requirement:

$$P_{\text{pcdu, out}} = P_{\text{sustained}} + P_{\text{recharge, extra}} = 120 + 77.2 \approx 197.2 \text{ kW} \tag{22}$$

EOL array power requirement (deducting bus losses, $\eta_{\text{bus}} = 0.94$):

$$P_{\text{array, eol}} = \frac{P_{\text{pcdu, out}}}{\eta_{\text{bus}}} = \frac{197.2}{0.94} \approx 209.8 \text{ kW} \tag{23}$$

BOL array power requirement (back-calculated from EOL to BOL, degradation factor = $\eta_{\text{eol}}/\eta_{\text{bol}}$):

$$P_{\text{array, bol}} = \frac{P_{\text{array, eol}}}{\eta_{\text{eol}}/\eta_{\text{bol}}} \tag{24}$$

Array area (solar constant $G = 1361.0$ W/m^2, packaging correction $f_{\text{pkg}} = 0.85$):

$$A_{\text{array}} = \frac{P_{\text{array, eol}}}{G \times \eta_{\text{eol}} \times f_{\text{pkg}}} \tag{25}$$

GaAs Primary Chain Substitution ($\eta_{\text{bol}} = 0.32$, $\eta_{\text{eol}} = 0.29$):

$$P_{\text{array, bol}} = \frac{209.8}{0.29/0.32} \approx 231.5 \text{ kW} \tag{26}$$

$$A_{\text{array, GaAs}} = \frac{209.8 \times 1000}{1361.0 \times 0.29 \times 0.85} \approx 625 \text{ m}^2 \tag{27}$$

Array mass (areal density $\sigma_{\text{array}} = 1.5$ kg/m^2, panel-only, deployment mechanisms accounted separately):

$$M_{\text{array, GaAs}} = 625 \times 1.5 = 938 \text{ kg} \tag{28}$$

HJT Parallel Route Substitution ($\eta_{\text{bol}} = 0.22$, $\eta_{\text{eol}} = 0.14$):

$$P_{\text{array, bol}} = \frac{209.8}{0.14/0.22} \approx 329.7 \text{ kW} \tag{29}$$

$$A_{\text{array, HJT}} = \frac{209.8 \times 1000}{1361.0 \times 0.14 \times 0.85} \approx 1295 \text{ m}^2 \tag{30}$$

Array mass (areal density $\sigma_{\text{array}} = 1.8$ kg/m^2):

$$M_{\text{array, HJT}} = 1295 \times 1.8 = 2330 \text{ kg} \tag{31}$$

C.2.4 PCDU Mass

PCDU rated power is taken as $\max(P_{\text{array, eol}}, P_{\text{peak}}) = \max(209.8, 150) = 209.8$ kW (EOL array output power is dominant because the array must simultaneously satisfy IT load + battery recharge):

$$M_{\text{pcdu}} = \frac{P_{\text{pcdu, rating}}}{\rho_{\text{pcdu}}} = \frac{209.8}{0.5} \approx 419.6 \text{ kg} \tag{32}$$

where $\rho_{\text{pcdu}} = 0.5$ kW/kg is the PCDU baseline specific power (Tier 4, Terma BepiColombo MTM flight product anchor 0.54 kW/kg as reference benchmark; 0.5 kW/kg adopted after Starlink "simplified power electronics" improvement).

The key anchor for PCDU specific power is the Terma flight product (0.54 kW/kg); this anchor significantly improves the reliability of the PCDU mass estimate.

C.3 Bottom-Up Compute Payload Mass Chain Derivation

C.3.1 Ground Hardware Baseline (Tier 2 / A1 Anchor)

The new chain for compute payload mass does not rely on any specific-power (kW/t) denominator, but instead starts from publicly disclosed NVIDIA ground hardware specifications:

Table 42: Ground Hardware Mass Anchors

Source	Hardware Unit	Mass Data	URL
Lenovo LP2357 Product Guide	GB300 NVL72 Compute Tray	29 kg/tray (4 GPU + 2 CPU)	lenovopress link
NVIDIA PCF Datasheet	HGX B200 Baseboard	32 kg (8 GPU)	NVIDIA PCF PDF
NVIDIA DGX B200 User Guide	DGX B200 Full System	142.4 kg/system	DGX B200 docs

Under a 150 GPU configuration (B200 TDP 1,000W/GPU), approximately 38 GB300 NVL72 compute trays are required:

$$M_{\text{ground baseline}} \approx 38 \times 29 \approx 1002\text{–}1102 \text{ kg} \tag{33}$$

This baseline already includes per-GPU liquid-cooling cold plates (ground GB300 NVL72 standard direct-to-chip liquid cooling), and excludes rack-level coolant distribution units (CDU), fans (not needed in space), and AC/DC PSUs.

C.3.2 Decomposition of Three Space-Hardening Increments

From bare ground hardware to space-hardened hardware, the increment is decomposed into three items (itemized

D Evidence Chains, Citation Chains, and Script Routing

This chapter serves as the executable traceability path from main-text conclusions back to original sources—the main text says "see Appendix D", and readers come here to trace layer by layer via the tables.

D.1 Five-Layer Routing Explanation

All key conclusions in this report can be traced back to the most original public sources via the following five-layer routing:

1. **Main-Text Conclusion** — Root-level deliverable (*Phase-2 Differential Analysis Preliminary Study.md*) or statistical figures in each results document
2. **Cross-Verification Document** — Per-Task gap closure status and cross-verification analysis (`stage2_gap_closure_verification.md`, etc.)
3. **Structured JSON Input** — Machine-readable parameter values (`stage2_mass_inputs_blackwell_hjt.json`, `stage2_cost_inputs_blackwell_hjt.json`)
4. **V Adjudication Records** — V1–V37 per-parameter adjudication records in the internal parameter adjudication document (containing derivation chains, source URLs, per-parameter adjudication)
5. **Original Source URL / Formula** — Independently verifiable public web pages, DOIs, or physical formulas

Usage: Find the conclusion you want to look up in the table → open the file in the corresponding column from left to right → each layer precisely points to more original supporting evidence. The rightmost column gives the final URL or formula, independently verifiable.
[3]

[3] All Python calculation scripts and structured input JSON files referenced in this report will be open-sourced to a GitHub research repository upon finalization of the report, at which point readers can independently reproduce all model outputs. The current-stage script routing is in §D.3.

D.2 Key Conclusion Quick-Lookup Routing Table

The following tables are extracted from `stage2_evidence_routing_guide.md`, covering all core conclusions. `{ROOT}` in paths denotes the project root directory. Due to the large number of parameters, the routing is divided into four tables: Power and Efficiency (D.2-A1), Areal Density and Battery (D.2-A2), Subsystem Physical Parameters (D.2-B1), and Cost Summary (D.2-B2).

Table 43: Key Conclusion Five-Layer Quick-Lookup Routing Table (A1: Power Baseline, Payload Mass, and PV Efficiency)

Conclusion	**Value**	**Tier**	**First Landing Point (Results Doc)**	**Second Landing Point (Cross-Verification Doc)**	**Third Landing Point (JSON / Adjudication)**	**Fifth Landing Point (URL / Formula)**
Power Orbital Baseline	Peak 150kW / Sustained 120kW / Orbit 600km	Tier 2	*Phase-2 Differential Analysis Preliminary Study.md* §1.3	`stage2_gap_closure_verification.md` §VII	`current-note.md` §V1–V3	Inkl / Tom's Hardware inkl.com / Knightli knightli.com / Indexbox indexbox.io three sources consistent
Compute Payload Mass (Baseline)	1,500 kg	Tier 2+3	Same as above §5.2	Same as above §III	`stage2_mass_inputs_blackwell_hjt.json` → `current-note.md` §V30	Lenovo LP2357, NVIDIA PCF, NVIDIA DGX B200 User Guide (URLs in Appendix C §C.3.1)
GaAs Efficiency BOL	32%	Tier 2	`stage2_power_system_results_blackwell_hjt.md` §III	`stage2_gap_closure_verification.md` §VII	`current-note.md` §V4	AzurSpace 4G32 product page https://www.azurspace.com/en/products/space-products/
GaAs Efficiency EOL	29%	Tier 2	Same as above	Same as above	`current-note.md` §V5	BOL 32% × (1−9.4% degradation); degradation rate from NASA NEPP report database III-V space cell public literature
HJT Efficiency BOL	22%	Tier 2	Same as above	Same as above §IV	`current-note.md` §V6	Huasun huasunsolar.com mass production >26%; Risen Energy prnewswire.com 26.61%; LONGi longi.com 27.81%; AM0 correction ×0.85→ conservatively take 22%
HJT Efficiency EOL	14%	Tier 2	Same as above	Same as above	`current-note.md` §V7	CEA/INES independent irradiation experiment https://www.cea.fr/cea-tech/liten/...; ISFH Caltec independent verification https://ines-solaire.org/

Table 44: Key Conclusion Five-Layer Quick-Lookup Routing Table (A2: Array Areal Density and Battery Parameters)

Conclusion	Value	Tier	First Landing Point (Results Doc)	Second Landing Point (Cross-Verification Doc)	Third Landing Point (JSON / Adjudication)	Fifth Landing Point (URL / Formula)
GaAs Array Areal Density	1.5 kg/m^2 (panel-only)	Tier 3	`stage2_power_system_results_blackwell_hjt.md` §III	`stage2_gap_closure_verification.md` §VII	`current-note.md` §V8	Industry range 1.5–2.5 kg/m^2 (space-grade rigid GaAs panels), taken at low end of range
HJT Array Areal Density	1.8 kg/m^2 (panel-only)	Tier 3	Same as above (HJT row)	Same as above §IV	`current-note.md` §V9	Starlink V2 Mini panel-level back-calculation $\sim$1.7–2.6 kg/m^2; 33FG Research silicon-based array range 1.0–3.0
Battery DoD (Three-Branch)	5yr 40% / 10yr Li-ion 25% / 10yr LTO 80%	Tier 3+4	*Phase-2 Differential Analysis Preliminary Study.md* §8.2	Same as above §V	`current-note.md` §V10	Ray Barsa (SpaceX Chief Battery Engineer) presentation https://ciclibattery.com/...; Saft LTO whitepaper https://saft.com/...
Battery Specific Energy	Li-ion NMC 190 Wh/kg / LTO 100 Wh/kg	Tier 3	Same as above	Same as above	`current-note.md` §V11	Starlink pack 230 Wh/kg (cell$\rightarrow$pack derating $\sim$12%); Saft LTO 30–50% lower than Li-ion

Table 45: Key Conclusion Five-Layer Quick-Lookup Routing Table (B1: Battery Cost and Subsystem Physical Parameters)

Conclusion	Value	Tier	First Landing Point (Results Doc)	Second Landing Point (Cross-Verification Doc)	Third Landing Point (JSON / Adjudication)	Fifth Landing Point (URL / Formula)
Battery NMC Cost	$200/kWh (fixed)	Tier 3	`stage2_economic_branches_blackwell.md` §IV	`stage2_gap_closure_verification.md` §V	`current-note.md` §V36	Direct Starlink transfer, zero-premium principle; Ray Barsa presentation + hundred-million-unit Panasonic 21700 procurement
Battery LTO Cost	$260/$350/$400/kWh	Tier 4	Same as above §VIII	Same as above	`current-note.md` §V37	Ground LTO/NMC price ratio 1.3–1.5× × $200/kWh; proportional compression method
Radiator Areal Density	4.0/5.0/6.5 kg/m^2	Tier 4	*Phase-2 Differential Analysis Preliminary Study.md* §4.3	Same as above §III	`current-note.md` §V20	Spacecraft thermal control structural areal density 3–8 kg/m^2; ISS 840m^2/1000kg≈1.2 is deployable design not applicable to rigid panels
Bus Coefficient	0.25/0.35/0.45	Tier 4	Same as above	Same as above §VII	`current-note.md` §V21	Starlink V2 Mini mass rough decomposition: all-up 740kg − PV 197 − battery 48 − propulsion 40 − comms 170 = structure/thermal/harness ∼285kg → ratio ∼0.38
Propulsion Mass	200/350/500 kg	Tier 4	Same as above §12.1	Same as above	`current-note.md` §V22	Starlink V2 Mini propulsion ∼5–6% of all-up mass (SpaceX in-house argon Hall thruster); AI1 6–8t-class → 5–8% → 300–640kg

Table 46: Key Conclusion Five-Layer Quick-Lookup Routing Table (B2: Communications / Power / Cost Summary)

Conclusion	Value	Tier	First Landing Point (Results Doc)	Second Landing Point (Cross-Verification Doc)	Third Landing Point (JSON / Adjudication)	Fifth Landing Point (URL / Formula)
Communications Mass	200/300/450 kg	Tier 4	`stage2_gap_closure_verification.md` §III	Same as above	`current-note.md` §V23+V28	Google Suncatcher (2025 arXiv) https://arxiv.org/html/2511.19468v1: space-based AI cluster requires tens of Tbps ISL
PCDU Specific Power	1.0/0.5/0.2 kW/kg	Tier 4	`stage2_power_system_results_blackwell_hjt.md` §III	Same as above §VII	`current-note.md` §V25	Terma PCDU flight product (satsearch): 15kW/28kg=0.54kW/kg
Compute Payload Cost	$38k/$60k/$100k/kW	Tier 3	`stage2_economic_branches_blackwell.md` §IV	Same as above §VII	`current-note.md` §V12	GB300 NVL72 ground $25–33k/kW awesomeagents.ai; H100 space prototype 17–33× culled.org; SpaceX COTS strategy 1.5–3×
GaAs Array Cost	$180/$200/$250/W_BOL	Tier 3	Same as above	Same as above	`current-note.md` §V14	AzurSpace product existence + previous-era estimate 200–400 USD/W + satsearch retail panel three anchors
10yr-GaAs Total Cost	$764.98M	Mixed	Same as above §III	Same as above §VI	Composite output of all JSON inputs	Results in Chapter 6 TCO table; script execution exit code 0
Deprecated kW/t Chain	70/55/45 kW/t → permanently deprecated	Deprecated	*Phase-2 Differential Analysis Preliminary Study.md* §5.1	Same as above §III	`deprecated_notice` field	Deprecation reason in Appendix E §E.1

D.3 Python Script Entry Points

All numerical calculations in this report are from the following two Python scripts. Both are independently Python-modeled, take structured JSON as input, and produce Markdown result documents; all values are reproducible and verifiable.

D.3.1 stage2_power_system_model.py

Function: Orbital eclipse time calculation, battery capacity/mass, array area/mass, PCDU mass. Fixed Blackwell primary line, GaAs/HJT parallel, battery branched (DoD selected per branch).

- **Input files**:
  `stage2_mass_inputs_blackwell_hjt.json`
  `stage2_cost_inputs_blackwell_hjt.json`

- **Output result file**:
  `stage2_power_system_results_blackwell_hjt.md`

- **Key formula chain**: Orbital period $\rightarrow$ eclipse time $\rightarrow$ eclipse-segment load energy $\rightarrow$ battery nominal capacity (/DoD/η)
  $\rightarrow$ recharge energy $\rightarrow$ sunlight-segment additional power $\rightarrow$ PCDU output
  $\rightarrow$ EOL array demand
  $\rightarrow$ array area ($/G/\eta_{\mathrm{eol}}/f_{\mathrm{pkg}}$)

- **Run command**: `pythonstage2_power_system_model.py`

D.3.2 stage2_mass_cost_model.py

Function: Bottom-up calculation of payload mass, all-up satellite 9-subsystem mass/cost per-item summation, three-branch TCO (10-year total window, including manufacturing cost + launch cost + insurance + O&M + decommissioning).

- **Input files**:
  `stage2_mass_inputs_blackwell_hjt.json`
  `stage2_cost_inputs_blackwell_hjt.json`

- **Output result file**:
  `stage2_power_system_results_blackwell_hjt.md`

- **Key formula chain**: Compute payload mass $\leftarrow$ `components[compute_payload].scenario_mass_kg` (direct read, no longer via kW/t denominator) $\rightarrow$ per-satellite mass = Σ(subsystem mass) $\rightarrow$ per-satellite manufacturing cost = Σ(subsystem cost) $\rightarrow$ total manufacturing cost = per-satellite cost $\times$ satellite equivalent $\times$ generations $\rightarrow$ total launch cost = per-satellite mass $\times$ satellite equivalent $\times$ launch unit price $\times$ generations $\rightarrow$ total cost = manufacturing + launch + insurance (8%) + O&M (\$5M/yr $\times$ 10) + decommissioning (\$3M/gen $\times$ generations)

- **Run command**: `pythonstage2_mass_cost_model.py`

D.4 Complete Reference Materials Inventory

The following lists all key public materials cited in this report, organized by topic. Format: name, URL or DOI, and where cited in the report.

D.4.1 Heat Dissipation Physics

- CODATA 2018 Stefan-Boltzmann constant: $\sigma = 5.670374419 \times 10^{-8}$ W m^{-2} K^{-4}. https://physics.nist.gov/cuu/Constants/
- Gilmore, D.G. (ed.) *Spacecraft Thermal Control Handbook*. The Aerospace Press. ISBN: 1-884989-11-X. (Main text Chapter 4, heat dissipation system areal density reference)

D.4.2 Orbital Environment

- WGS84 Earth parameters: $R_{\oplus} = 6378.137$ km, $\mu = 3.986004418 \times 10^{14}$ m^3/s^2.
- ASTM E-490 AM0 Solar Constant: $G = 1361.0$ W/m^2. (Main text Chapter 5, Appendix C §C.2)

D.4.3 Launch Cost

- SpaceX Starship Payload User's Guide (2024). https://www.spacex.com/vehicles/starship/ (Main text Chapter 6, launch capacity baseline)
- FAA Starship/Super Heavy Launch License environmental assessment documents. (Launch frequency constraint source)

D.4.4 GPU Hardware

- Lenovo LP2357 Product Guide: lenovopress (Appendix C §C.3.1)
- NVIDIA HGX B200 PCF Summary: NVIDIA PDF (Same as above)
- NVIDIA DGX B200 User Guide: docs.nvidia.com (Same as above)
- NVIDIA Rubin Space-1 Platform (GTC 2026). (Main text Chapter 3)

D.4.5 Power System

- AzurSpace 4G32 Product Page: AzurSpace Official Site (Appendix C §C.2, GaAs BOL efficiency)
- CEA/INES Irradiation Experiment: CEA/INES News Page (HJT EOL efficiency)
- Ray Barsa (SpaceX Chief Battery Engineer): Battery Power Online (Battery DoD / specific energy / cost)
- NASA Starlink Batteries Reliability Workshop (2025): NASA PDF (Battery specific energy)
- Saft LTO Technical Whitepaper: Saft Whitepaper (LTO DoD / specific energy)

- Terma PCDU Flight Product (BepiColombo MTM): satsearch; satnow (PCDU specific power)
- Huasun Solar (HJT): Huasun Official Site
- Risen Energy HJT 26.61%: PR Newswire
- LONGi HJT 27.81%: LONGi Official Site

D.4.6 Commercial / Economic

- Google Suncatcher (2025 arXiv): arXiv HTML (Communications system mass/cost inference)
- 33FG Research Starlink V2 Mini Analysis: 33FG Research (HJT array areal density range)
- AwesomeAgents.ai GB300 NVL72 Hardware Cost: AwesomeAgents.ai (Compute payload ground cost anchor)
- SpaceX AI1 Public Release (2026-06-08). Multi-source relay: Inkl / Tom's Hardware, Knightli, Indexbox. (Power / orbit / wingspan parameters)
- NVIDIA GTC 2026 Keynote, Space-1 Rubin Platform. (Main text Chapter 4)

D.4.7 Databases

- satsearch.co — Space product database (battery, PCDU, PV panel price lookup)
- NASA NEPP Report Database — III-V space cell radiation degradation rate statistics

E Deprecated Derivation Chains and Parameter Constraints

This chapter records derivation paths that were superseded during the analysis process, parameter values that were corrected, and the applicable physical boundaries or transfer ranges of each parameter. When comparing against previous versions or external references, readers should note that the following numbers are no longer valid.

E.1 kW/t Back-Calculation Chain Deprecation

E.1.1 Deprecated Content

The previous-era derivation path computed payload mass using the following formula:

$$M_{\text{payload}} = \frac{P_{\text{sustained}}}{s_{\text{kW/t}}} = \frac{150 \text{ kW}}{(70/55/45) \text{ kW/t}} \tag{34}$$

Three-scenario values: 2,143 kg (70 kW/t) / 2,727 kg (55 kW/t) / 3,333 kg (45 kW/t). This derivation chain was permanently deprecated in the third round of this report (`close-stage-two-evidence-gaps`, Task 2).

E.1.2 Three Root Causes

1. **Denominator sourced from B1-level second-hand relay (not official NVIDIA/SpaceX)**: 70/55/45 kW/t originates from second-hand media relay or speculation about "SpaceX AI1 specific power", not from official NVIDIA GPU specifications nor from official SpaceX public documentation. Under the four-tier evidence system, B1-level relay does not possess sufficient evidentiary strength to independently support key model inputs.

2. **Fundamental ambiguity in the domain of definition**: What mass the "t" (ton) in the denominator corresponds to has never been locked down by primary-source material. If it corresponds to all-up satellite mass, then compute payload mass = all-up satellite mass—which is physically impossible (would exclude the existence of power, thermal, structure, propulsion, and all other subsystems). If it corresponds to the compute payload subsystem, then the specific power value of 70 kW/t needs independent verification, but no public anchor is available for verification.

3. **Circular reference risk**: Two usages simultaneously appeared in previous-era materials: (a) back-calculating compute payload mass from 70 kW/t; (b) using the back-calculated mass to prove 70 kW/t is reasonable. This logically forms a closed loop—first assuming the denominator, then using the calculation result to verify that same denominator. Circular reference is a fundamental defect in an evidence chain.

   Supplementary judgment: The above three root causes are not issues "only discovered in the third round." The kW/t back-calculation chain used "B1-level second-hand relay" as its source from the first round (`produce-stage-two-differen`
   and was flagged as "denominator definition suspect / P0 Gap 2" in the second round (`refine-stage-two-blackwell-hjt-model`). The first

two rounds retained the chain as a temporary placeholder "pending an alternative"—this was an explicit execution judgment, not a missed detection. After Task 2 of the third round completed the bottom-up mass chain reconstruction, a verifiable alternative existed, and the deprecation formally took effect.

E.1.3 Replacement Path

kW/t back-calculation $\longrightarrow$ bottom-up ground hardware anchor + three space-hardening increments
New chain methodology detailed in Appendix C §C.3. New values (V30 per-parameter adjudication): optimistic 1,200 kg / baseline 1,500 kg / conservative 2,000 kg. Evidentiary strength upgraded from B1 second-hand relay to Tier 2 (ground hardware A1 anchor) + Tier 3 (space-hardening increment inference).

Previous-era analysis numbers (2,143/2,727/3,333 kg) are archived in this appendix solely as deprecation reasons. Compute payload mass in the main text uses the V30 adjudicated values (1,200/1,500/2,000 kg).

E.2 Derivation Path Replacement Relationship Table

Table 47: Derivation Path Replacement Relationship Master Table (1/2)

Previous-Era Path / Value	**Deprecation Reason**	**This Report's Method / Value**	**This Report's Method Section**
`compute_specific_power_kw_per_t` = 70/55/45 kW/t	B1-level second-hand relay, domain-of-definition ambiguity, circular reference	Direct read of `components[compute_payload].scenario_mass_kg`, no longer via kW/t denominator	Appendix C §C.3
Compute Payload Mass = 2,143/2,727/3,333 kg	Derived from kW/t back-calculation chain; denominator deprecated	1,200/1,500/2,000 kg (V30 bottom-up adjudication)	Appendix C §C.3.3
`battery_cost_per_kwh` = $400/kWh	∼225× order-of-magnitude contradiction with public space-grade battery list prices	NMC: $200/kWh fixed (V36 Starlink transfer); LTO: $260/350/400/kWh (V37 proportional compression)	Appendix B §B.2
HJT "can only be downgraded" / "prohibited from entry into primary model"	Route positioning contradicted by Task 3 evidence package (17 URLs + 12 physical chain closures)	"Parallel maturation route"—physical chain fully comparable, cost chain closure separately noted	Appendix B §B.4
`stage2_mass_cost_results.md`	Previous-era aggregate model results based on kW/t chain	`stage2_economic_branches_blackwell.md`: three-branch per-item expansion, each subsystem independent mass + cost, per-item traceable	—
`stage2_power_system_results.md`	Did not include HJT parallel route, did not include battery branching	`stage2_power_system_results_blackwell_hjt.md`: Blackwell primary line, GaAs/HJT parallel, battery three-branch	—

Table 48: Derivation Path Replacement Relationship Master Table (2/2)

Previous-Era Path / Value	Deprecation Reason	This Report's Method / Value	This Report's Method Section
`stage2_architecture_reconciliation.md`	Rubin downgraded to branch-line trace, no longer enters primary model	No direct replacement; Rubin architecture efficiency remapping retained only as external reference direction, does not enter primary model	—
PCDU Specific Power = 10.0 kW/kg	Terma flight product anchor revealed overestimation by $\sim 18\times$ (originally 150kW/10.0=15kg, corrected to 209.8kW/0.5≈420kg)	1.0/0.5/0.2 kW/kg (V25, Tier 4)	Appendix C §C.2.5
Bus Coefficient = 0.18/0.22/0.28	Starlink V2 Mini mass rough decomposition reference ratio ∼0.38; original values too low	0.25/0.35/0.45 (V21, Tier 4)	Appendix B §B.3
Propulsion Mass = 180/220/350 kg	Starlink V2 Mini propulsion fraction reference suggests upside room	200/350/500 kg (V22, Tier 4)	Same as above
Communications Mass = 150/180/300 kg	4 rounds of search + Google Suncatcher confirms high bandwidth demand	200/300/450 kg (V23, Tier 4)	Same as above
Battery DoD = single 40% (no lifetime/chemistry split)	Research team held that one-size-fits-all should not be applied; required split by electrochemistry measurements	5yr NMC 40% / 10yr Li-ion 25% / 10yr LTO 80% (V10, Tier 3+4)	Appendix B §B.2; Appendix C §C.2.2

E.3 Superseded Parameters and Current Value Comparison

The following parameters or formulations were superseded or corrected during the analysis process. The purpose of listing them is: (1) to enable readers to know which numbers are no longer valid when comparing against previous versions or external studies; (2) to exhibit key divergence points identified and corrected during the analysis process.

1. **The kW/t back-calculation chain (70/55/45 kW/t) has been deprecated.** This chain used second-hand relay as denominator and had domain-of-definition ambiguity and circular reference risk (see §E.1 for details). Current compute payload mass is instead derived directly from the ground hardware anchor (Appendix C §C.3).

2. **Previous-era compute payload mass values (2,143/2,727/3,333 kg) have been replaced.** Current values are 1,200/1,500/2,000 kg (V30 bottom-up

adjudication); the previous three values are archived solely as deprecation reasons in §E.1.

3. **The JSON parameter `compute_specific_power_kw_per_t` has been removed.** In the Python scripts, compute payload mass is now read directly from the payload mass field in JSON, without going through the kW/t intermediate quantity.

4. **The battery cost item_id `battery_cost_per_kwh` has been split into two items.**
   Current item_ids are `battery_cost_per_kwh_nmc` (NMC $200/kWh, Tier 3) and `battery_cost_per_kwh_lto` (LTO $260/350/400/kWh, Tier 4). The previous unified value $400/kWh had a $\sim$225$\times$ contradiction with the satsearch price anchor and has been deprecated.

5. **HJT route positioning has been corrected from the previous-era "downgraded / prohibited" to "parallel maturation route."** HJT and GaAs physical chains are fully comparable (Appendix C §C.2.4), with cost chain closure separately noted. HJT does not satisfy the AI1 70m wingspan geometric constraint (70m $\times$ 18.5m not realizable) and therefore is not a feasible primary route—this is a conclusion from the geometric constraint, not a negation of the technology route.

6. **The following result documents have been superseded:**

   - `stage2_mass_cost_results.md`
     → `stage2_economic_branches_blackwell.md`
   - `stage2_power_system_results.md`
     → `stage2_power_system_results_blackwell_hjt.md`
   - `stage2_architecture_reconciliation.md` (Rubin downgraded to branch-line trace, no longer enters primary model)

7. **Parameters marked as "external reference only" do not constitute primary chain closure conclusions.** This applies to LTO battery cost (Tier 4), HJT full TCO (currently only directional estimate provided), and all 14 Tier 4 parameters including bus coefficient / propulsion mass / communications mass / PCDU cost, etc.

8. **The HJT backfill estimate $\sim$\$1,005M is a directional indicator, not a closed precise TCO.** This value indicates the directional conclusion that HJT cost does not outperform GaAs; the main text labels it as "backfill estimate" and separately declares the degree of closure.

E.4 Parameter Four-Tier Classification and DoD Boundary Constraints

E.4.1 Parameter Four-Tier Classification Constraints

All 47 parameters were re-classified item-by-item in Task 4, with the following statistics:

Table 49: Parameter Four-Tier Classification Statistics

Tier	Count	Fraction	Citation Convention in Main Text
Tier 1 (Physical Derivation Chain)	5	11%	Can be directly cited as main-text conclusions; no additional uncertainty statement required
Tier 2 (Authoritative Source)	7	15%	Can be cited; recommend noting source type (multi-source cross-verified or single authoritative source)
Tier 3 (Transferable Evidence)	17	36%	Can be cited; must include transfer limitation note ("this value comes from XX transfer, basis is YY")
Tier 4 (Reasonable Inference)	14	30%	Can be cited; must include inference basis and failure risk statement; does not enter executive summary
Prohibited from Model Entry	1	2%	Does not enter main-text analysis; related analysis uses the replacement new value
Deprecated	3	6%	Archived only in this appendix; does not enter main text
Total	**47**	**100%**	

Tier 4 parameters accounting for 30% (affecting $\sim$35% of manufacturing cost items) is an objective consequence of the limited public disclosure of AI1 information, not an execution deficiency. All 14 Tier 4 parameters carry inference basis and failure risk statements (see internal parameter adjudication document for details), but should not be cited as confirmed figures.

E.4.2 DoD Boundary Constraints

The system boundary of DoD (Depth of Discharge) was defined in Task 5:

- DoD is an independent parameter of the battery energy storage subsystem, determining the battery nominal capacity $E_{\text{battery}} = P_{\text{sustained}} \times t_{\text{eclipse}} / (\text{DoD} \times \eta_{\text{discharge}})$, thereby affecting battery mass and cost.

- The relationship between DoD and GaAs/HJT is an indirect coupling: the photovoltaic array must meet battery recharge requirements, but DoD itself does not vary with the PV route. The battery configuration of the same satellite does not change with the PV cell chemistry system. DoD is uniformly expressed in the report as "the DoD value of the battery subsystem (shared across GaAs and HJT routes)."
- DoD indirectly affects all-up satellite mass and launch cost through the mass chain (DoD$\downarrow$ $\rightarrow E_{\text{battery}} \uparrow \rightarrow M_{\text{battery}} \uparrow \rightarrow M_{\text{power}} \uparrow \rightarrow M_{\text{total}} \uparrow \rightarrow C_{\text{launch}} \uparrow$); this is the penetrating impact of the battery subsystem on the entire satellite.

E.5 Parameter Tier Reading Guide

All key numbers in this report carry evidence tier labels (Tiers 1–4, plus "Prohibited from Model Entry" and "Deprecated"). The following is a reference guide for readers to judge the reliability of each number when reading the main text.

Tier 1 (Physical Derivation Chain) Can be accepted directly. Source is physical law or engineering necessity constraint (e.g., S-B equation-derived radiator area, orbital-mechanics-derived eclipse time). No additional uncertainty statement required.

Tier 2 (Authoritative Source) Can be accepted, but recommend attention to source type. A single authoritative source (e.g., AzurSpace 4G32 BOL efficiency) and multi-source cross-validation (e.g., three media outlets' consistent relay of AI1 power parameters) have differing evidentiary strength, which has been noted in the main text.

Tier 3 (Transferable Evidence) Can be referenced, but the transfer process has limitations. Typical example: Starlink battery data transferred to AI1 scenario—the physical logic of the transfer holds (both are LEO space batteries), but mission profile differences (communications vs. compute) may introduce bias. In the main text, Tier 3 parameters all carry transfer basis and limitation statements.

Tier 4 (Reasonable Inference) Used only to fill model completeness; must not independently serve as conclusion support. Involves 14 parameters (radiator areal density, bus coefficient, propulsion/communications mass and cost, etc.), each annotated with inference basis and primary failure risks. Tier 4 parameters are not used as core conclusions in the main-text executive summary.

Prohibited from Model Entry / Deprecated Refuted by public evidence or unsupported, or derivation path permanently superseded by this report. Archived only in this appendix; does not appear in main-text analysis.

Tier 4 accounts for 30% of the current 47 parameters (14/47), affecting $\sim$35% of manufacturing cost items. This is an objective consequence of the limited public disclosure of AI1 information: under public evidence constraints, all 14 parameters lack the conditions for upgrade to higher tiers, but each has been embedded with inference basis and failure risk statements (see internal parameter adjudication document for details). Readers citing numbers from this report should pay special attention to the inferential nature of these 14 parameters.

E.6 Key Parameter Constraint Conditions

The following constraint conditions arise from physical boundaries or the applicable range of transfer sources; extrapolation beyond range may lead to biased conclusions:

- **Radiator Areal Density (4.0/5.0/6.5 kg/m^2)**: Applicable to rigid panel designs. If AI1 adopts non-conventional solutions such as deployable film radiators or droplet radiators, actual areal density may deviate by more than 50%.
- **Bus Coefficient (0.25/0.35/0.45)**: Based on Starlink V2 Mini mass rough decomposition (ratio $\sim$0.38), applicable to $\sim$8t-class satellites. Must not be linearly extrapolated to hundred-tonne-class or distributed-fragmented-cluster architectures.
- **GaAs Array Areal Density (1.5 kg/m^2, panel-only)**: Includes cells + cover glass + interconnects + substrate only; excludes deployment mechanisms (hinges/motors/hold-down release). Deployment mechanism mass is accounted separately in the model.
- **HJT Route Overall Positioning**: Physical chain closure is comparable to GaAs, but does not satisfy the AI1 70m wingspan geometric constraint (required $\sim$18.5m average width not realizable). Its role in the report is as a parallel technology reference, used to demonstrate the opportunity cost of not adopting GaAs.